Fourth Year Pharmacy Students – Assiut University
Elective Medicinal Chemistry Course – First Semester

# Drug Design and Discovery
# Lecture 1

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# DRUG DESIGN AND DEVELOPMENT

## Stages

1) Identify target disease
2) Identify drug target
3) Establish testing procedures
4) Find a lead compound
5) Structure Activity Relationships (SAR)
6) Identify a pharmacophore
7) Drug design- optimising target interactions
8) Drug design - optimising pharmacokinetic properties
9) Toxicological and safety tests
10) Chemical development and production
11) Patenting and regulatory affairs
12) Clinical trials

# Why drug design?

The drug should have a good selectivity for its target

2. The drug should have a good level of activity for its target

3. The drug should have minimum side effects

4. The drug should be easily synthesized

5. The drug should be chemically stable

6. The drug should have acceptable pharmacokinetics properties

7. The drug should be non-toxic

# Aspects in drug design

**1.Pharmacodynamics properties:**

To improve the drug's ability to reach its target

**2.Pharmacokinetics properties:**

To have acceptable life time .

# Improving Pharmacodynamics properties:

To optimise binding interactions with target and induce selectivity

## Strategies:

1-Vary alkyl substituents.

2-Vary aryl substituents.

3-Extension.

4-Chain extensions / contractions.

5-Ring expansions / contractions.

6-Ring variation.

7-Isosteres.

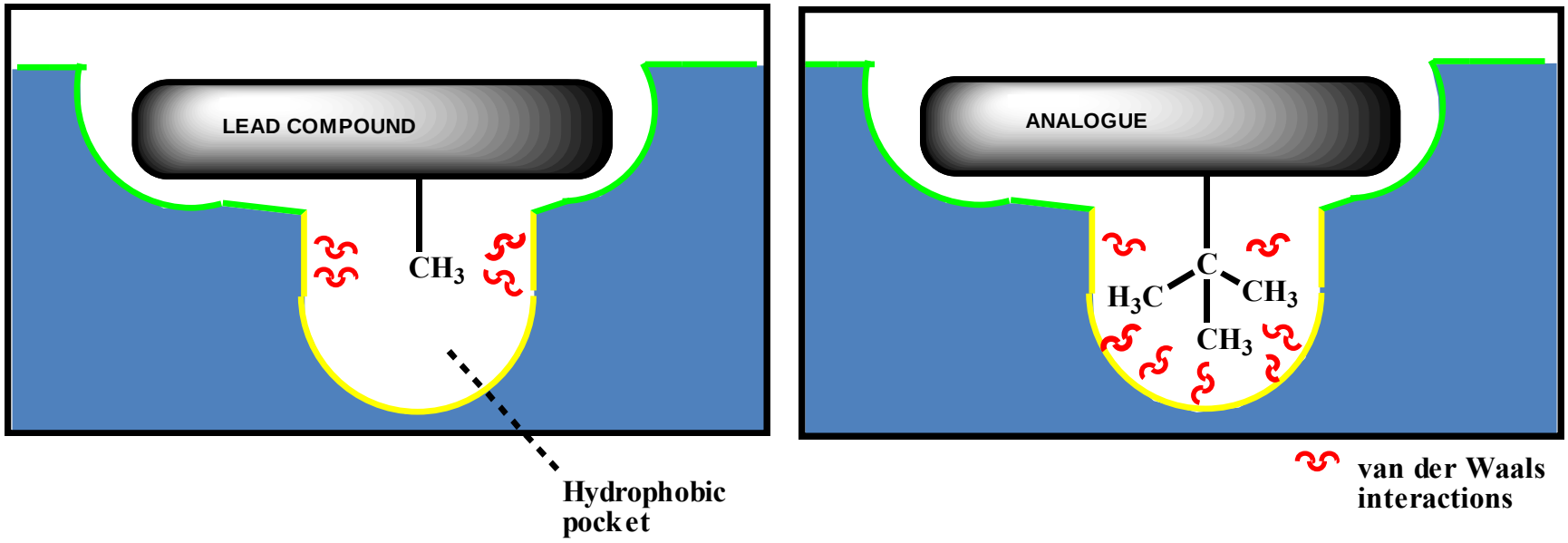
8-Simplification.

9-Rigidification.

# 1 - Vary Alkyl Substituents

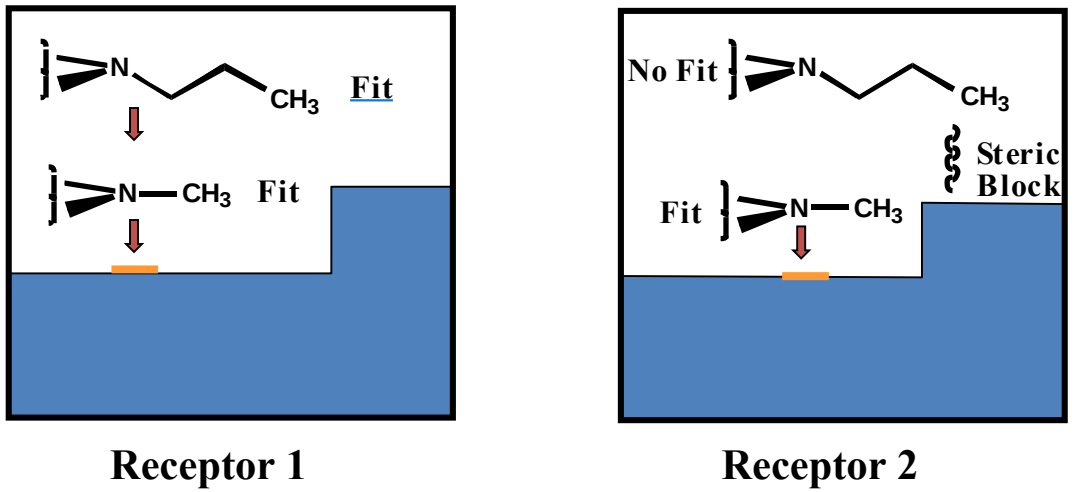
# 1-To optimise interaction

Alkyl group in lead compound may interact with hydrophobic region in binding site

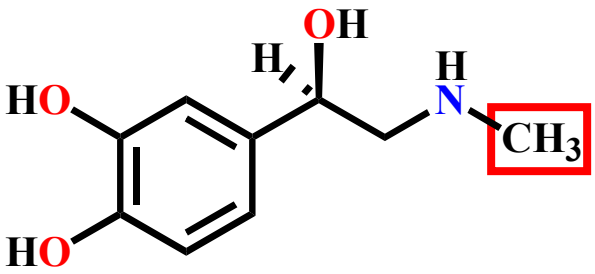


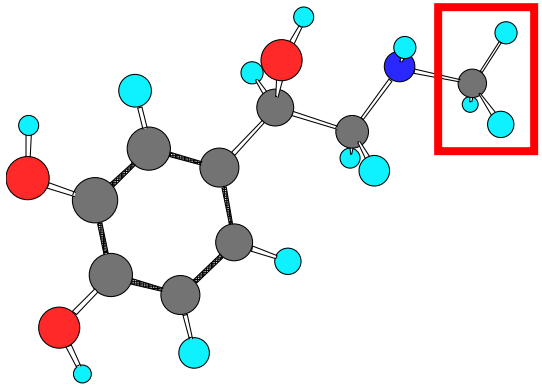
# 2-To introduce selectivity

Selectivity of adrenergic agonists and antagonists for b-adrenoceptors over α-adrenoceptors

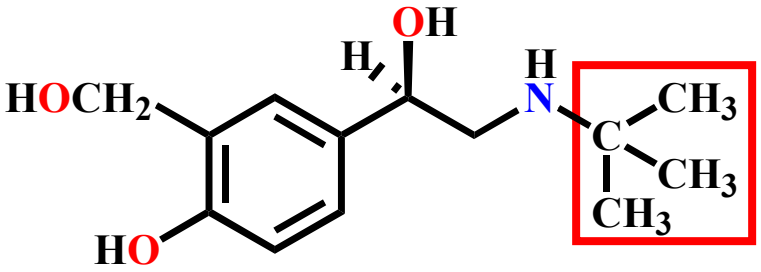


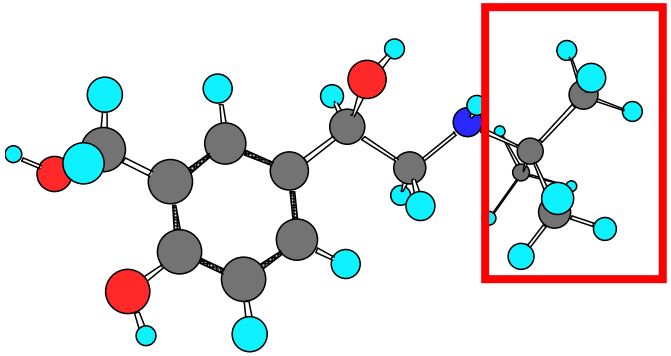
**Adrenaline**



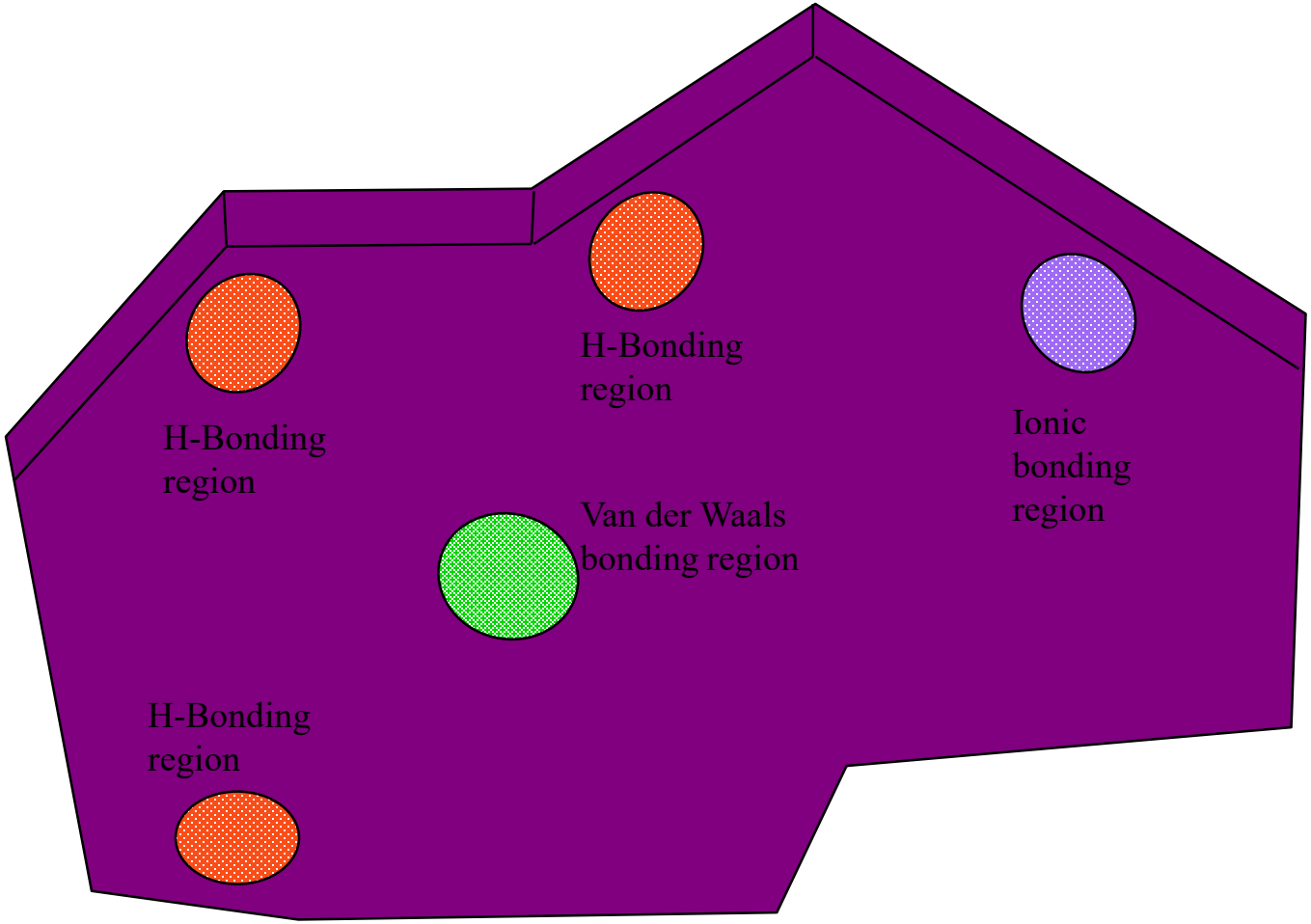



**Salbutamol (Ventolin)**
**(Anti-asthmatic)**

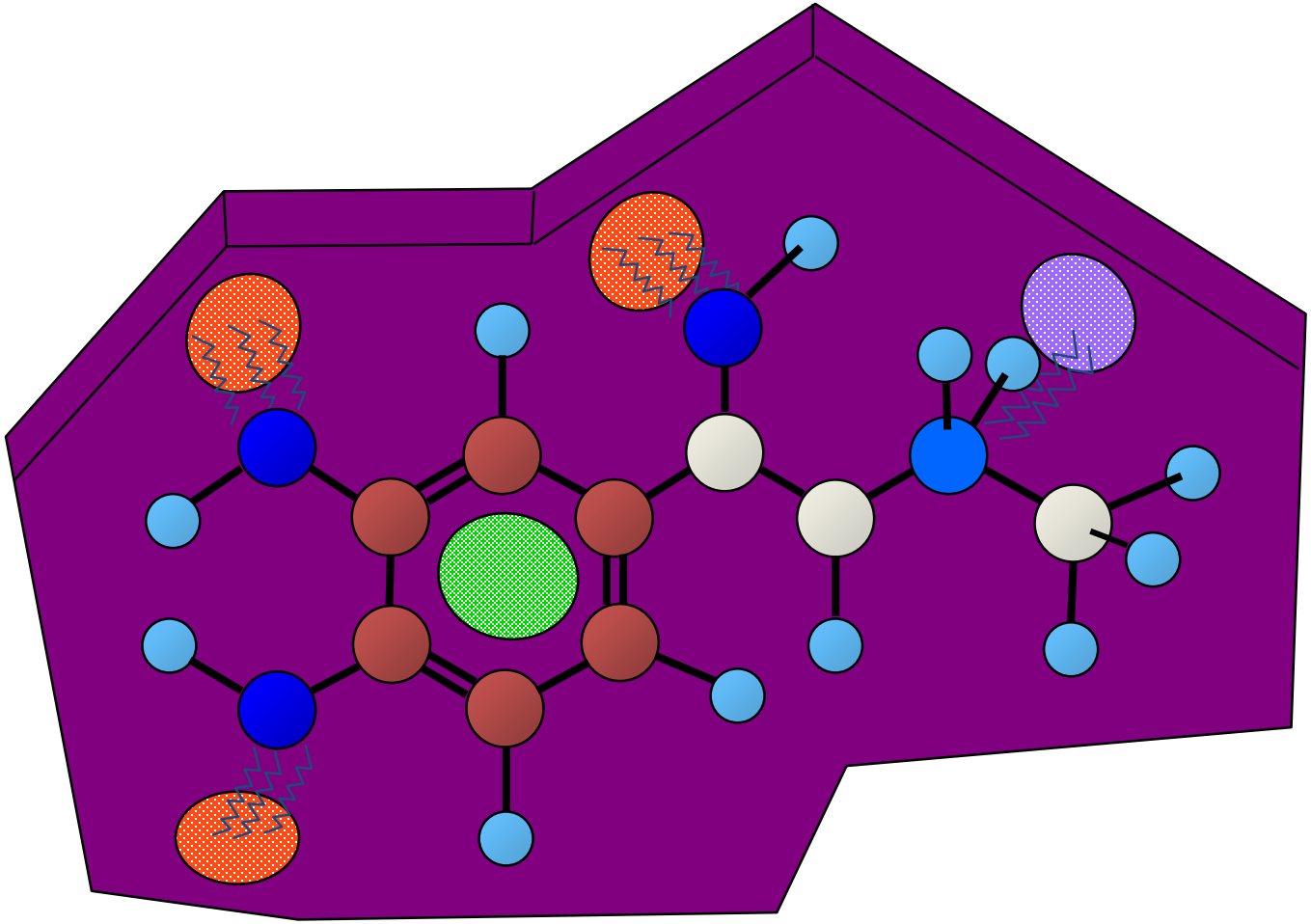


# α-Adrenoceptor

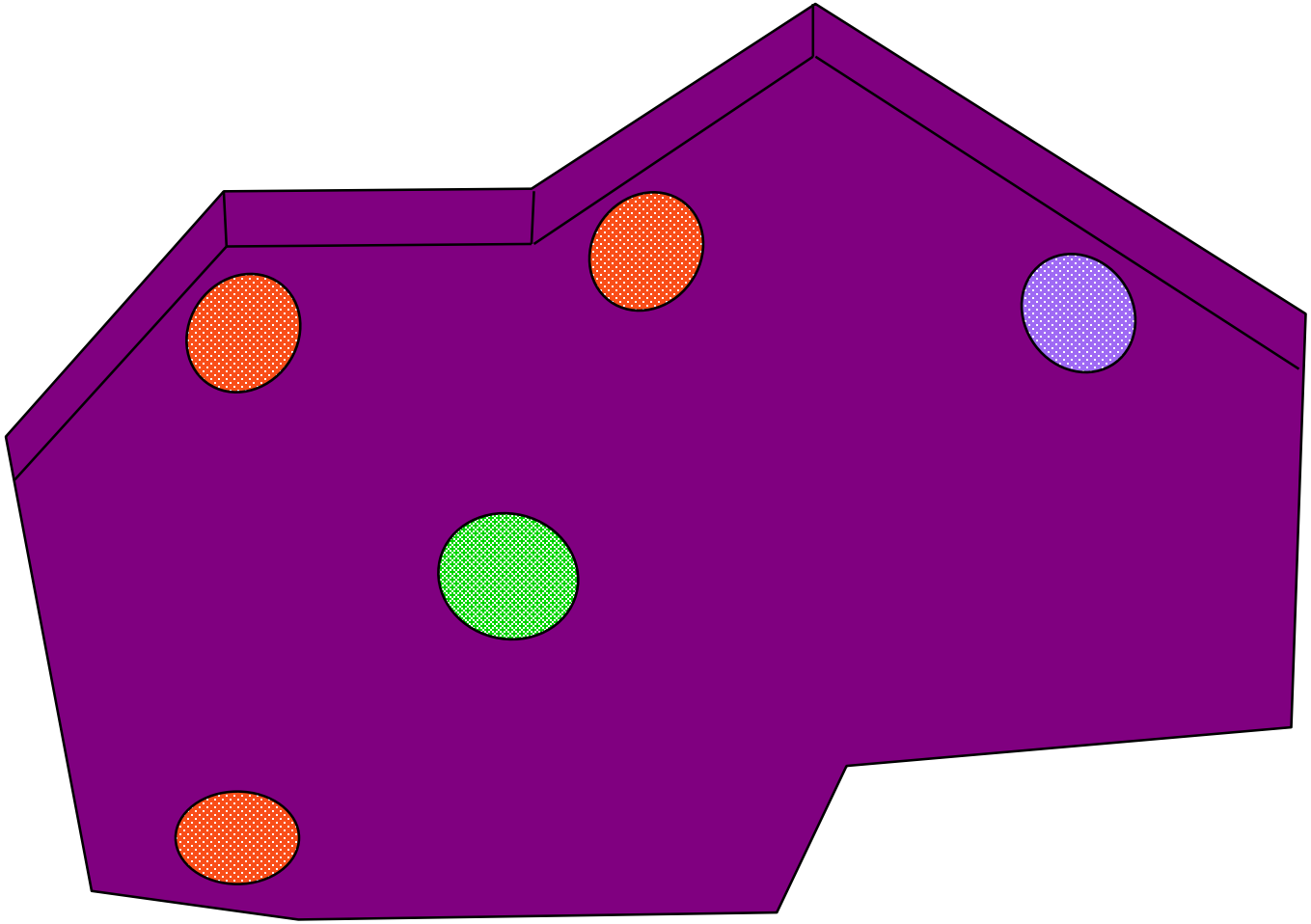


# α-Adrenoceptor

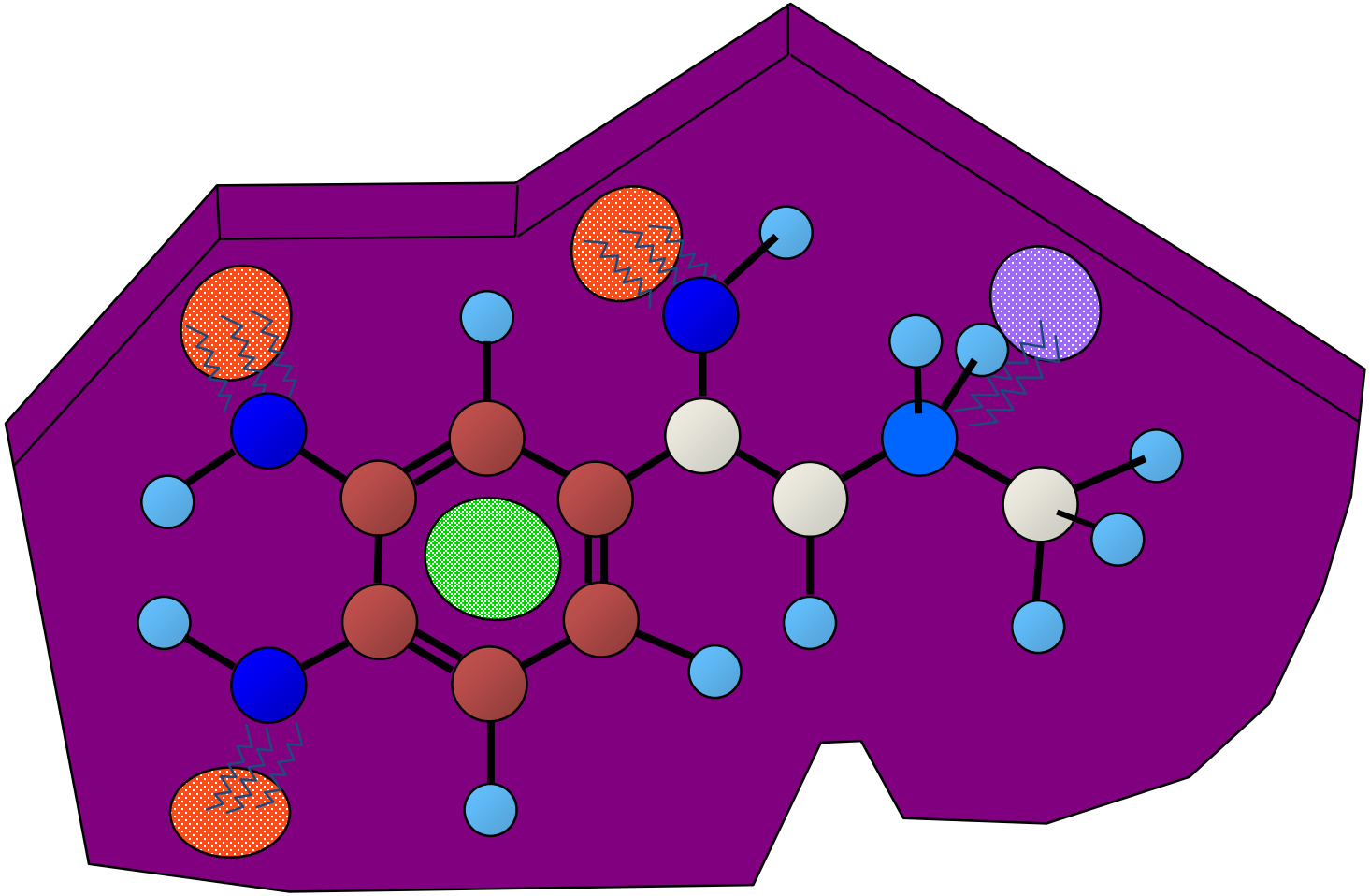


**ADRENALINE**

# α-Adrenoceptor

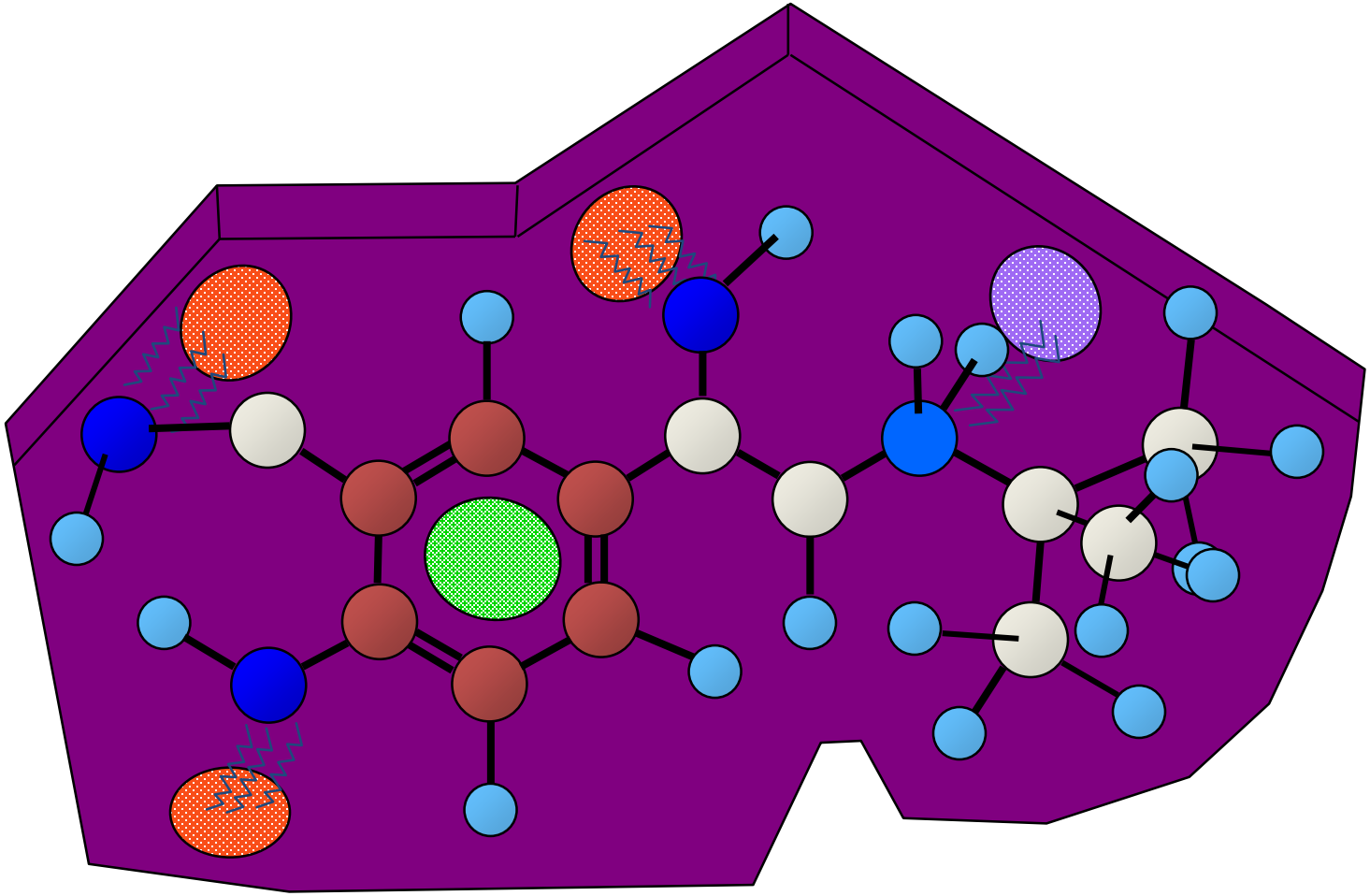


# β-Adrenoceptor



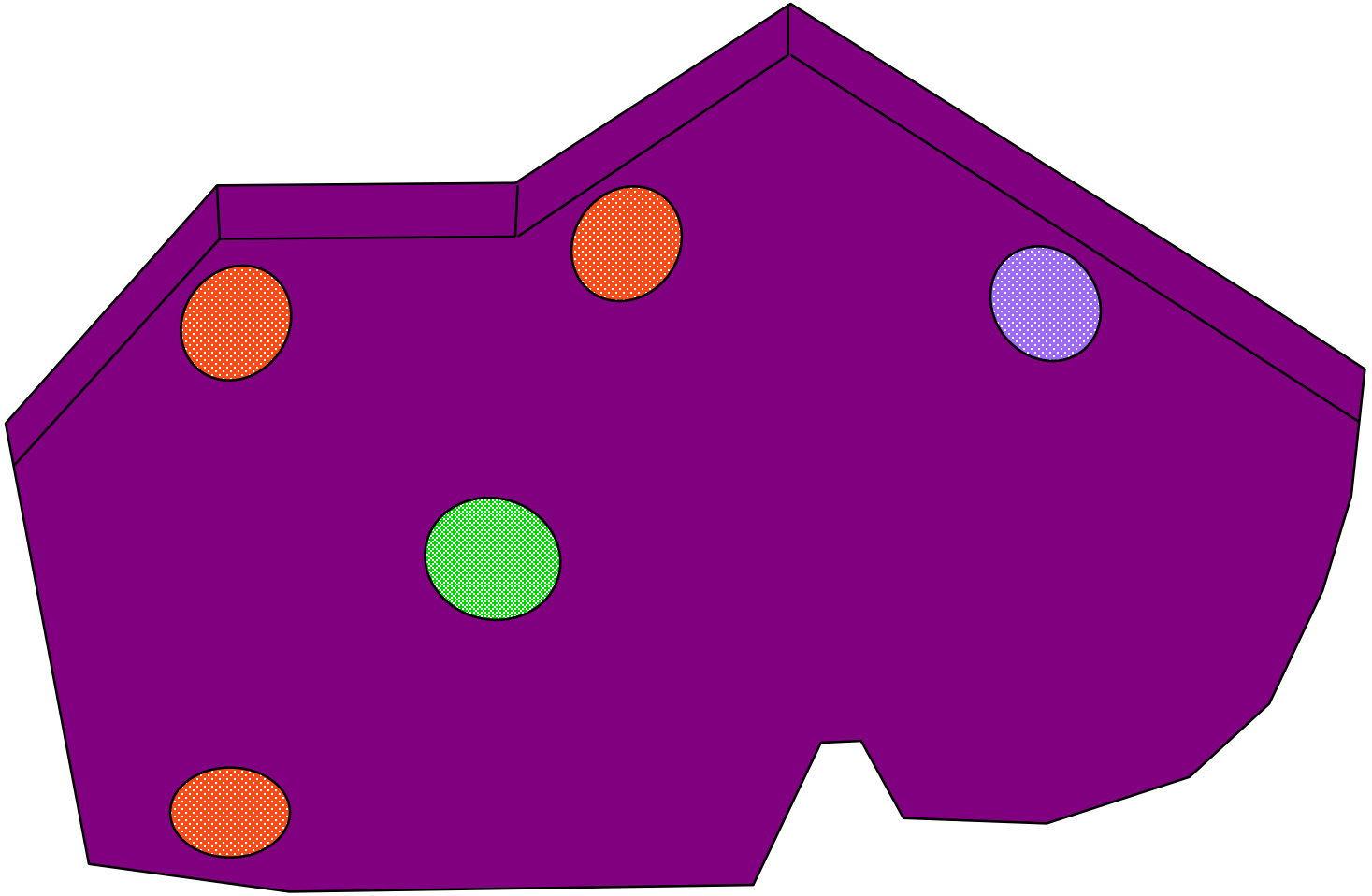
**ADRENALINE**

# β-Adrenoceptor

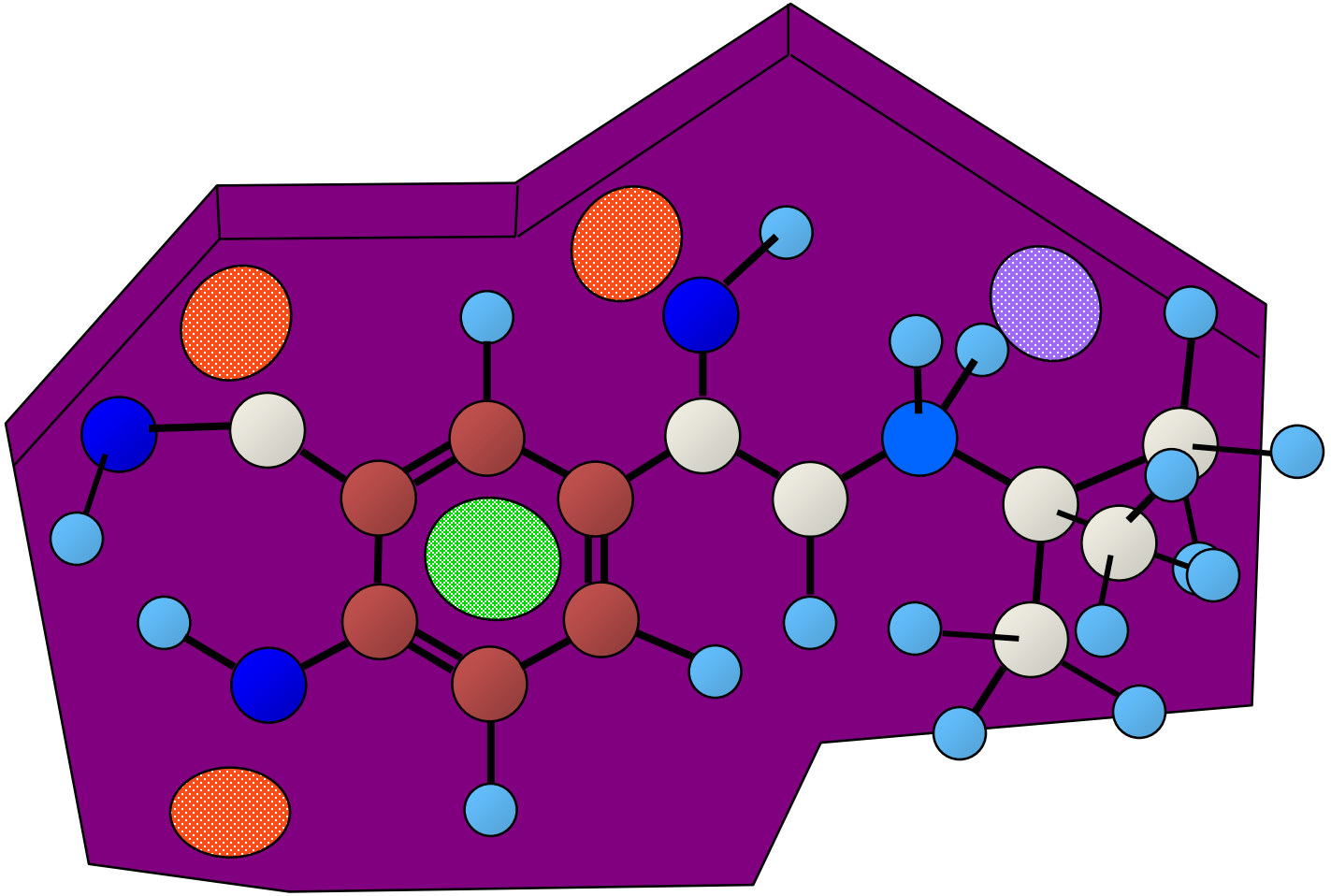


**SALBUTAMOL**

# β-Adrenoceptor

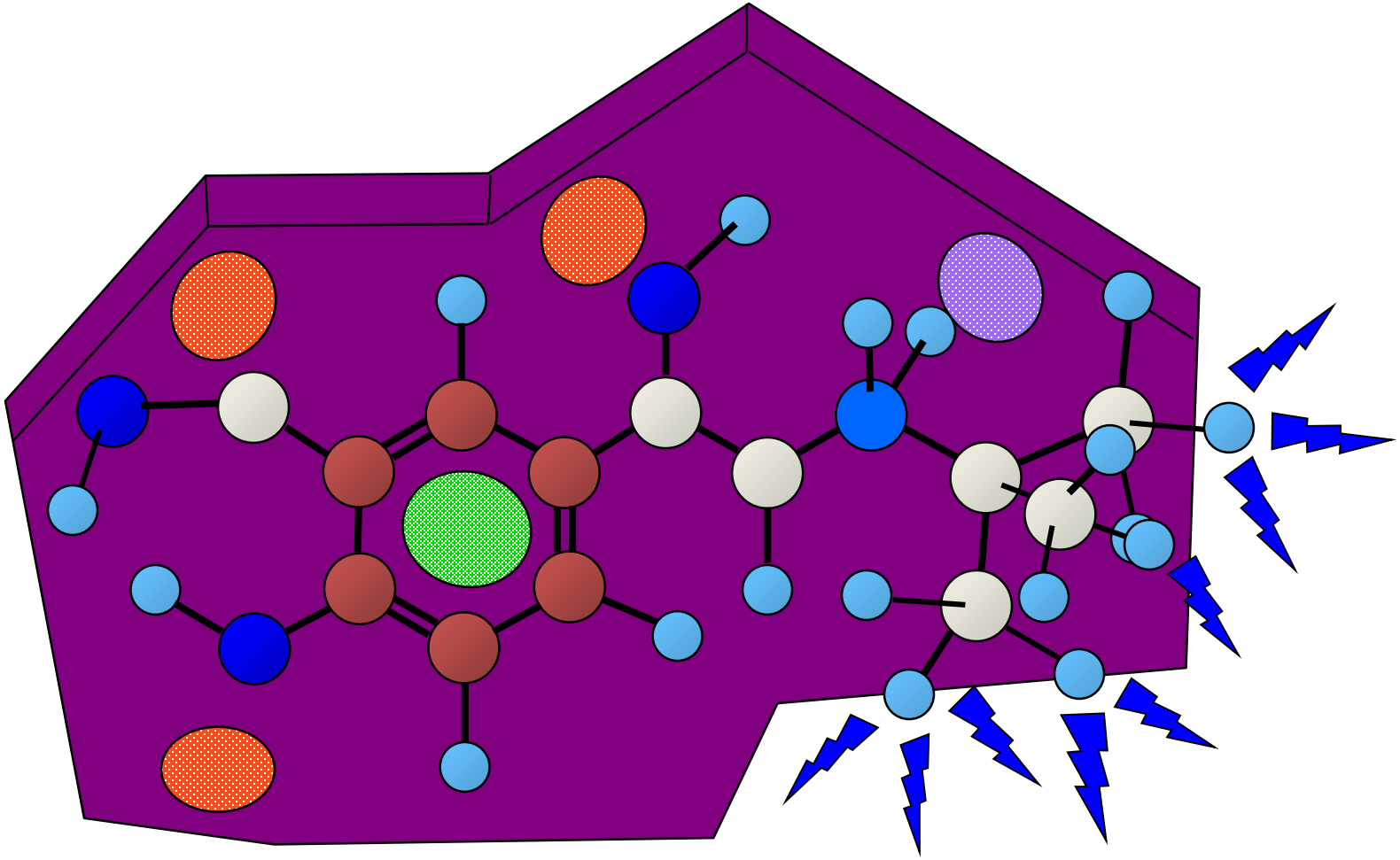


# α-Adrenoceptor



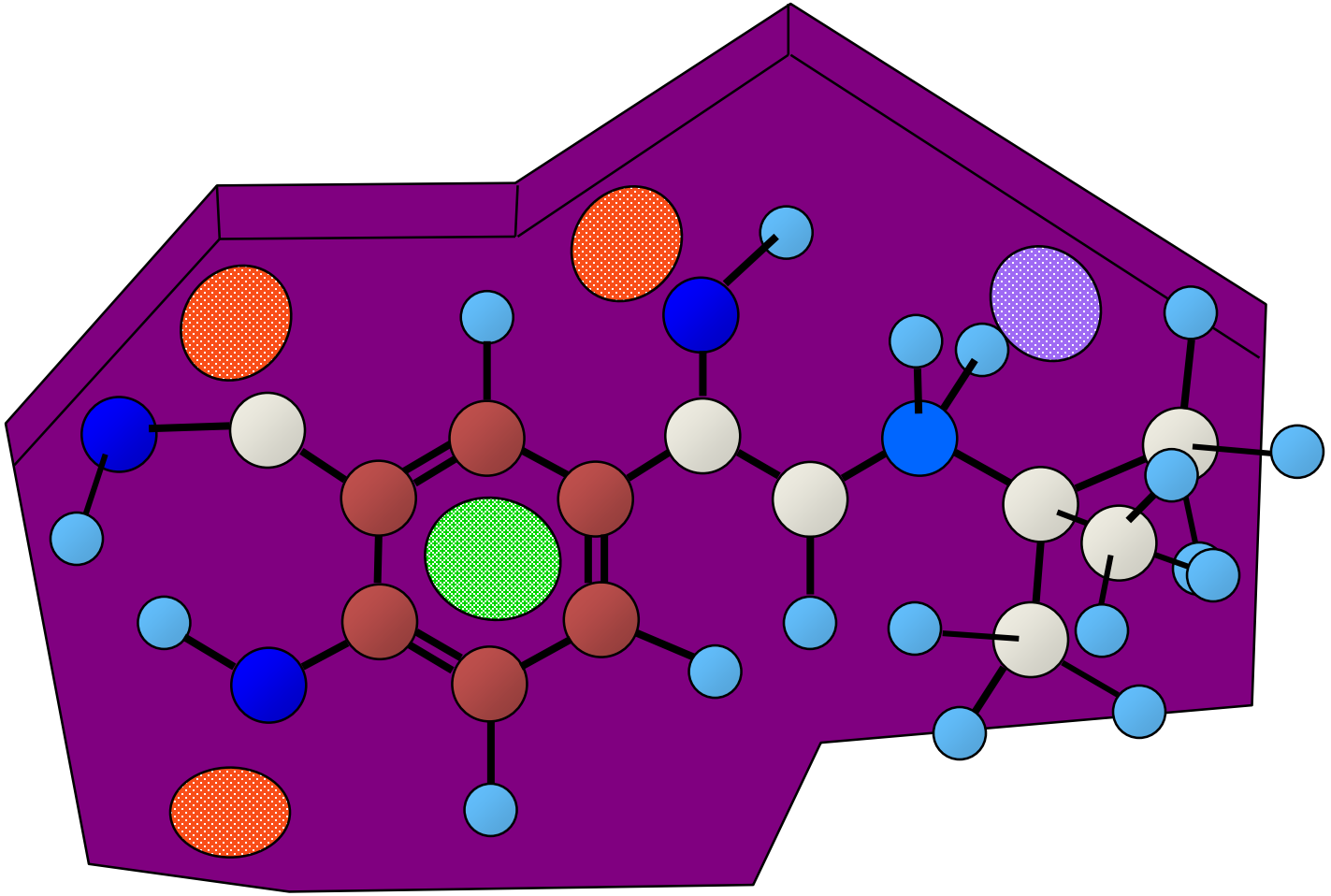
**SALBUTAMOL**

# α-Adrenoceptor



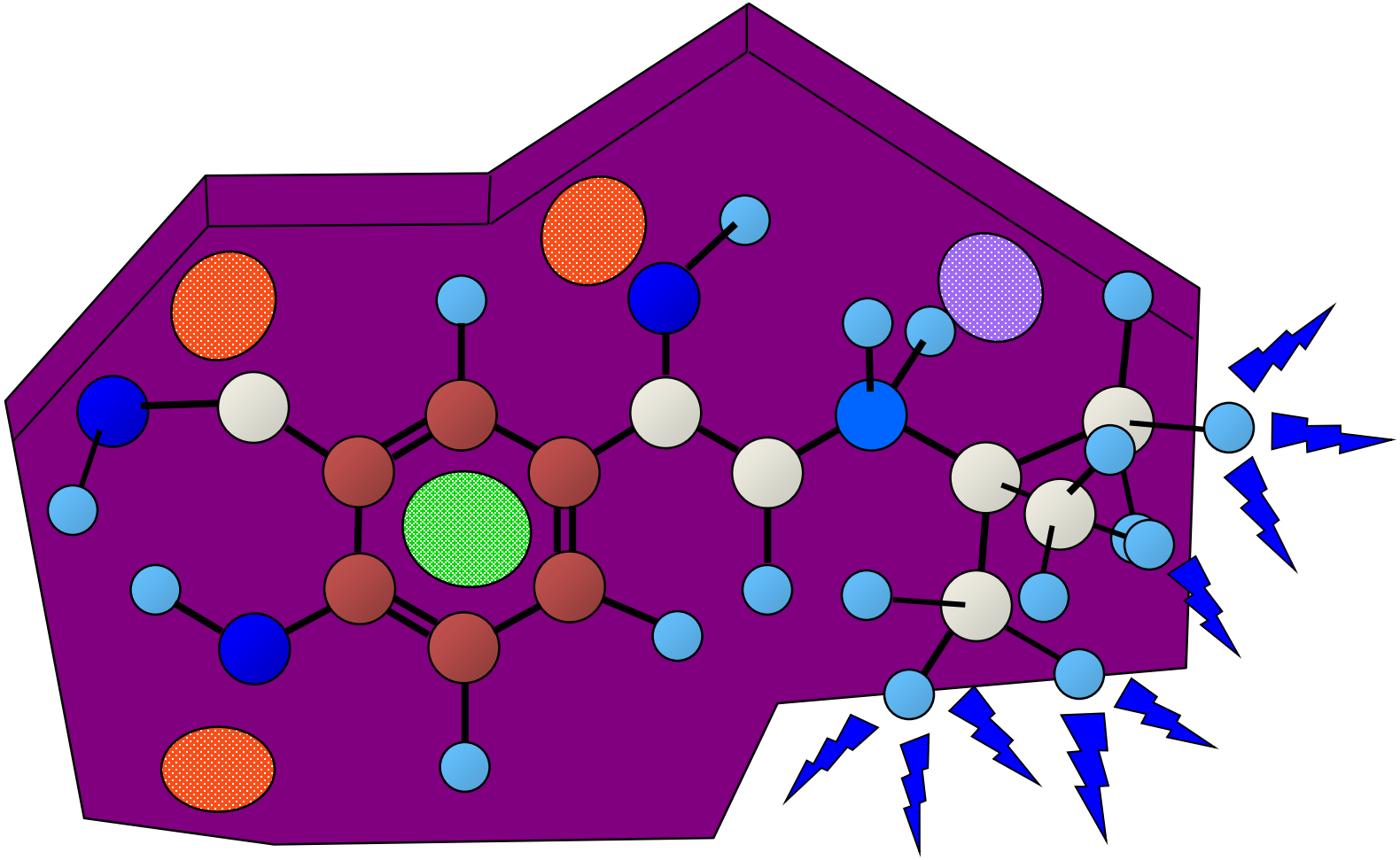
**SALBUTAMOL**

# α-Adrenoceptor



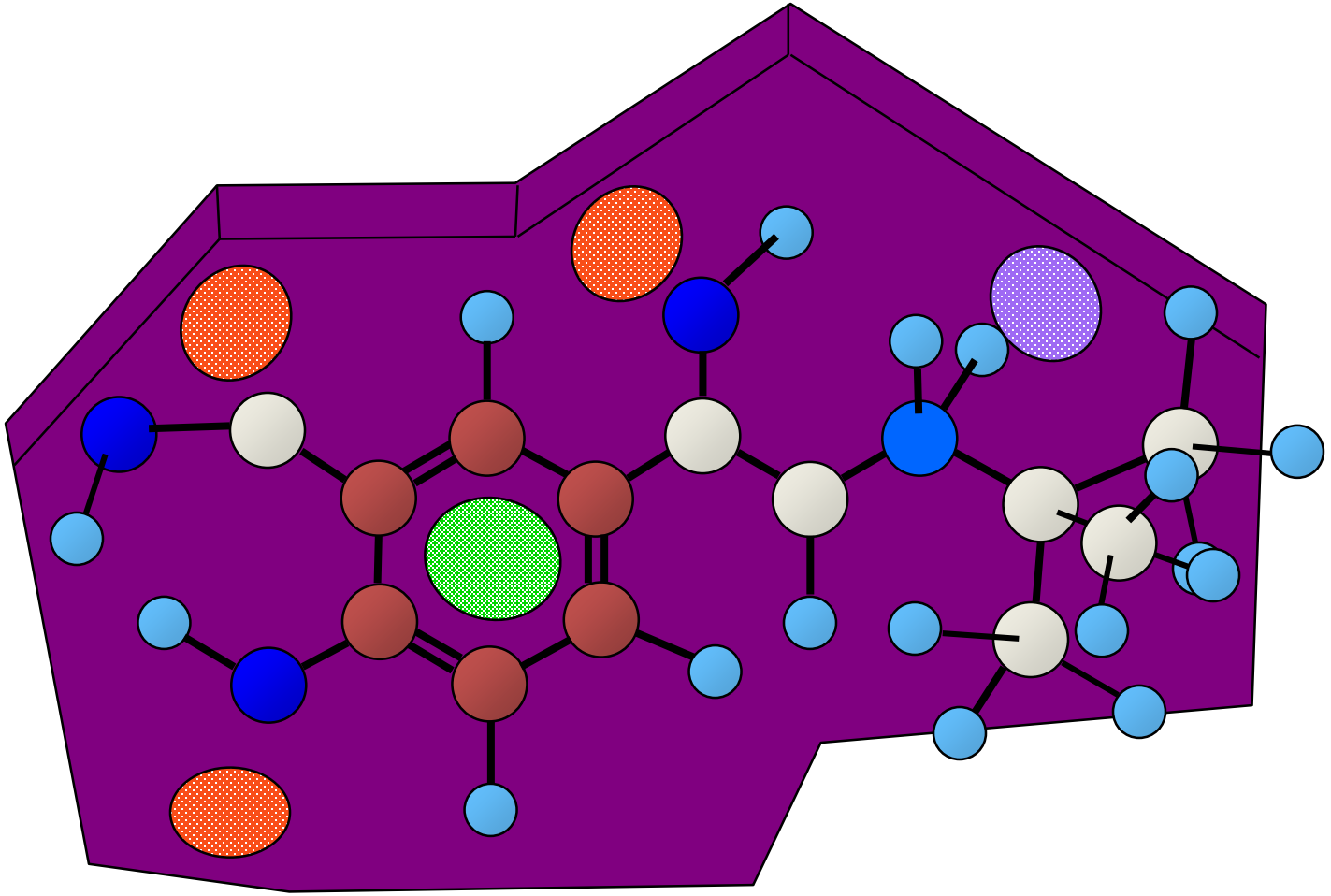
**SALBUTAMOL**

# α-Adrenoceptor



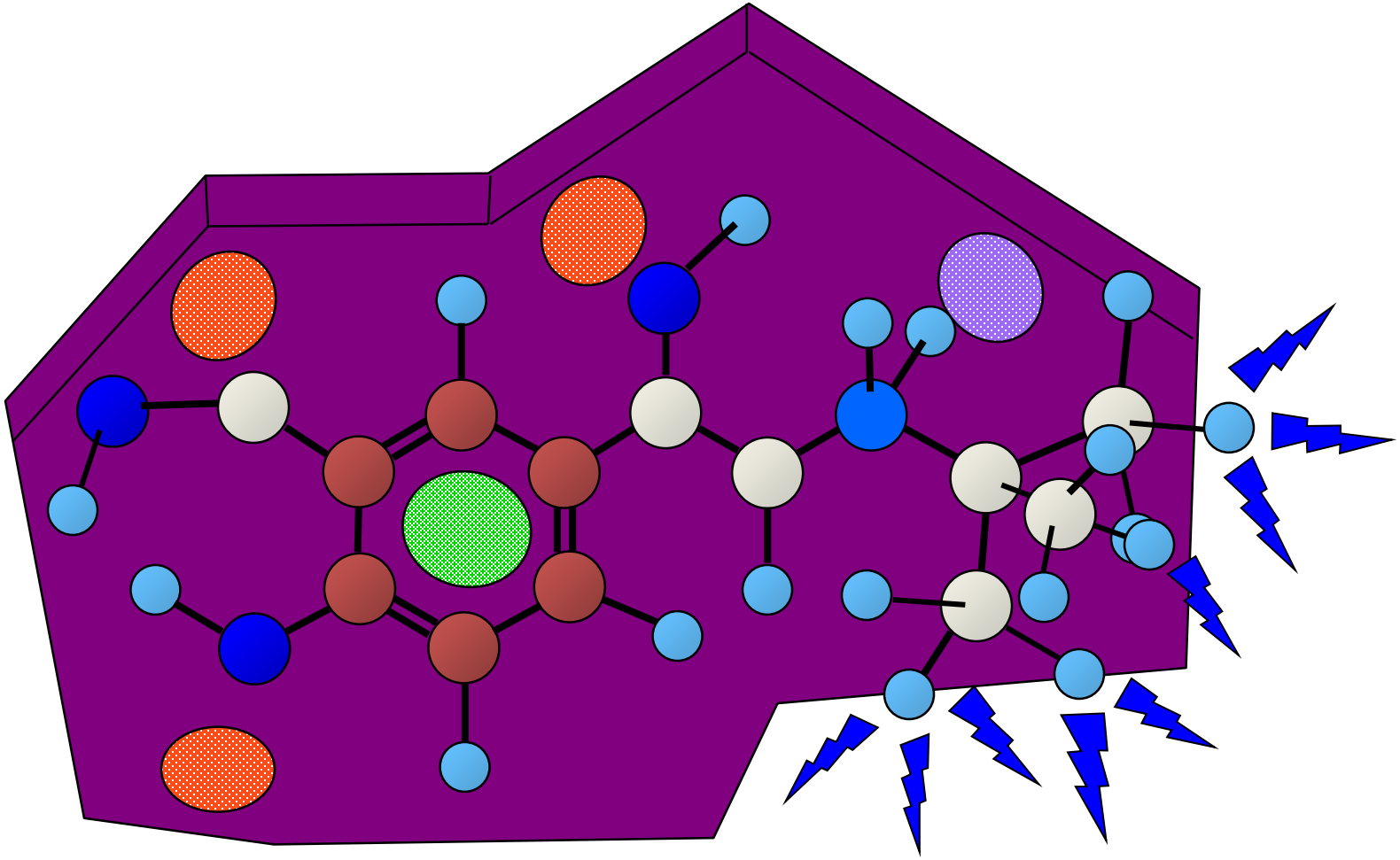
**SALBUTAMOL**

# α-Adrenoceptor



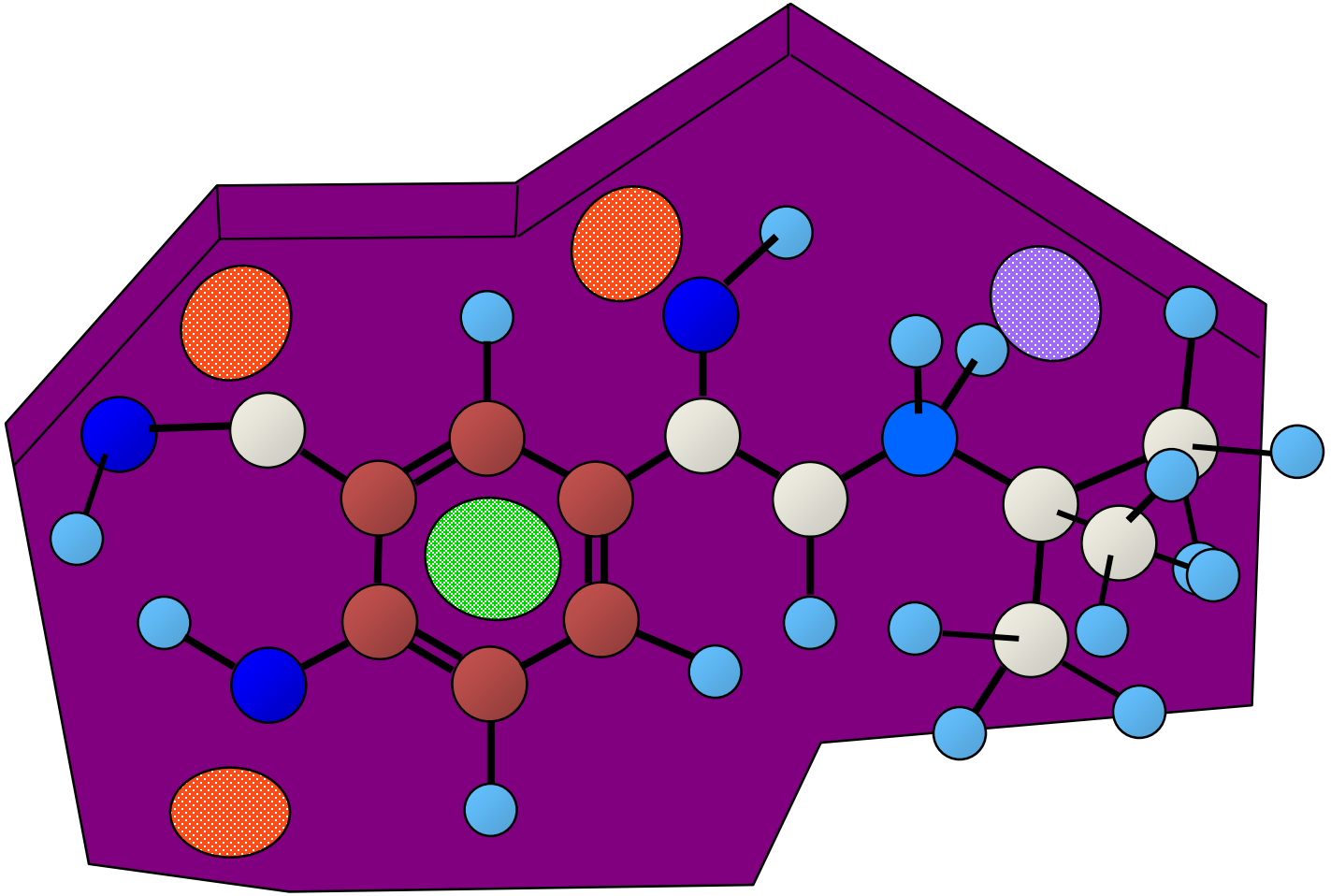
**SALBUTAMOL**

# α-Adrenoceptor



**SALBUTAMOL**

# α-Adrenoceptor



**SALBUTAMOL**

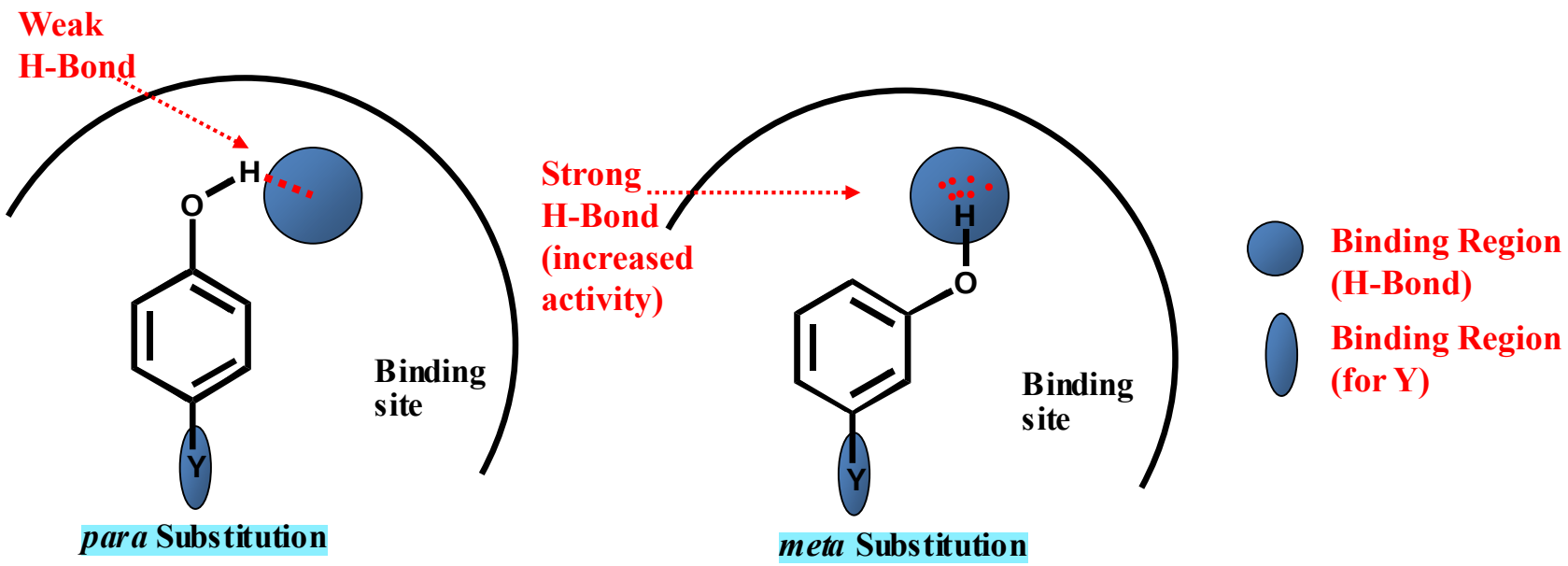
## Notes on synthetic feasibility of analogues

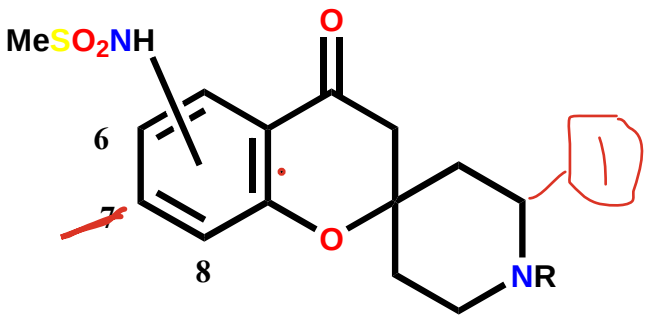
- Feasible to remove alkyl substituents on heteroatoms and replace with other alkyl substituents

- Difficult to modify alkyl substituents on the carbon skeleton of a lead compound. Full synthesis is usually required

# 2-Vary aryl substituents

# 1-Vary substitution pattern to enhance binding interactions

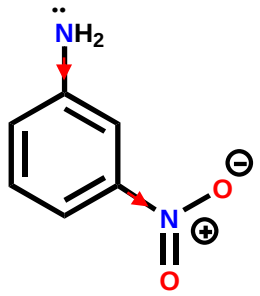




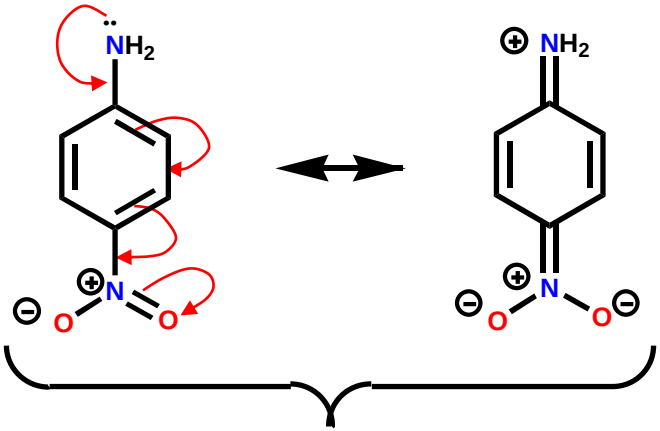

Benzopyrans

Anti-arrhythmic activity best when substituent is at 7-position

# 2-To enhance binding strength indirectly(electronic effects)




*Meta* (Inductive electron withdrawing effect)



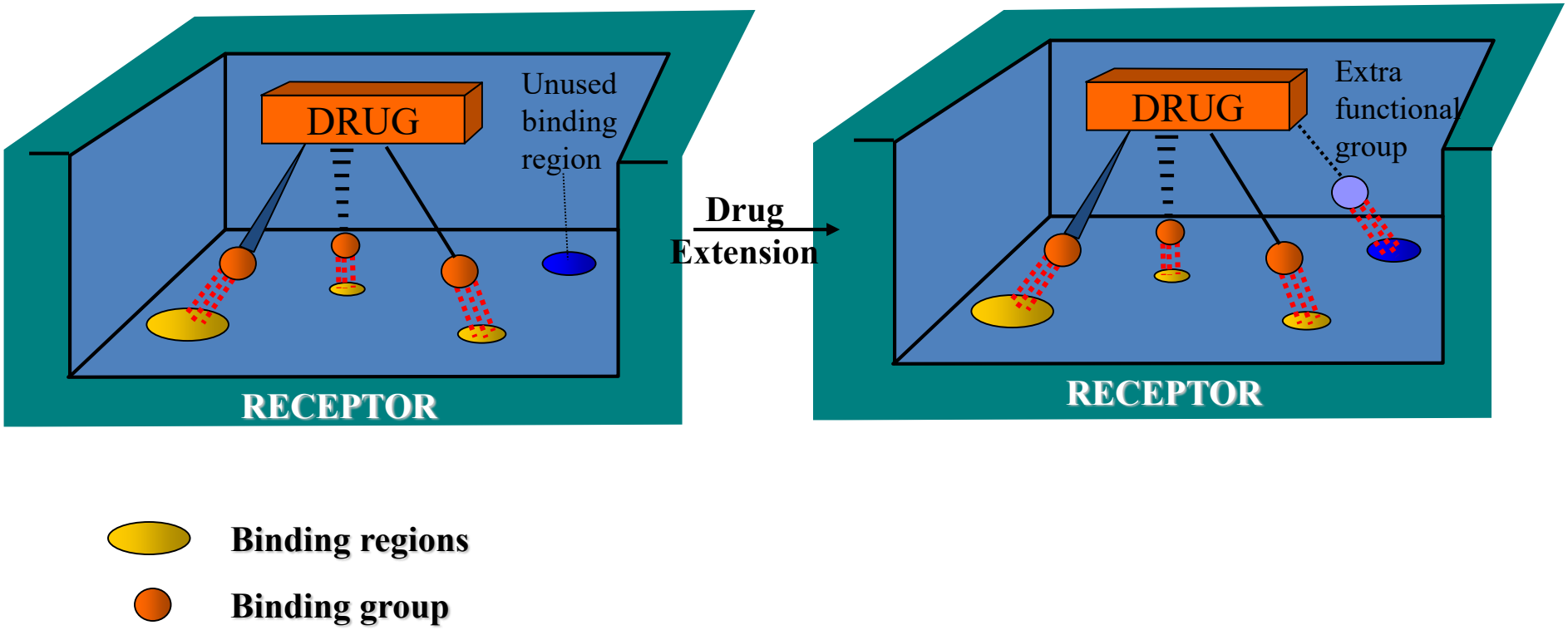

*Para* (Electron withdrawing effect due to resonance and inductive effects leading to a weaker base)

# Vary substituents

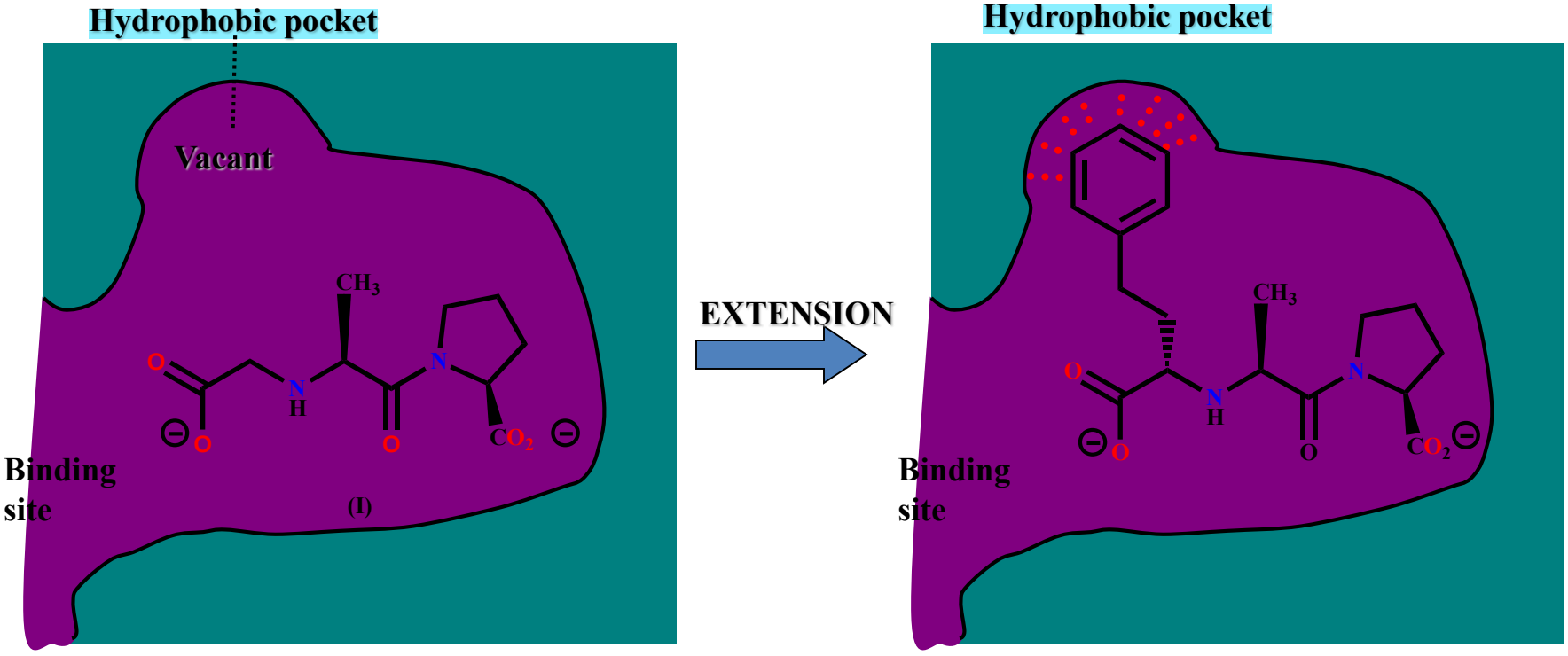
If the substitution pattern is ideal, then we can try varying the substituents themselves.

# 3-Chain Extension / Contraction

# 1-To explore target binding site for further binding regions to achieve additional binding interactions



# Example : ACE Inhibitors




Lead compound

enaliprilate
1000 fold increase in activity

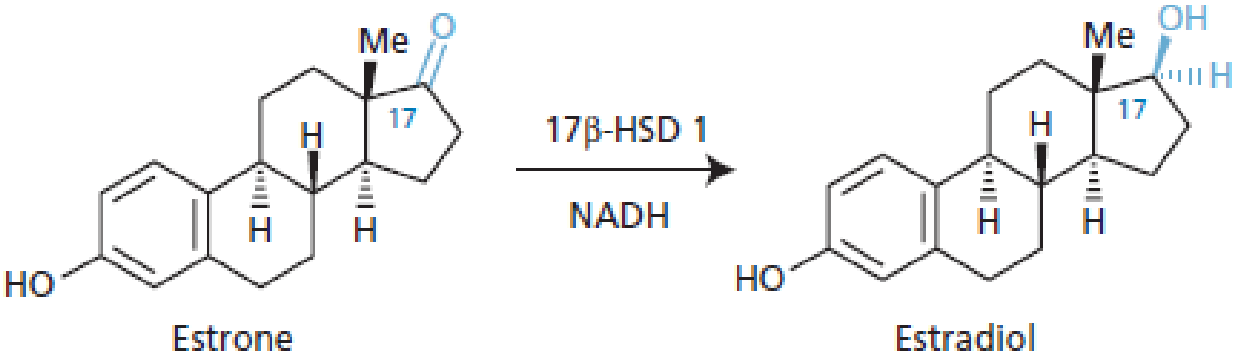



**FIGURE 1** The enzyme-catalysed conversion of estrone to estradiol.

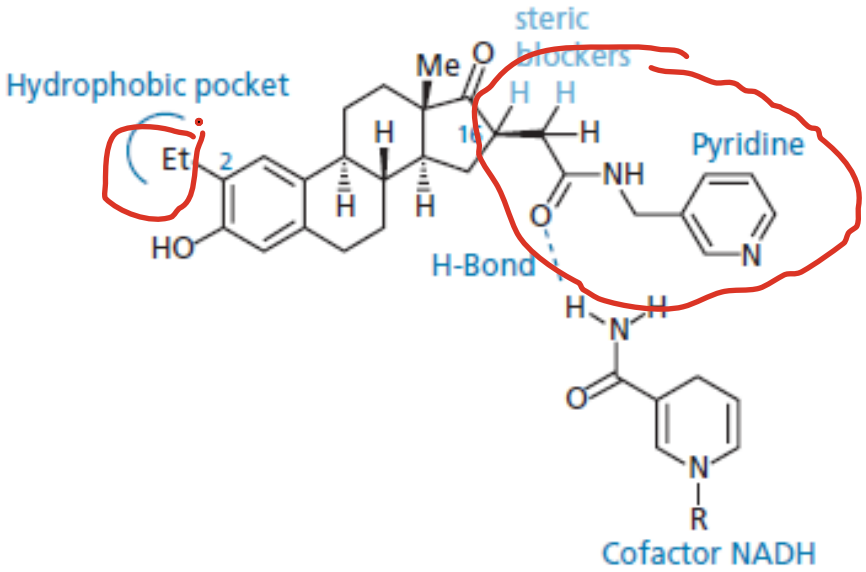
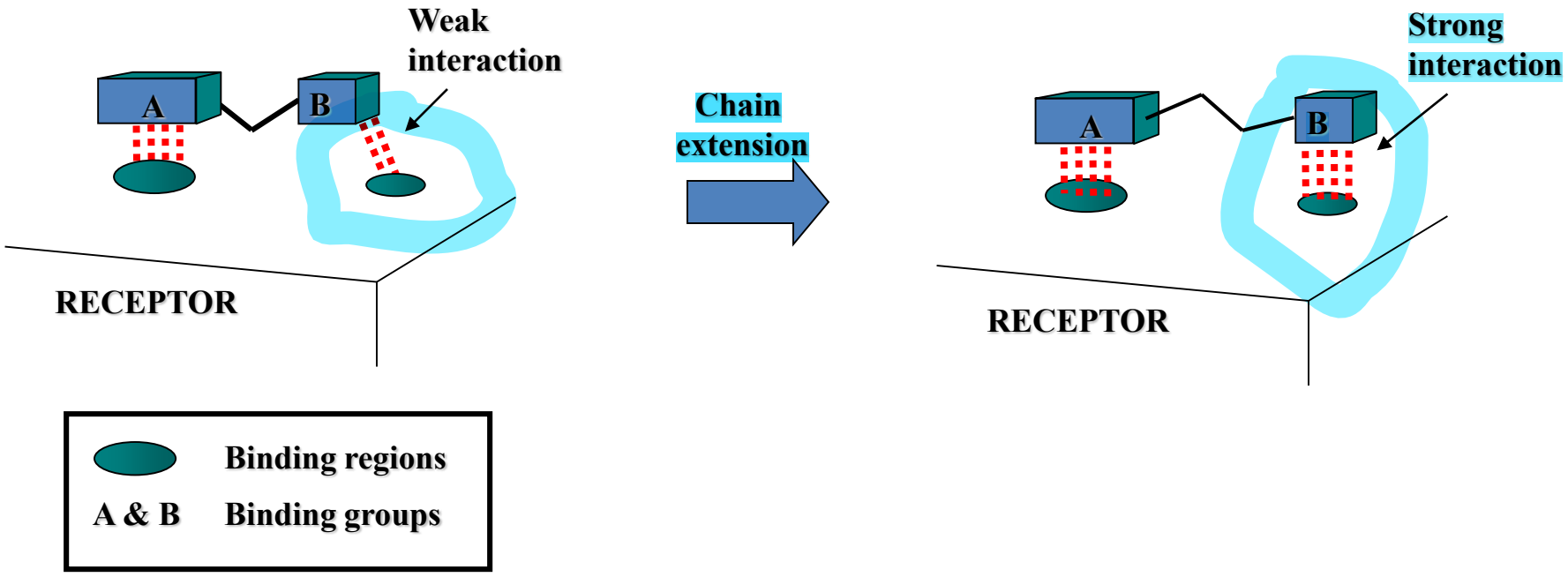


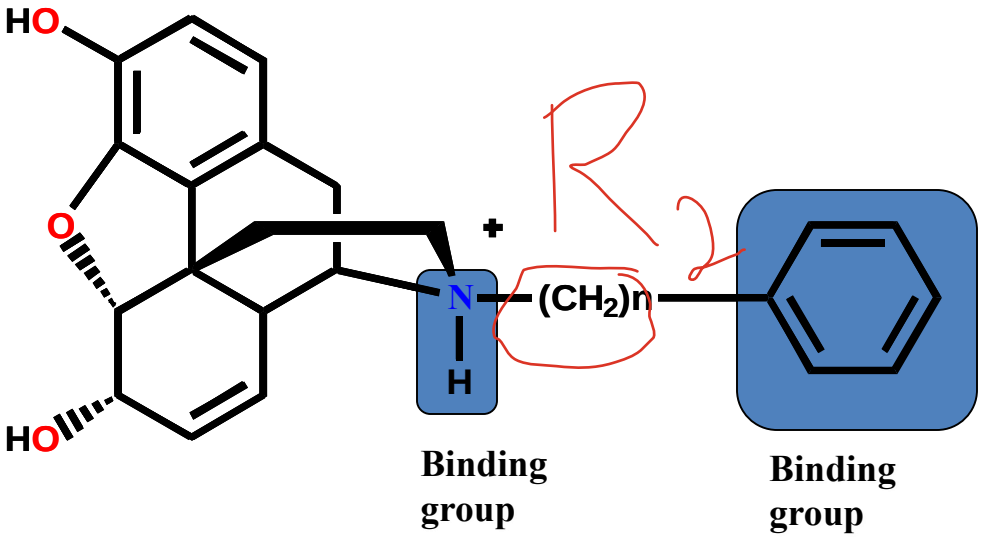

**FIGURE 2** Extra binding interactions resulting from the extension strategy.

# 2- Vary length of chain to optimise interactions

Useful if a chain is present connecting two binding groups

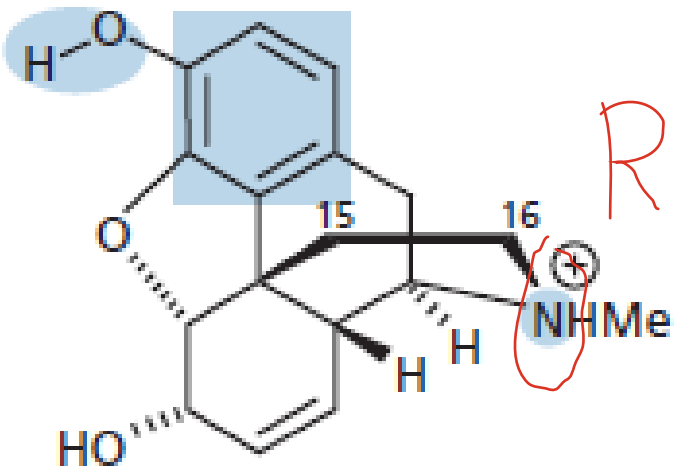


# **Example** : *N*-Phenethylmorphine




**Optimum chain length = 2**

14 fold increase in activity

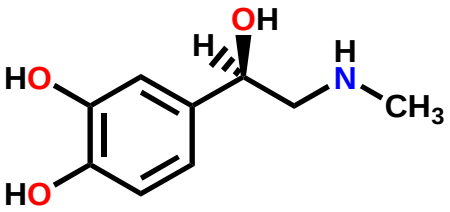



morphine

| $R^2$ = | Me Et Pr | Bu | Pentyl, Hexyl | $CH_2CH_2Ph$ |
|---|---|---|---|---|
| | Agonism decreases<br>Antagonism increases | Zero activity | Agonists | 14 × activity of morphine |

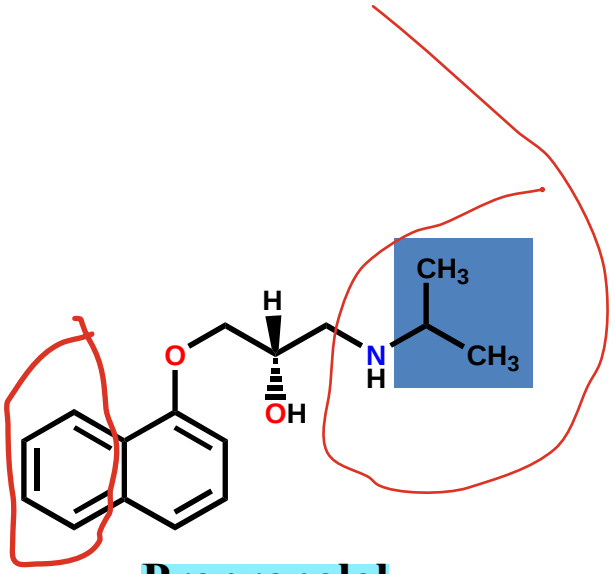
**FIGURE 24.10** Change in activity with respect to alkyl group size.

# 3- Antagonists from agonists

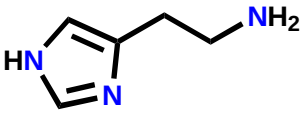



**Adrenaline**



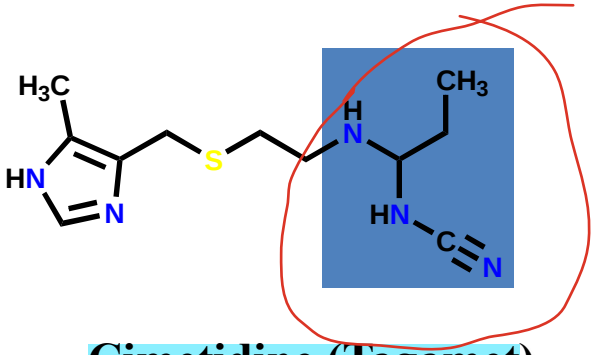



**Propranolol**
**(β-Blocker)**




**Histamine**

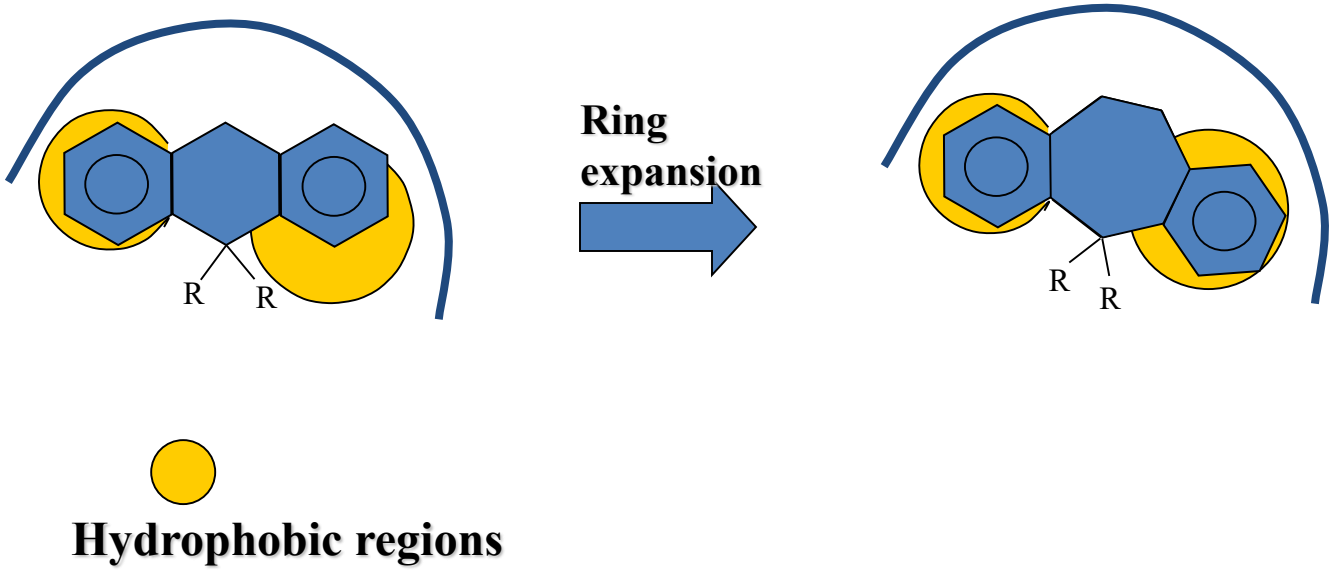


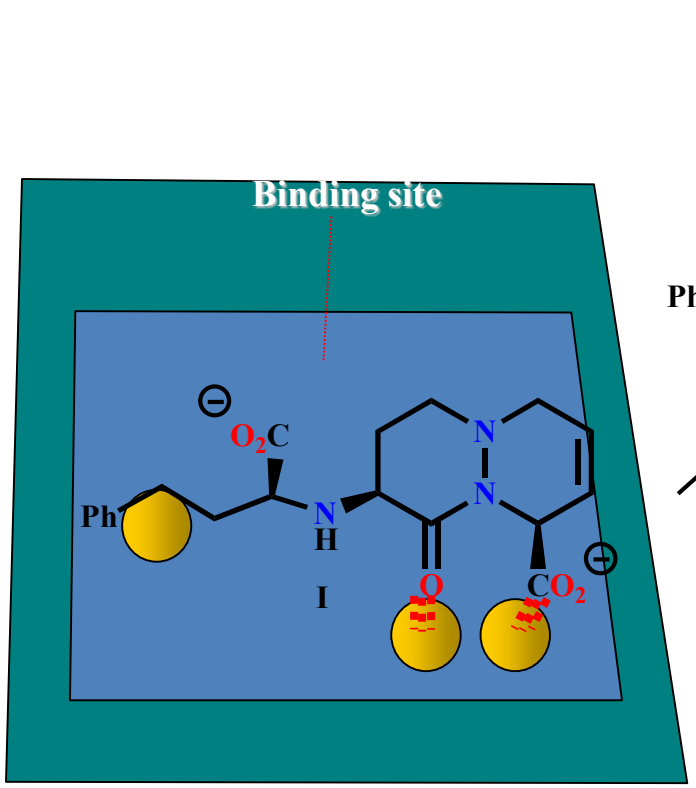



**Cimetidine (Tagamet)**
**(Anti-ulcer)**

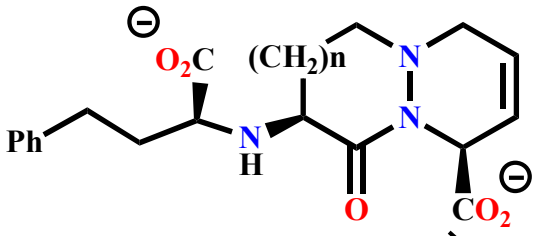
# 4-Ring Expansion / Contraction

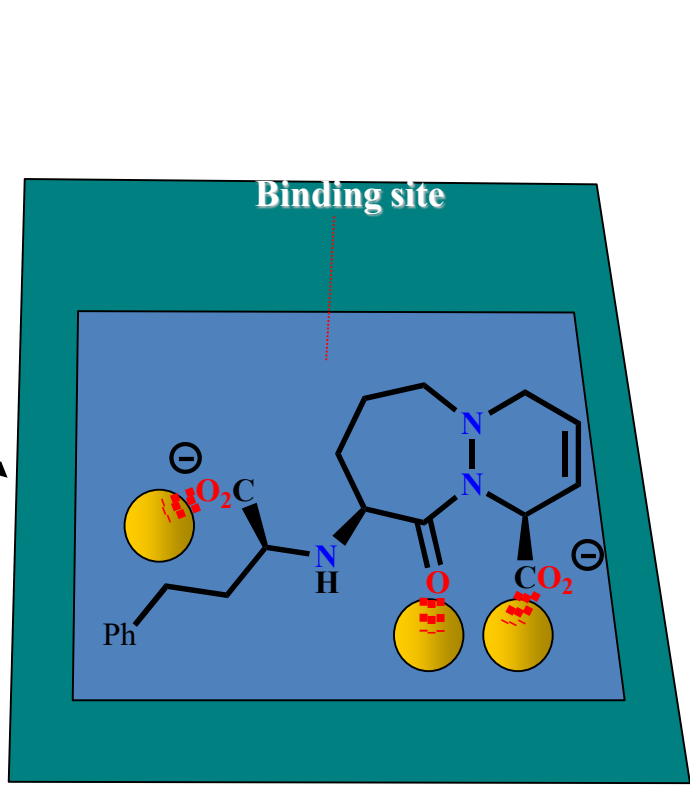
# 1-To improve overlap of binding groups with their binding regions






Vary n
to vary
ring size



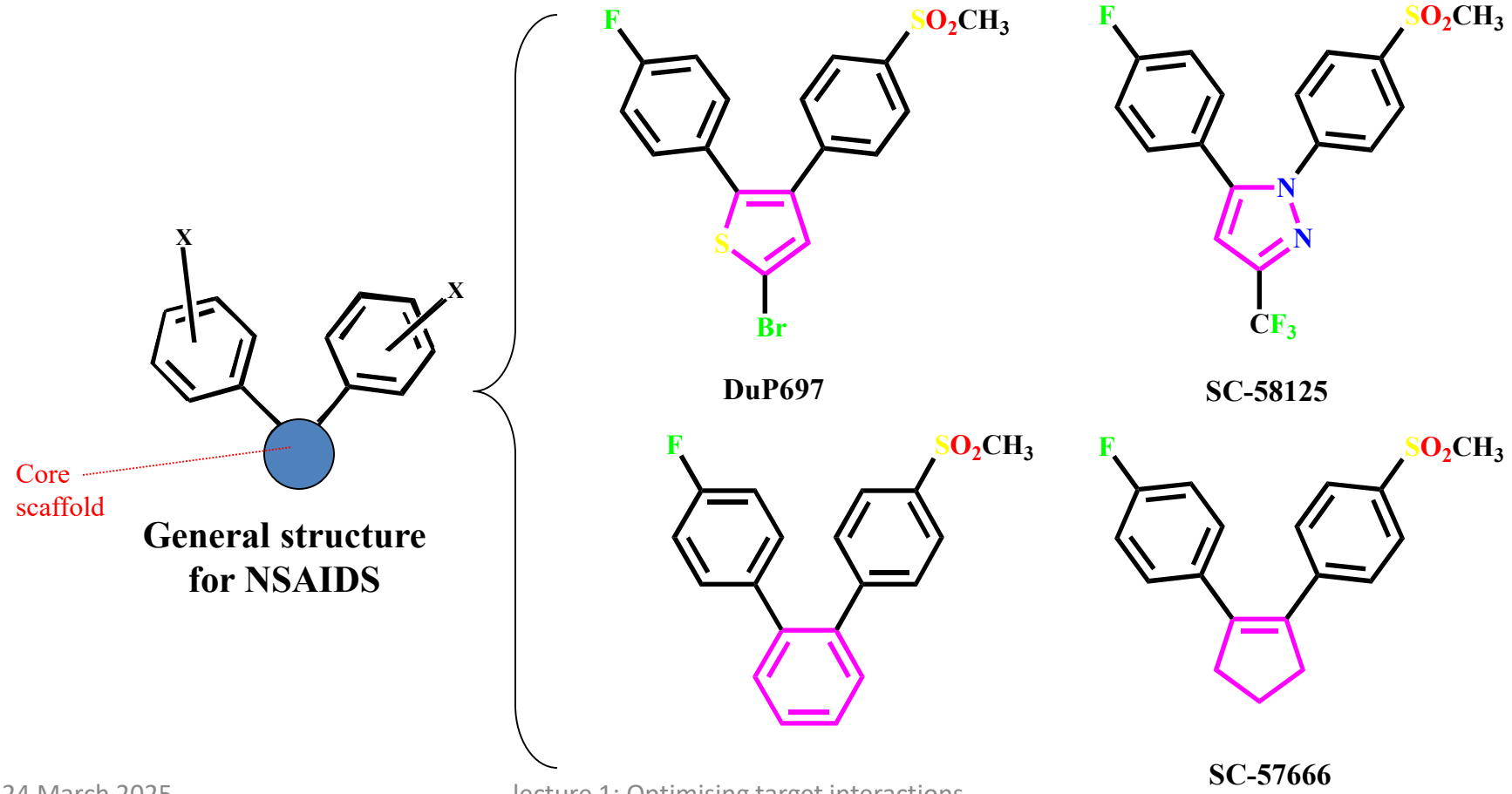




**cilazaprilat**
(ACE inhibitor)

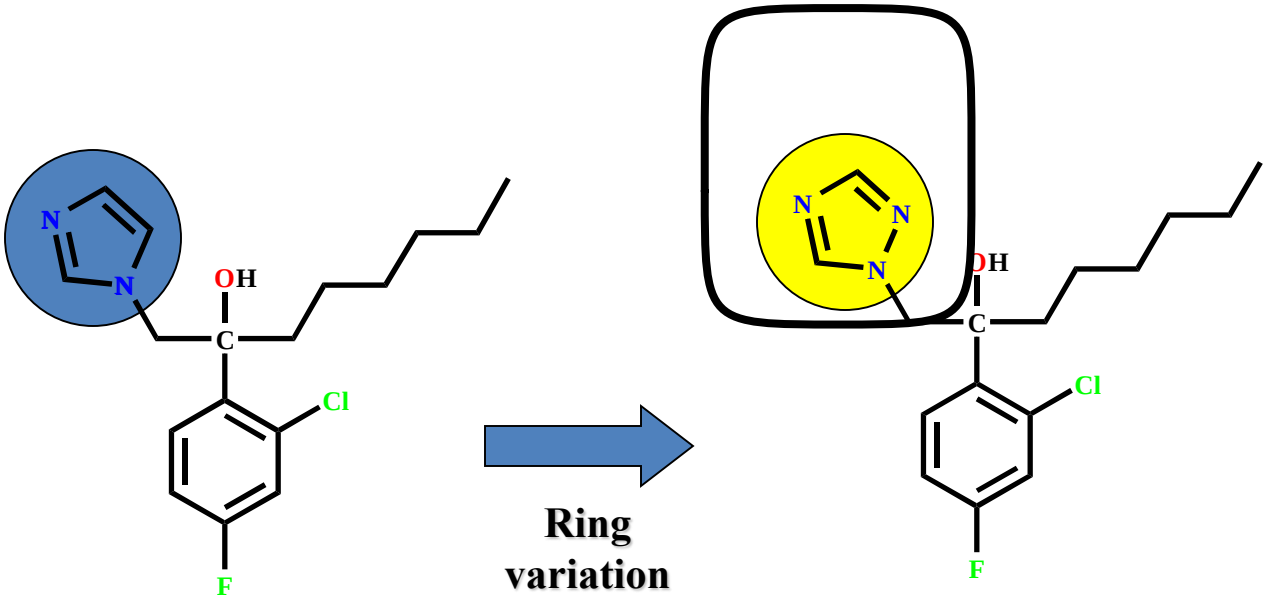
# 5-Ring Variation

# 1-Often done for patent reasons

Replace aromatic/heterocyclic rings with other ring systems



# 2-Sometimes results in improved selectivity



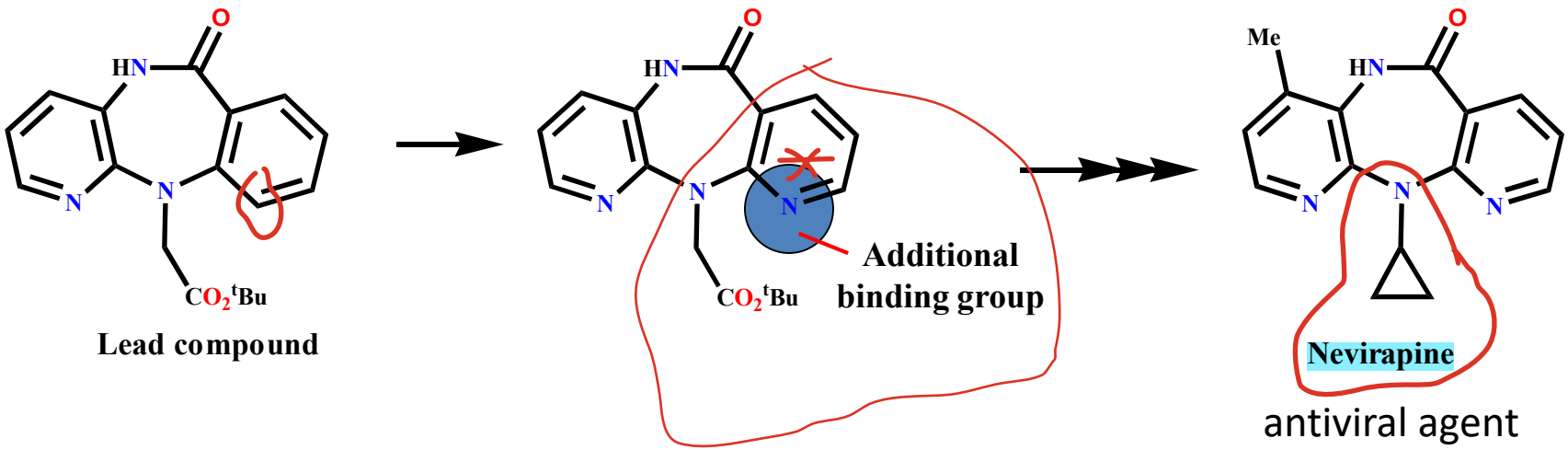

**Structure I**

**Antifungal agent**

**UK-46245**

**Improved selectivity vs. fungal enzyme**

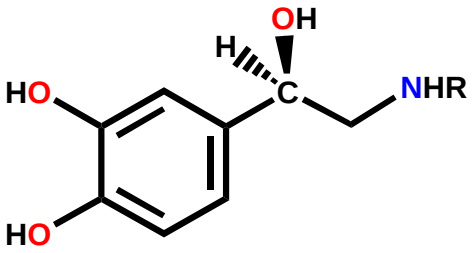
# 3- Introduces the possibility of an extra hydrogen bonding interaction with the binding site, should a suitable binding region be available



# 6-Ring fusions

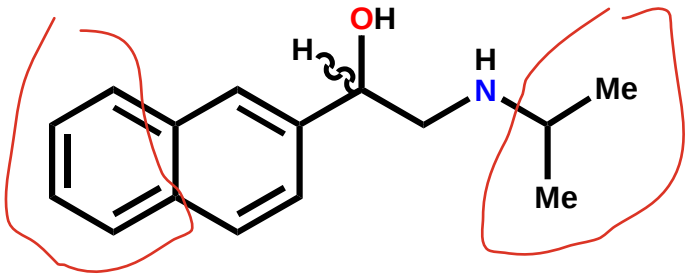
# 1- increased interactions or increased selectivity

## Pronethalol (β-blocker)




R = Me Adrenaline
R = H Noradrenaline






Pronethalol

Selective for b-receptorsover α-receptors

- larger van der Waals binding area
- Naphthalene ring system is sterically too big for the α-receptor

# 7-Isosterism

**Isosteres :** are atoms or groups of atoms which share the same valency and which have chemical or physical similarities

**Non-classical** isosteres: are groups that do not obey the steric and electronic rules used to define classical isosteres, but which have similar physical and chemical properties.

**Bioisostere** is used in drug design and includes both classical and non-classical isosteres. A bioisostere is a group that can be used to replace another group while retaining the desired biological activity

# Bioisosterism: Classification



**I. Classic isosteres:**

A. Monovalent atoms or groups

B. Divalent atoms or groups

C. Trivalent atoms or groups

D. Tetravalent atoms

E. Ring equivalents

**II. Non-classical isosteres:**

A. Cyclic vs. Non cyclic

B. Non classic bioisosterism of functional groups

B.1. Carboxylic group bioisosteres

B.2 Hydroxyl group bioisosteres

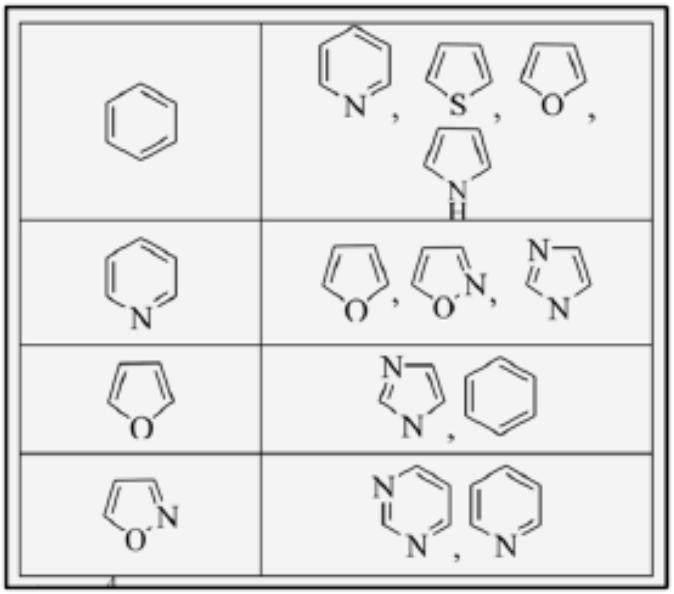
B.3. Amide group bioisosteres

B.4. Halogen bioisosteres.

# Classic Bioisosteres:



| Monovalent | Divalent | Trivalent | Tetravalent |
|---|---|---|---|
| —OH, —$NH_2$, —$CH_3$, —OR | —$CH_2$— | =CH— | =C= |
| —F, —Cl, —Br, —I, —SH, —$PH_2$ | —O— | =N— | =Si= |
| —$Si_3$, —SR | —S— | =P— | $=N^+=$ |
| | —Se— | =As— | $=P^+=$ |
| | —Te— | =Sb— | $=As^+=$ |
| | | | $=Sb^+=$ |



# Non- Classic Bioisosteres



| | | | | | | |
|---|---|---|---|---|---|---|
| —CO— | —COOH | $—SO_2NH_2$ | —H | —CONH— | —COOR | $—CONH_2$ |
| $—CO_2—$ | $—SO_3H$ | $—PO(OH)NH_2$ | —F | —NHCO— | —ROCO— | $—CSNH_2$ |
| $—SO_2—$ | —tetrazole | | | | | |
| $—SO_2NR—$ | $—SO_2NHR$<br>$—SO_2NH_2$ | | —OH<br>$—CH_2OH$ | | —catechol | |
| —CON— | —3-hydroxyisoxazole | | | | —benzimidazole | |
| —CH(CN)— | —2-hydroxychromones | | $—NHCONH_2$ | | | $C_4H_4S$ |
| R—S—R | | | $—NH—CS—NH_2$ | | | $—C_5H_4N$ |
| (R—O—R) | =N— | | | | | $—C_6H_5$ |
| R—N(CN)— | C(CN)=R′ | | $—NH—C(=CHNO_2)—NH_2$<br>$—NH—C(=CHCN)—NH_2$ | | | |
| —halide | | | | | | $—C_4H_4NH$ |
| | $—CF_3$ | | | | | |
| | —CN | | | | | |
| | $—N(CN)_2$ | | | | | |
| | $—C(CN)_3$ | | | | | |

# Why isosterism

1- To determine whether a particular group is an important binding group

2- Isosteric groups could be used to determine whether a particular group is involved in hydrogen bonding.

Replacing O with CH2…… little difference to the size

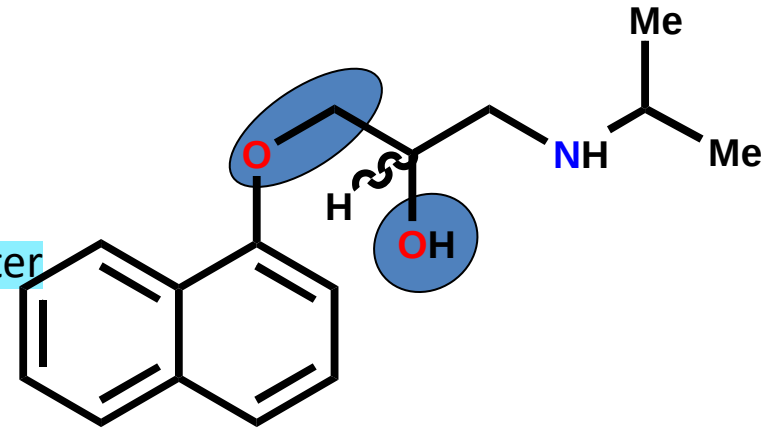
polarity, electronic distribution, and bonding

Replacing OH with the larger SH …… the electronic character

steric factors

Replacing OH with CH3……. eliminate hydrogen bonding

Replacing OH with NH2




**Propranolol (β-blocker)**

3- The presence of fluorine in place of an enzymatically labile hydrogen can also disrupt an enzymatic reaction, as C–F bonds are not easily broken.

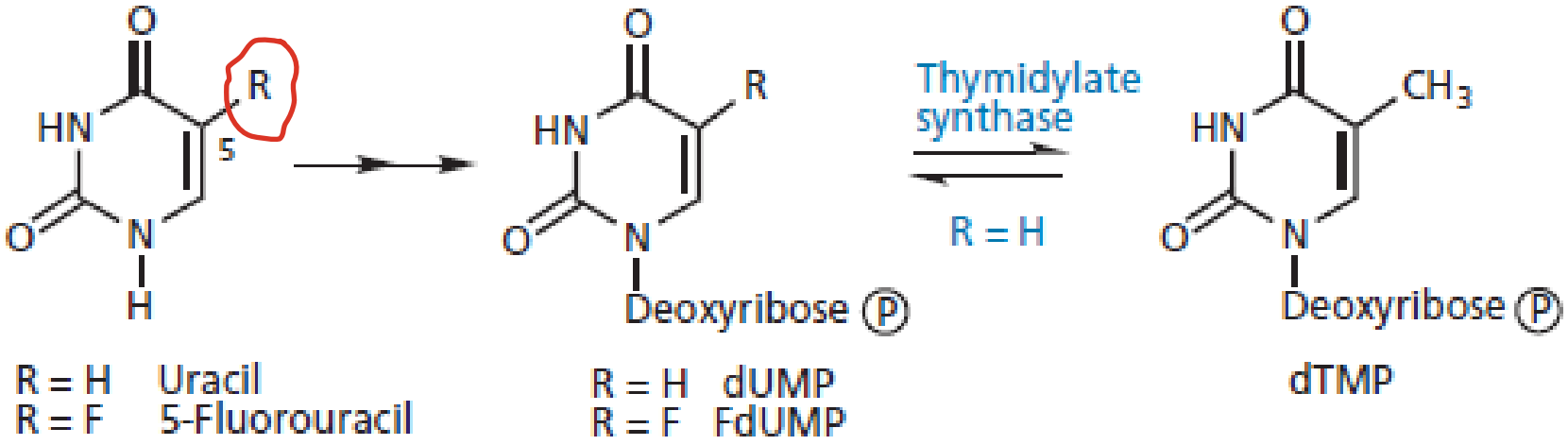



**FIGURE 21.19** Biosynthesis of dTMP. Ⓟ = phosphate.

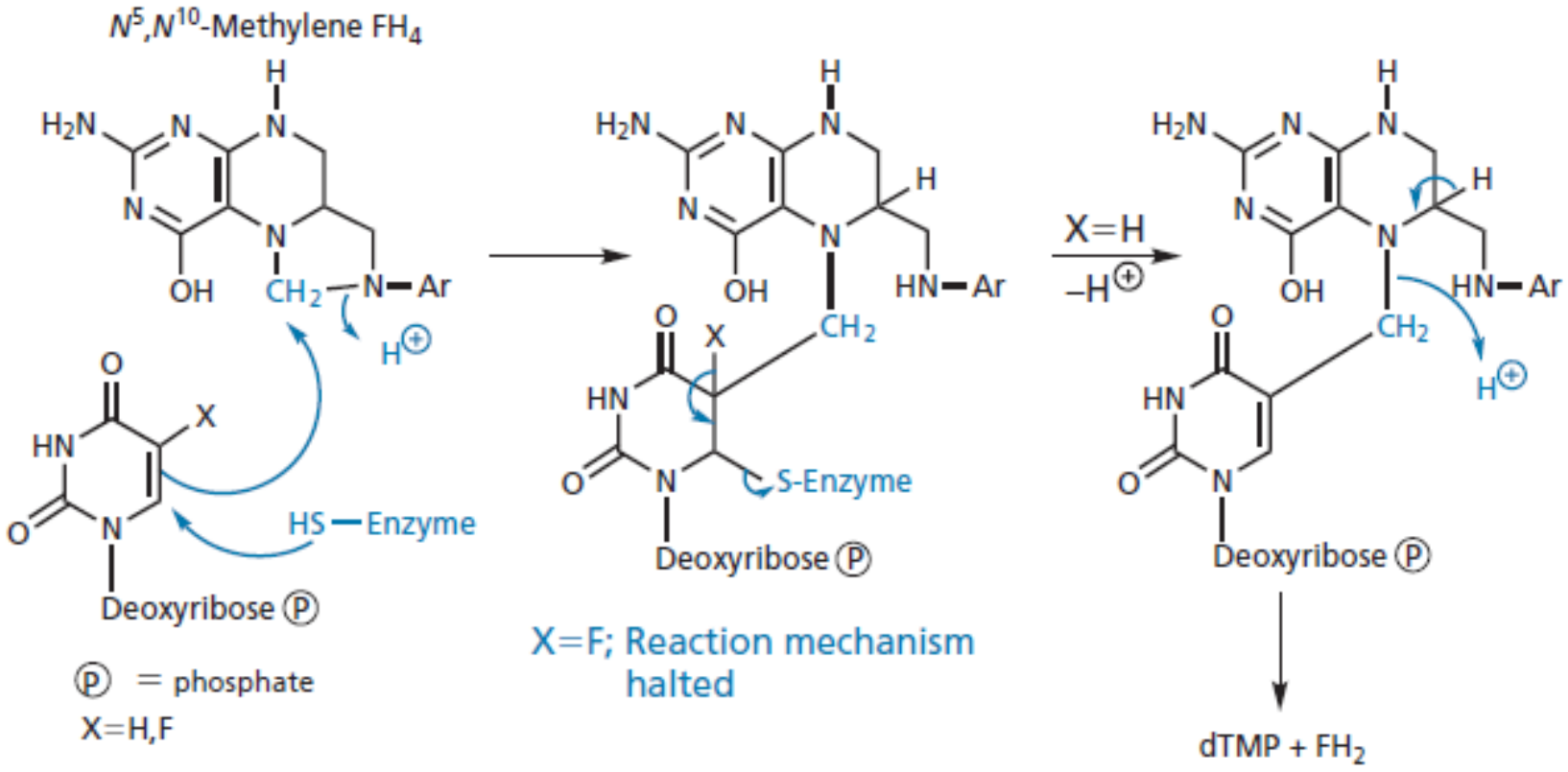
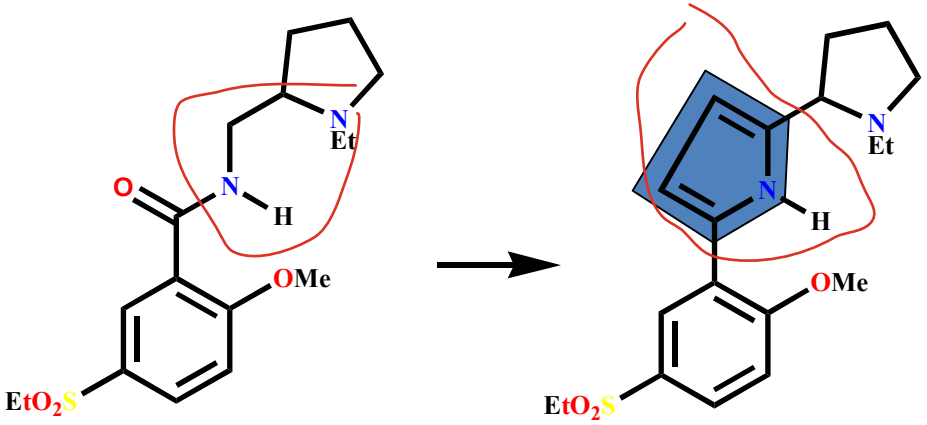



**FIGURE 21.20** Use of 5-fluorouracil as a prodrug for a suicide substrate.

# 4- Improve activity and selectivity

- Replace a functional group with another group which retains the same biological activity
- Not necessarily the same valency

**antipsychotics**




**Sultopride**

dopamine antagonist

**DU 122290**

**Improved selectivity for $D_3$ receptor over $D_2$ receptor**

**Pyrrole ring = bio-isostere for amide group**

# 8-Simplification

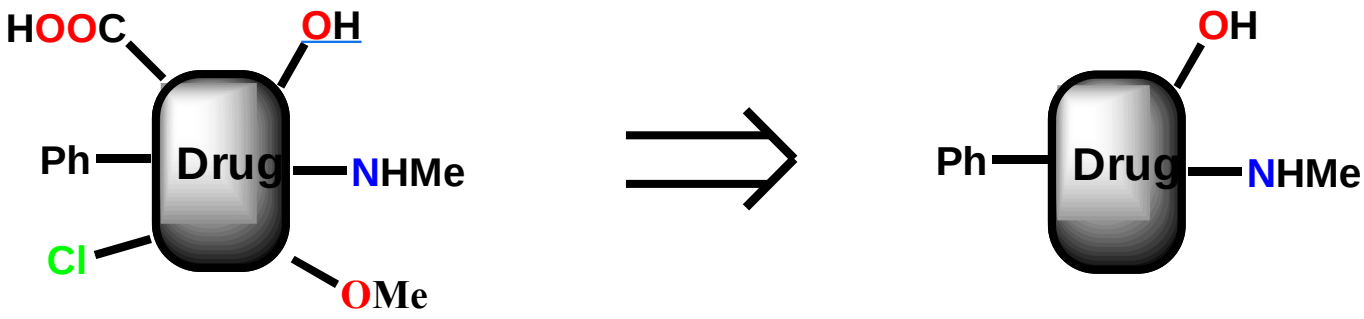
## Rationale :

- Lead compounds from natural sources are often complex and difficult to synthesise.
- Simplifying the molecule makes synthesis of analogues easier, quicker and cheaper.
- Simpler structures may fit binding site easier and increase activity.
- Simpler structures may be more selective and less toxic if excess functional groups removed

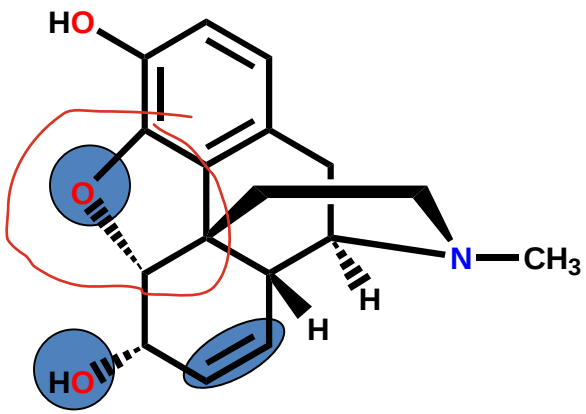
# Methods

Retain pharmacophore

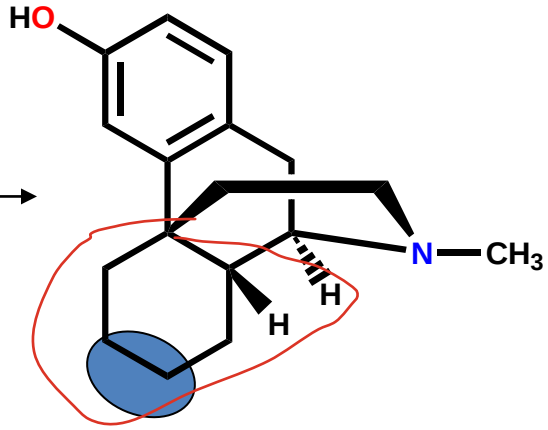
Remove unnecessary functional groups



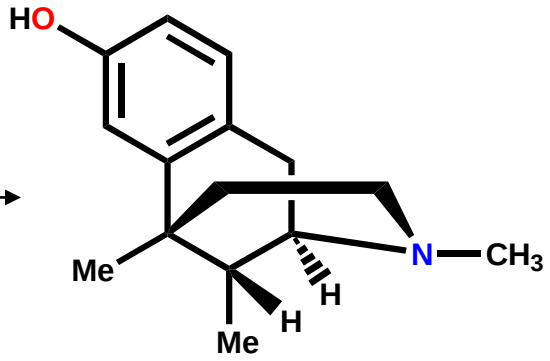
# 1- Remove excess rings




Morphine




Levorphanol




Metazocine

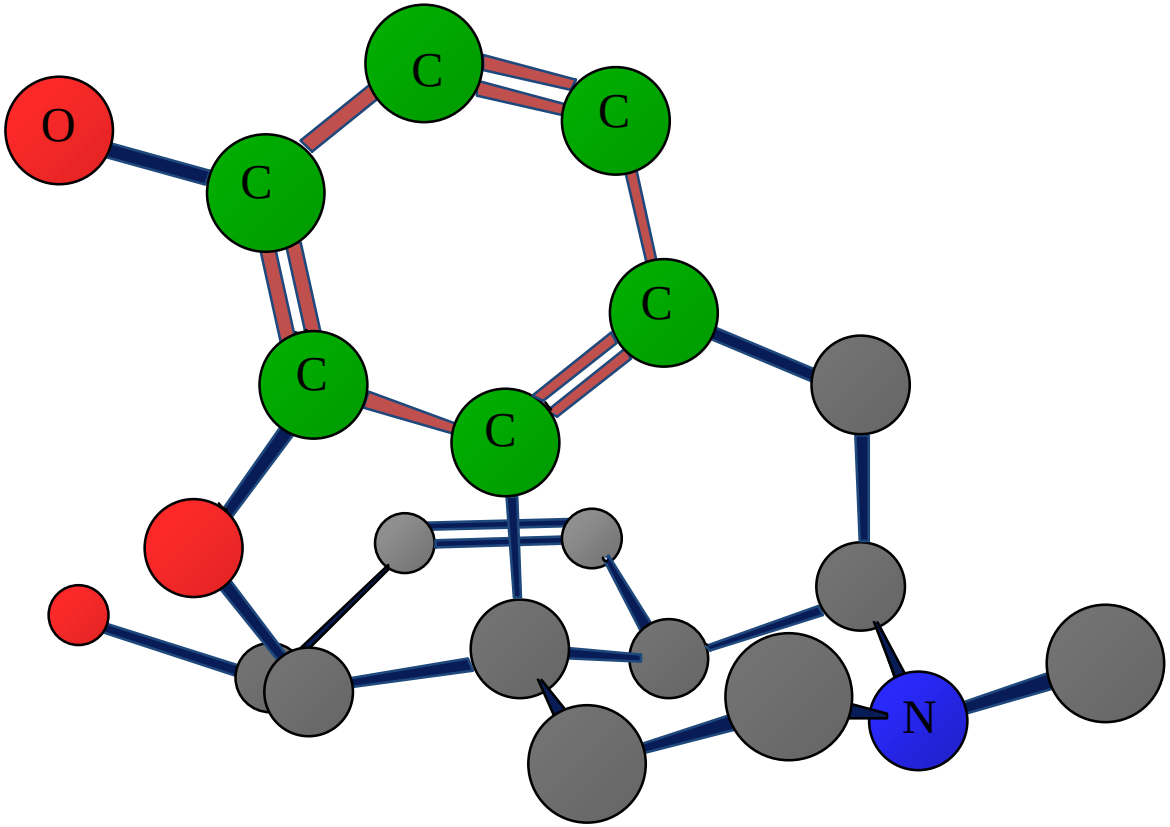
Excess functional groups

5X more potent than morphine

same potency as morphine

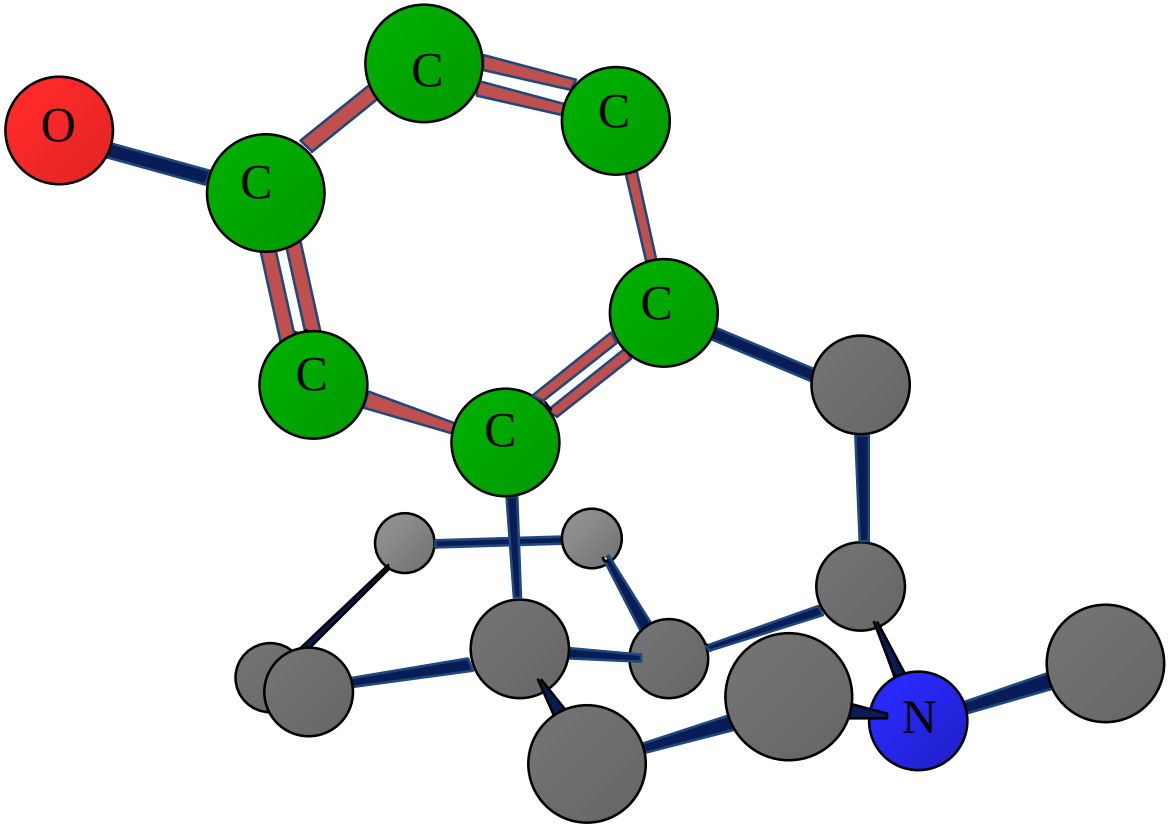
Excess ring

# MORPHINE



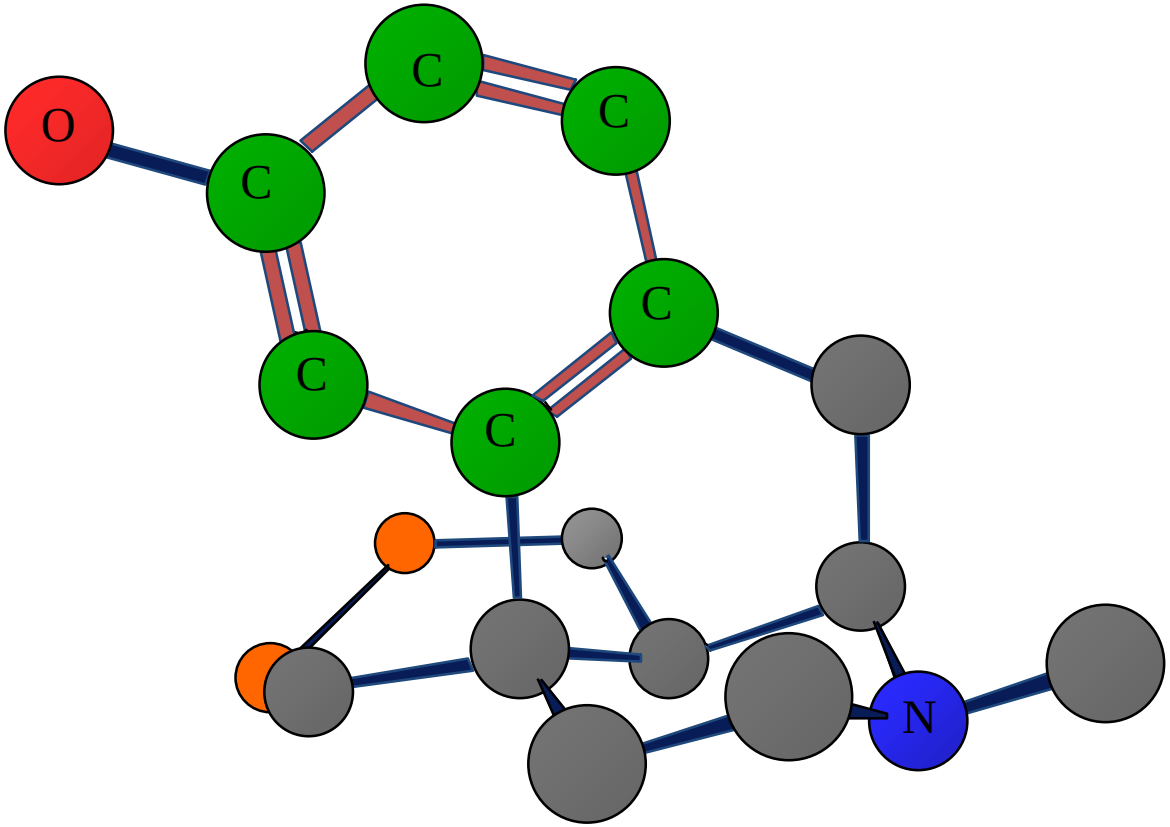

# SIMPLIFICATION

# LEVORPHANOL



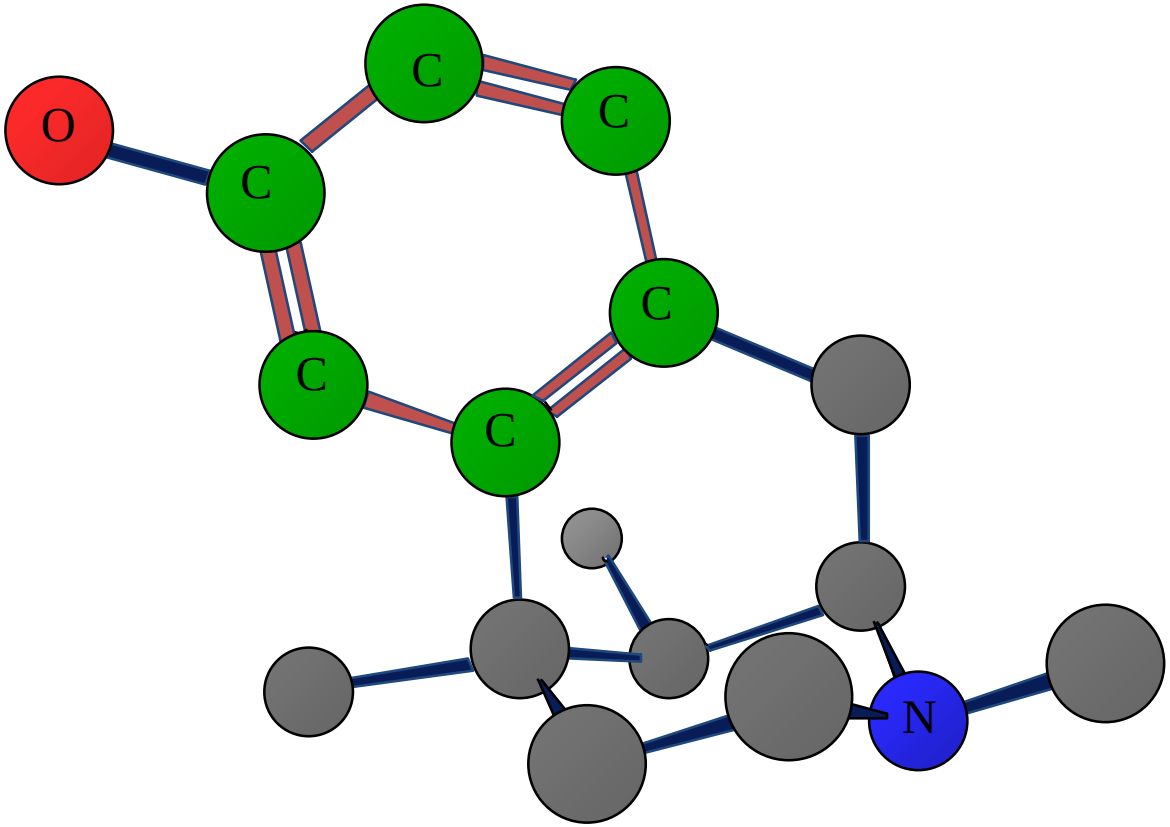

# SIMPLIFICATION

# LEVORPHANOL

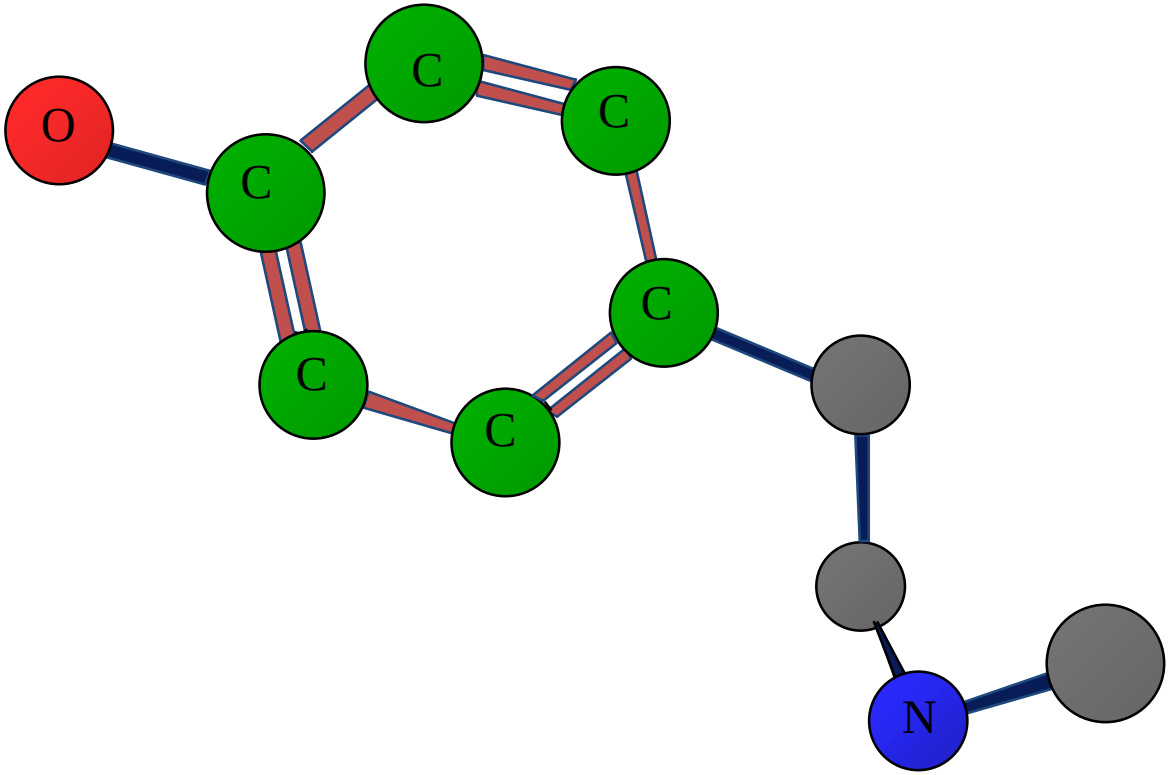



# SIMPLIFICATION

# METAZOCINE

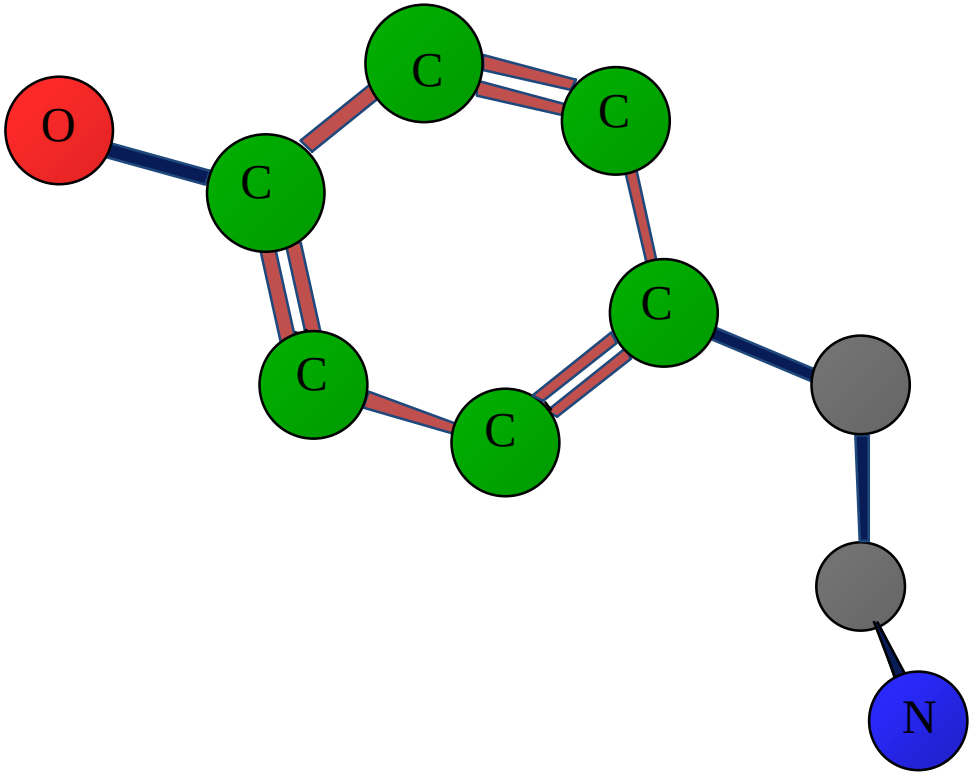



# SIMPLIFICATION



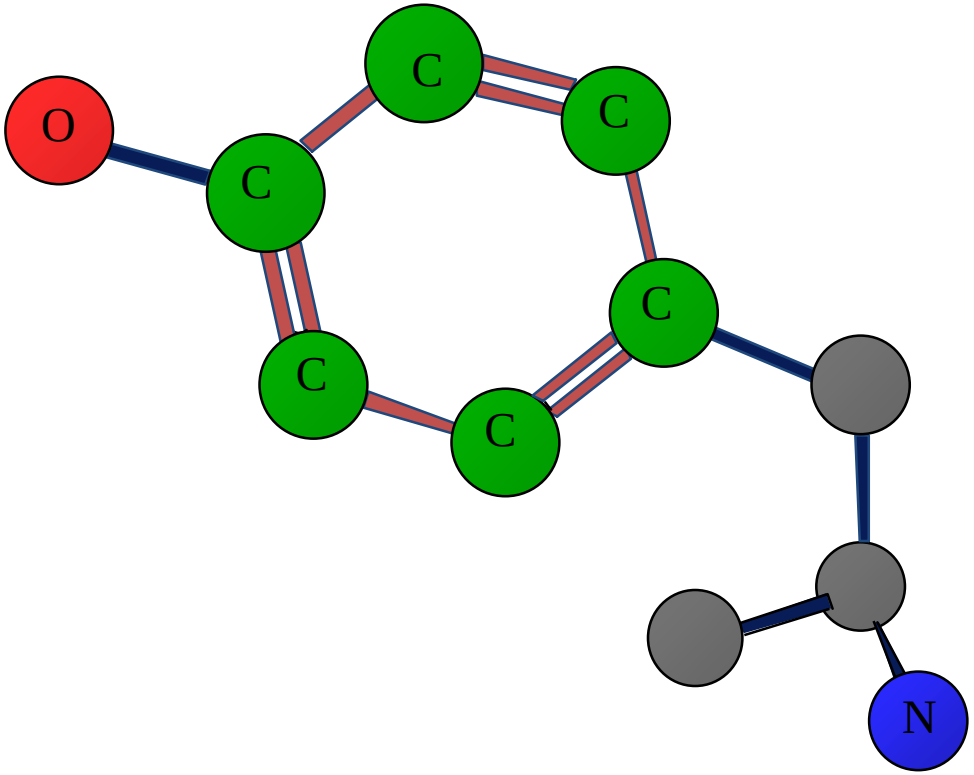

**OVERSIMPLIFICATION**

# TYRAMINE

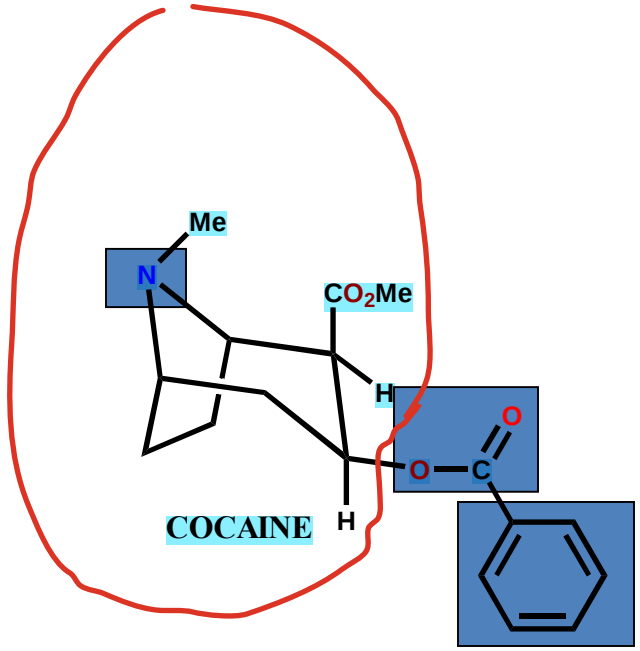


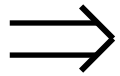

## OVERSIMPLIFICATION

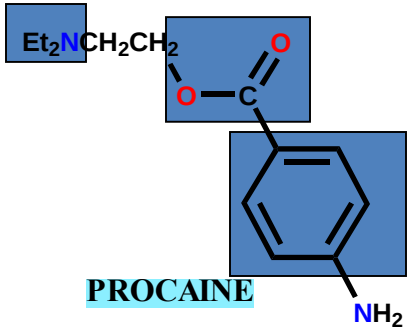
# AMPHETAMINE




## OVERSIMPLIFICATION



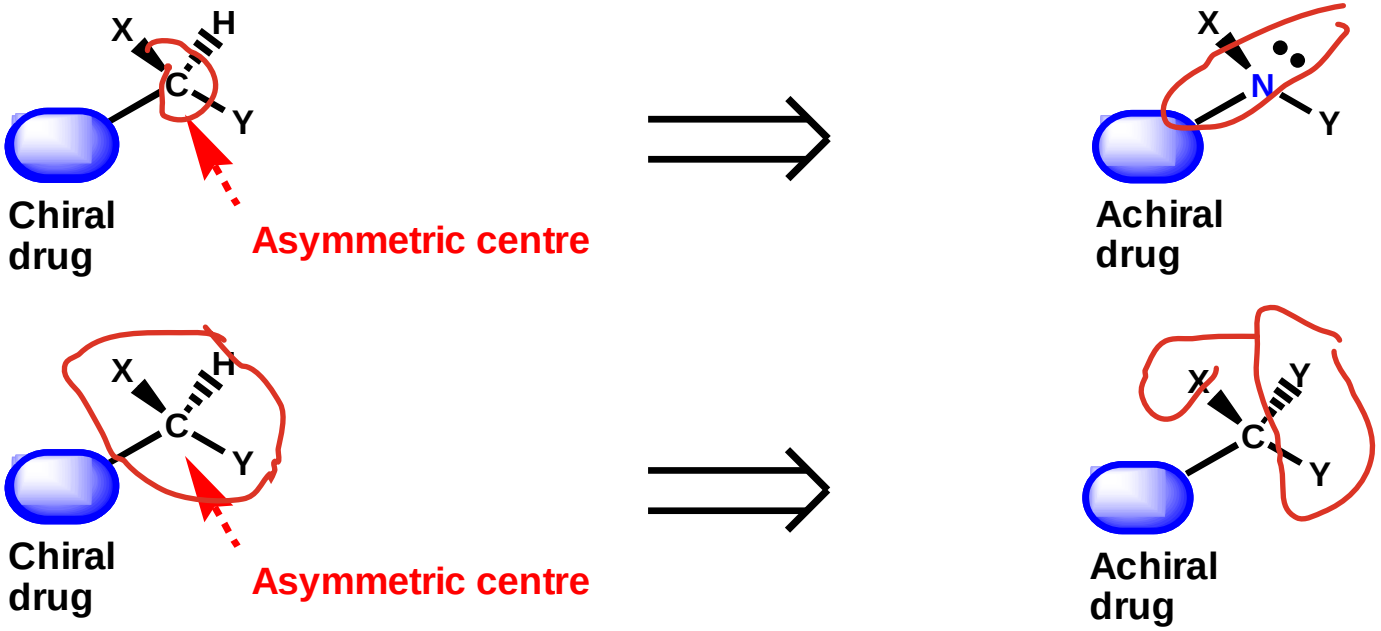






local anaesthetics

Pharmacophore

# 2-Remove asymmetric centres



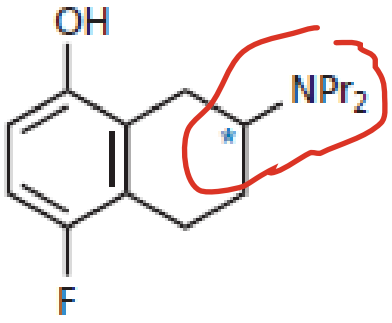
**Why the use of racemates is discouraged and it is preferable to use a pure enantiomer ?**

1-High cost of synthesis and seperation.

2-Both enantiomers then have to be tested for their activity and side effects, doubling the number of tests that have to be carried out.

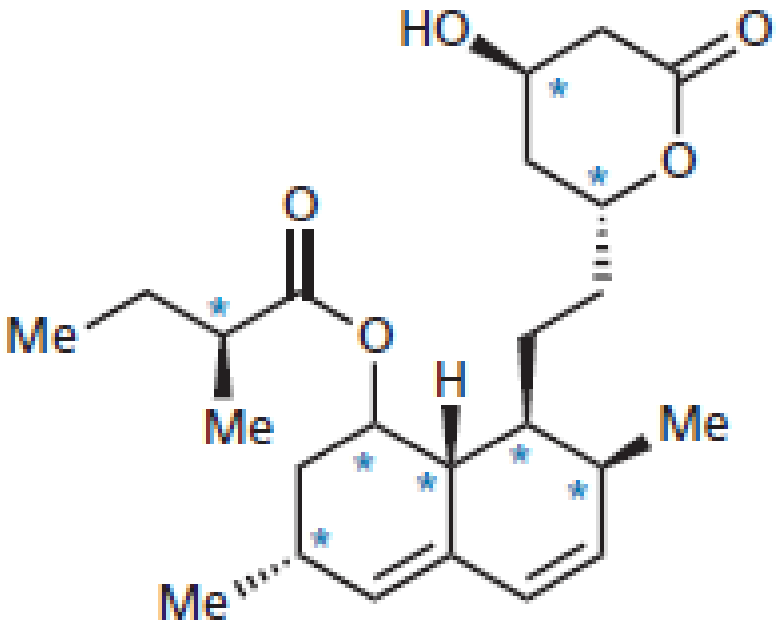
3-different enantiomers can have different activities.

Enantiomers have opposing agonist and antagonist activity at the serotonin receptor (5-HT 1A)

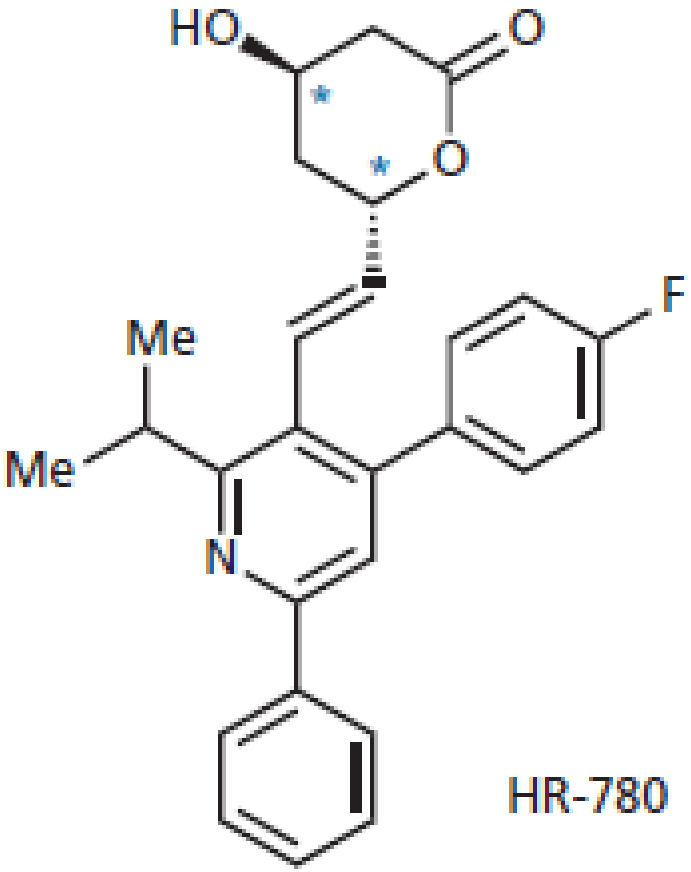



UH-301

Inactive as a racemate




Mevinolin (lovastatin)




HR-780

cholesterol-lowering agents

**Disadvantages of simplification:**

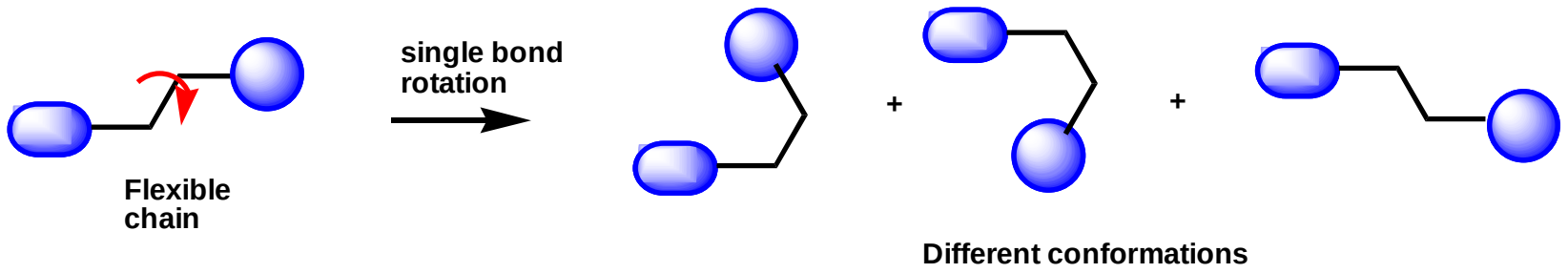
- Oversimplification may result in decreased activity and selectivity. e.g morphine over simplication
- Simpler molecules have more conformations bind in a different way than the lead compound.
- More likely to interact with more than one target binding site more side effects.

# 9- Rigidification

**Rationale :**

Endogenous lead compounds often simple and flexible (e.g. adrenaline)

Fit several targets due to different active conformations (e.g. Adrenergic receptor types and subtypes)

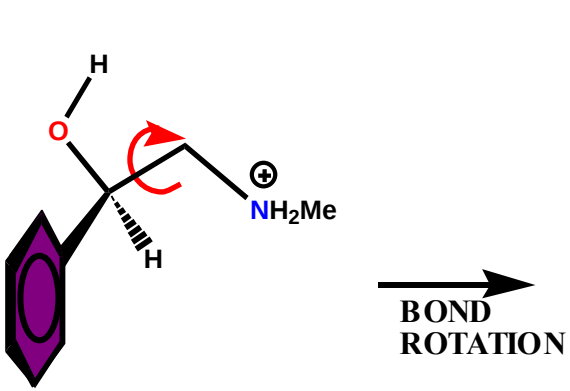



- Rigidify molecule to limit conformations - conformational restraint
- Increases activity (more chance of desired active conformation)
- Increases selectivity (less chance of undesired active conformations)

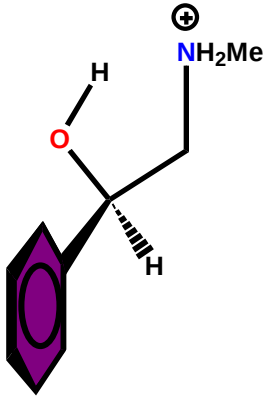
Disadvantage:

- Molecule more complex and may be more difficult to synthesise

# The body's own neurotransmitters

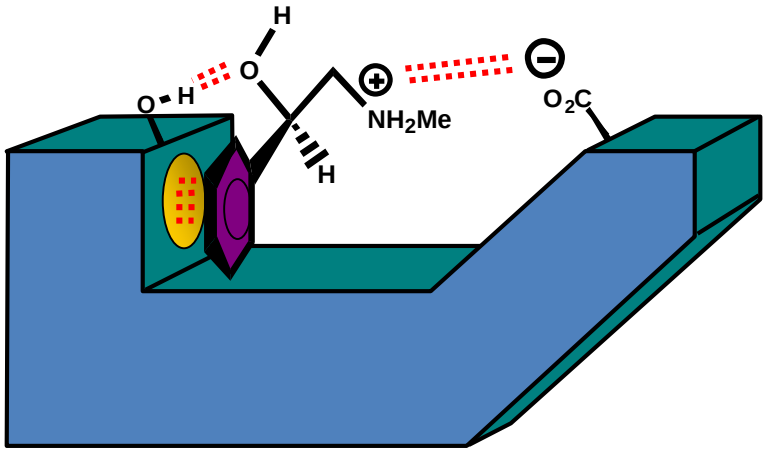



BOND ROTATION

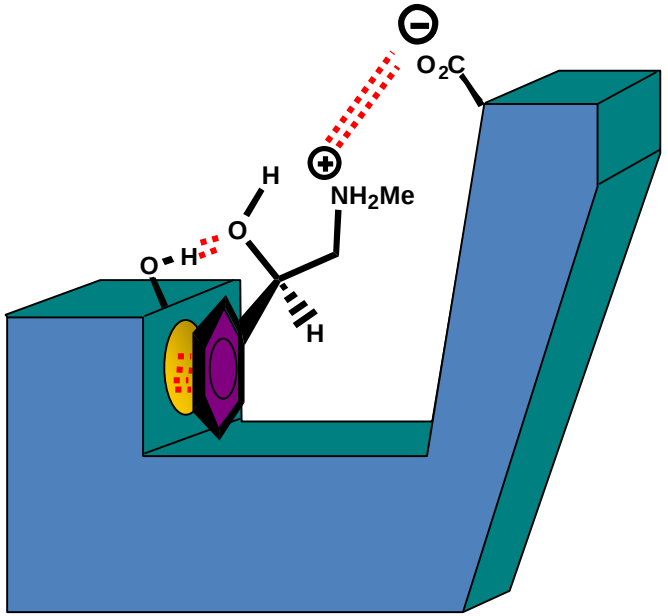



I

II



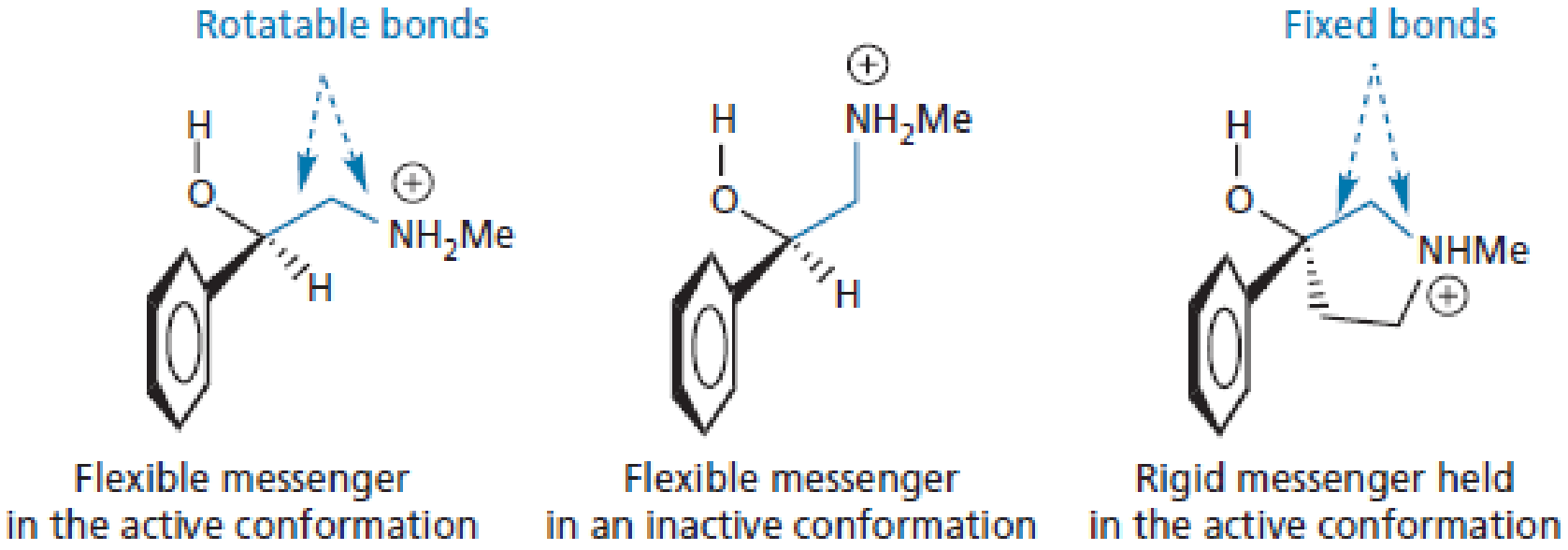

RECEPTOR 1



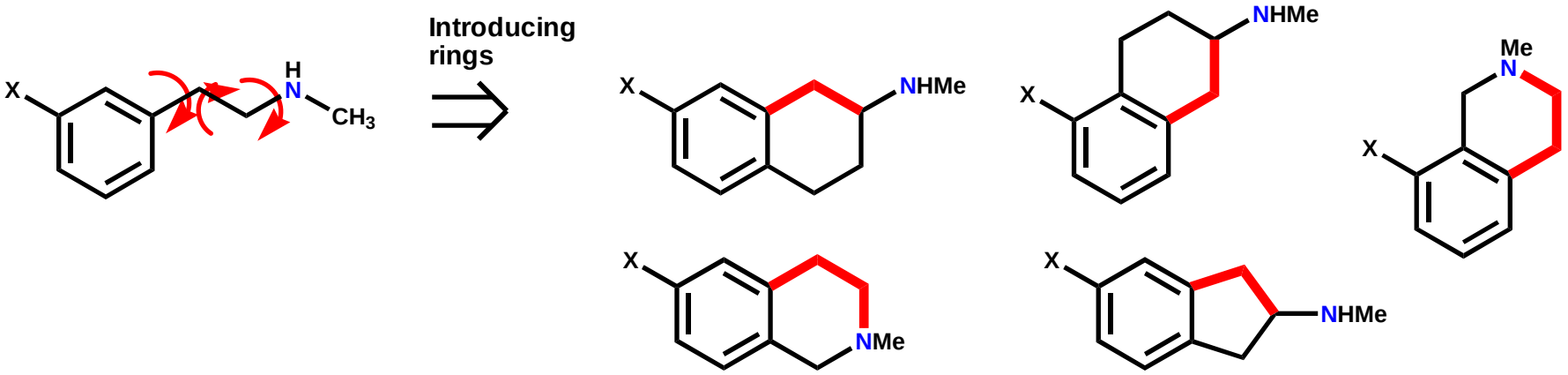

RECEPTOR 2

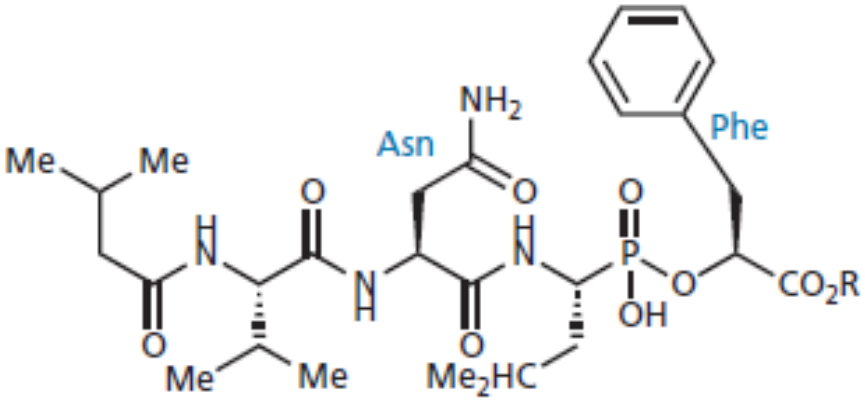
# 1- Introduce rings

Bonds within ring systems are locked and cannot rotate freely

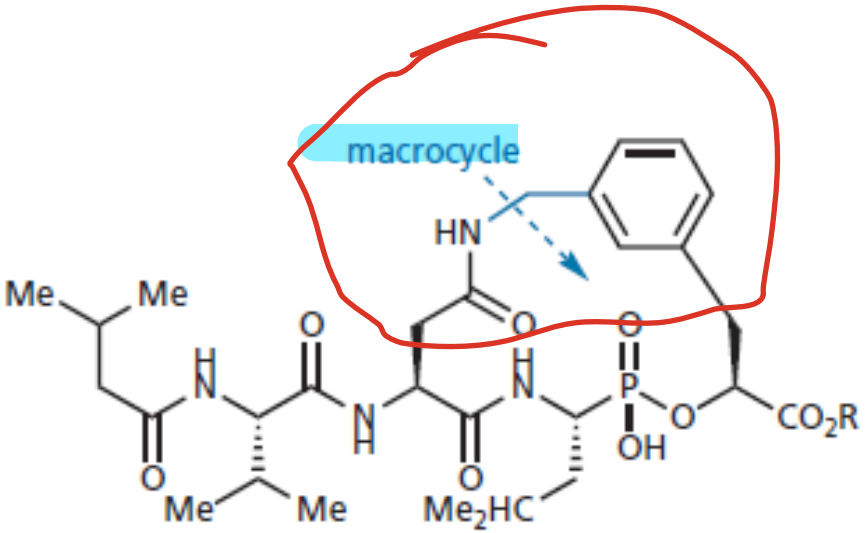


# Test rigid structures to see which ones have retained active conformation






Acyclic pentapeptide ($K_i$ 42 nM)

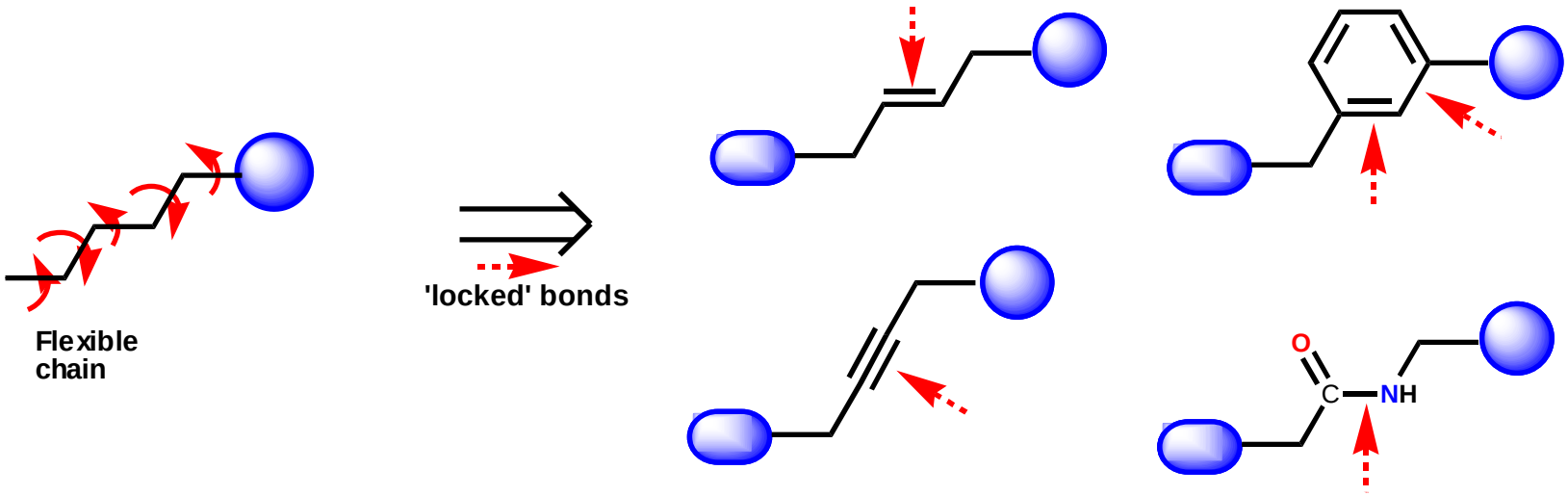



Rigidified pentapeptide ($K_i$ 0.1 nM)

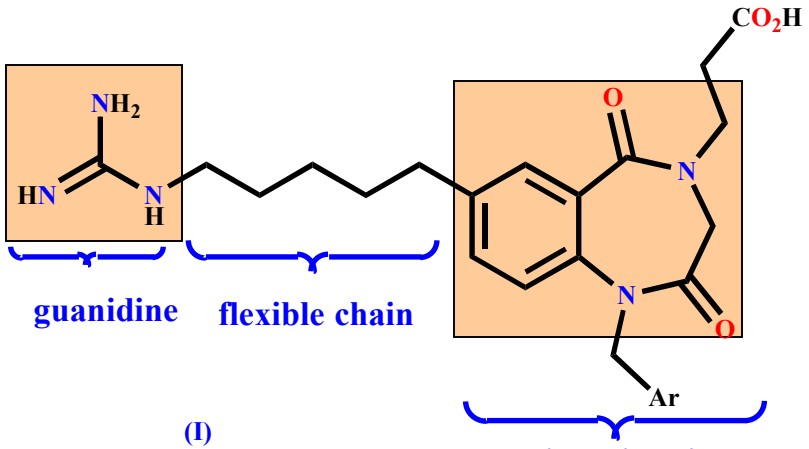
inhibitor of a proteolytic enzyme

400-fold increase in activity

# 2-Introduce rigid functional groups



Important binding groups

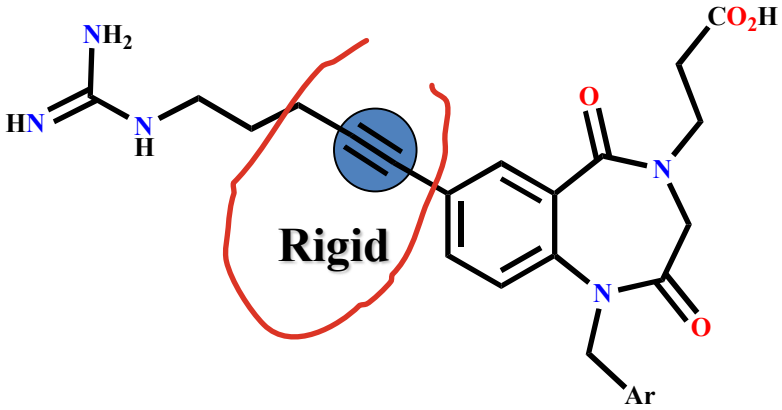



(I)

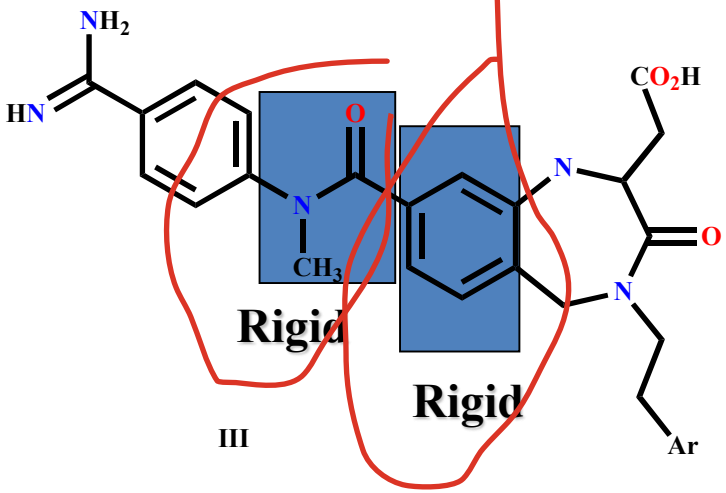
diazepin

Inhibits platelet aggregation

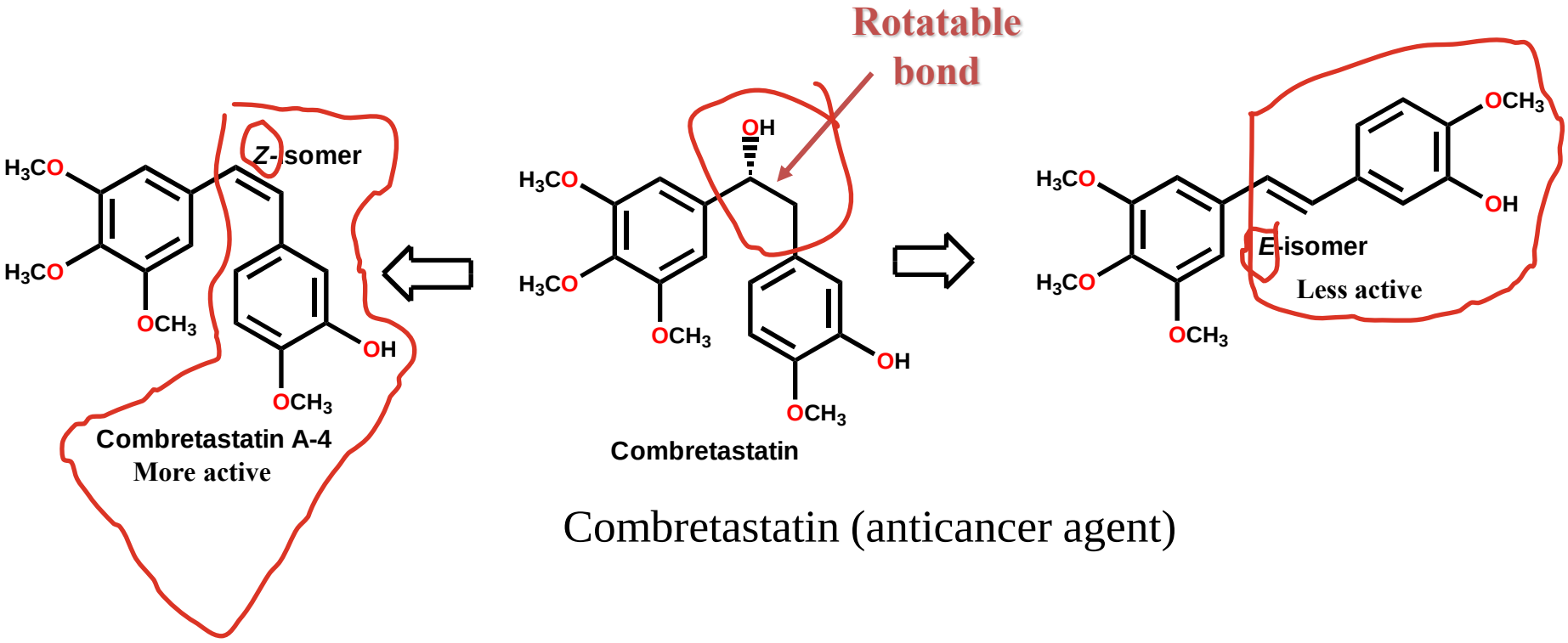
## Analogues




II



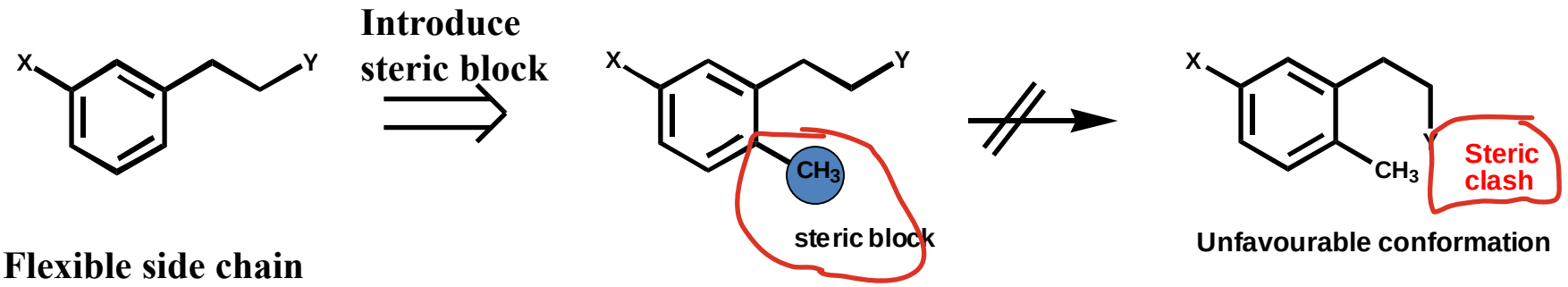

III

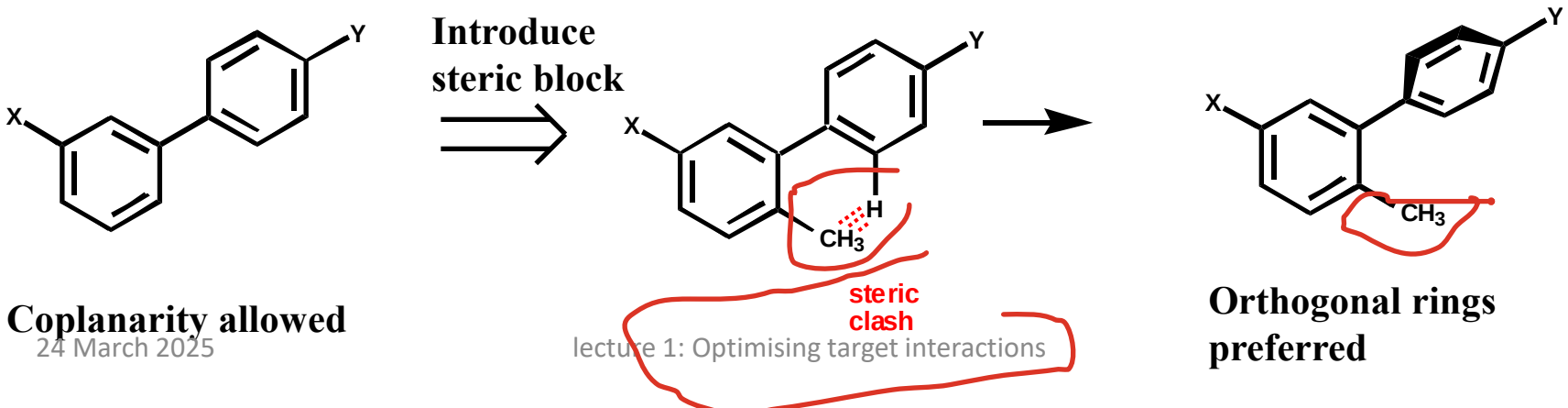


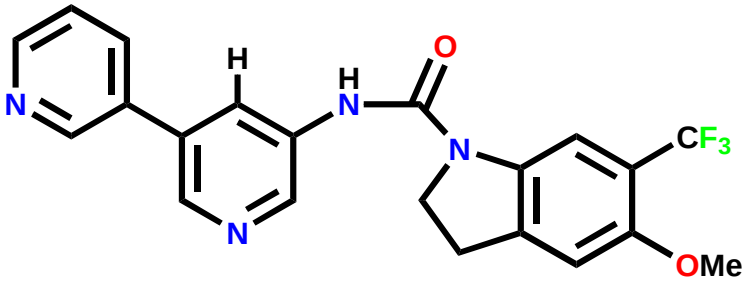

Combretastatin (anticancer agent)

# 3- Steric blockers

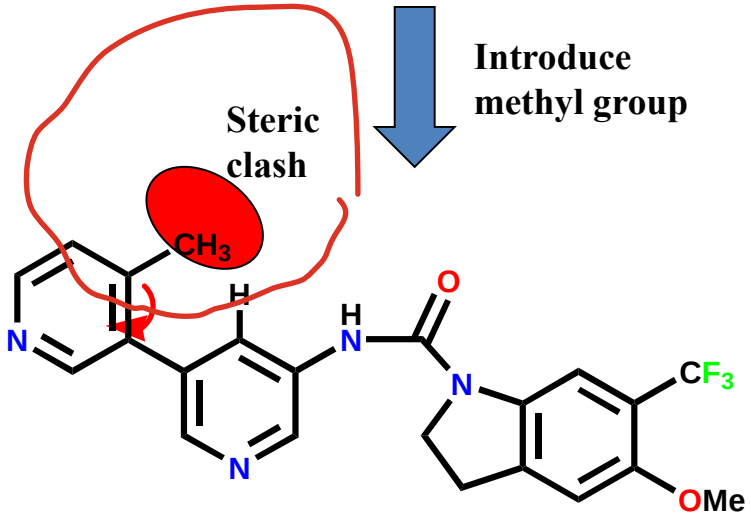
In certain situations, a quite simple substituent can hinder the free rotation of a single bond.



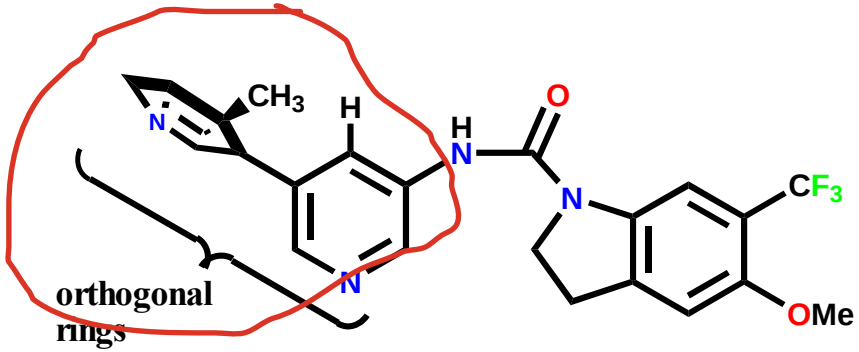





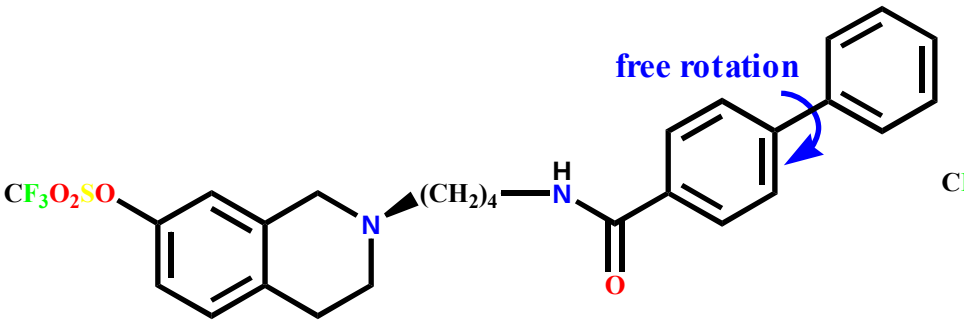

**Serotonin antagonist**




**Introduce methyl group**

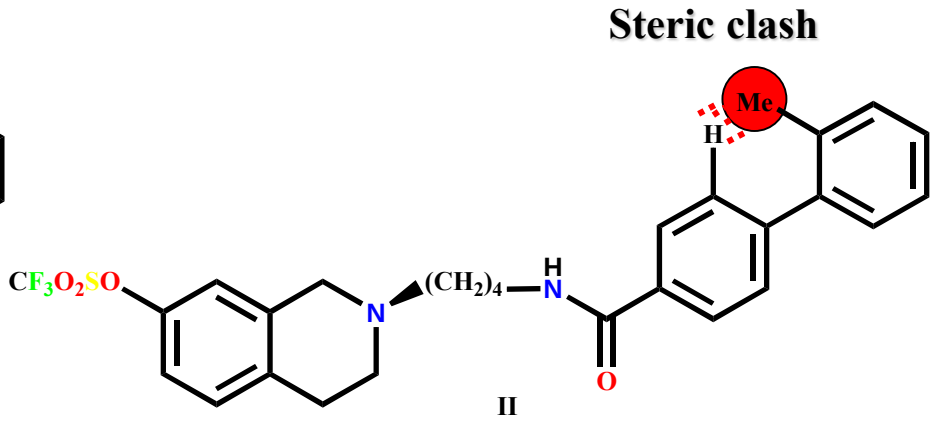



**Increase in activity**
**Active conformation retained**




I

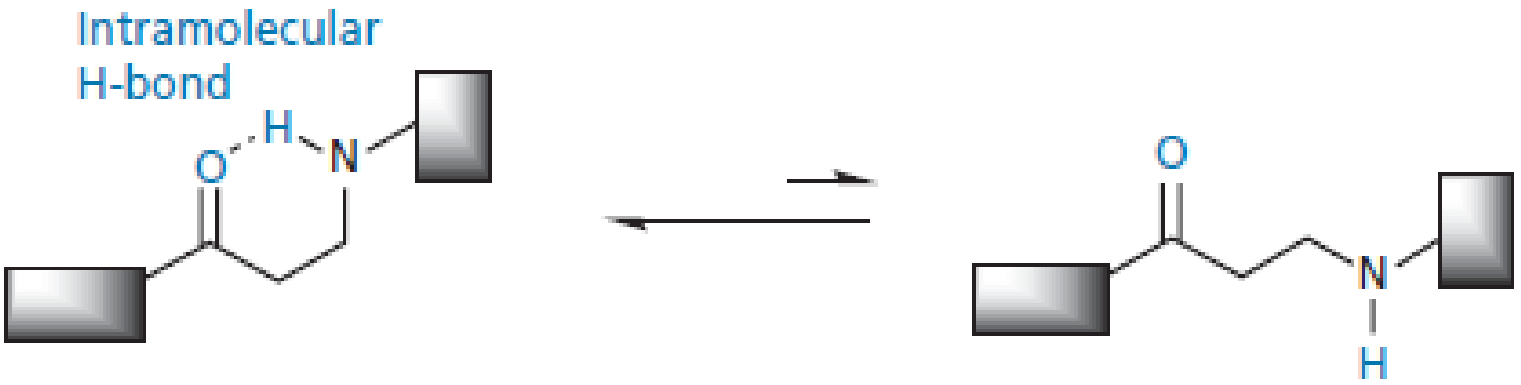
**$D_3$ Antagonist**



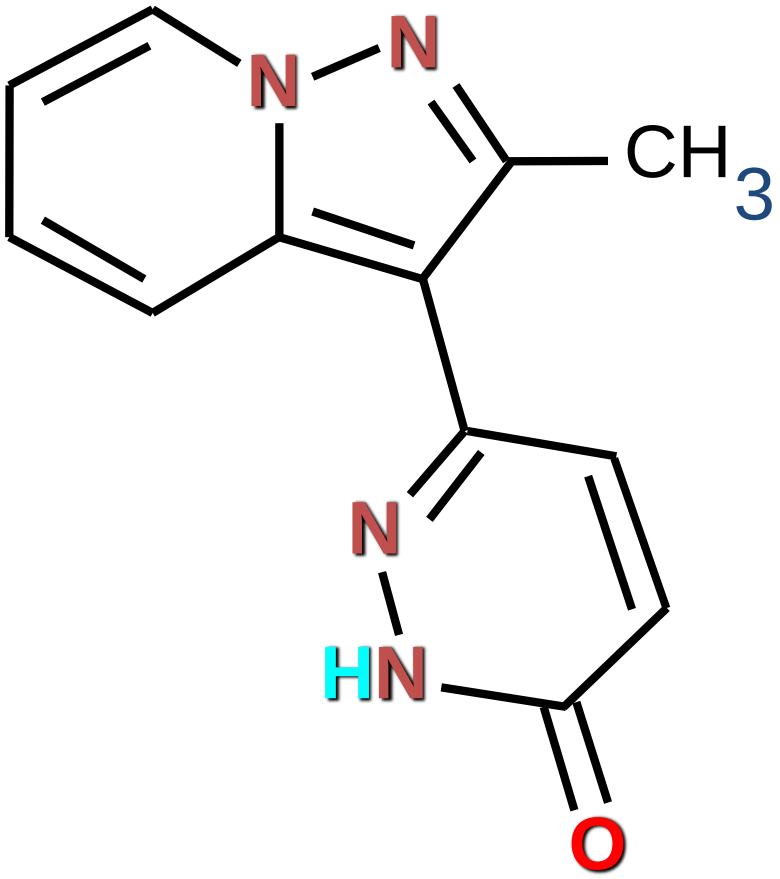

II

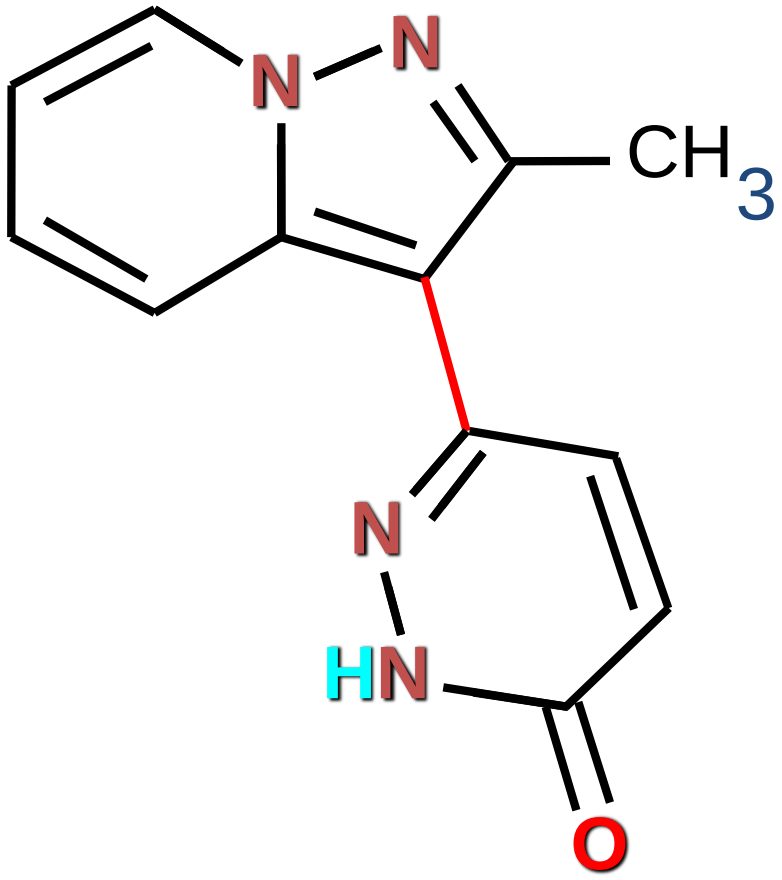
**Inactive**
**Active conformation disallowed**

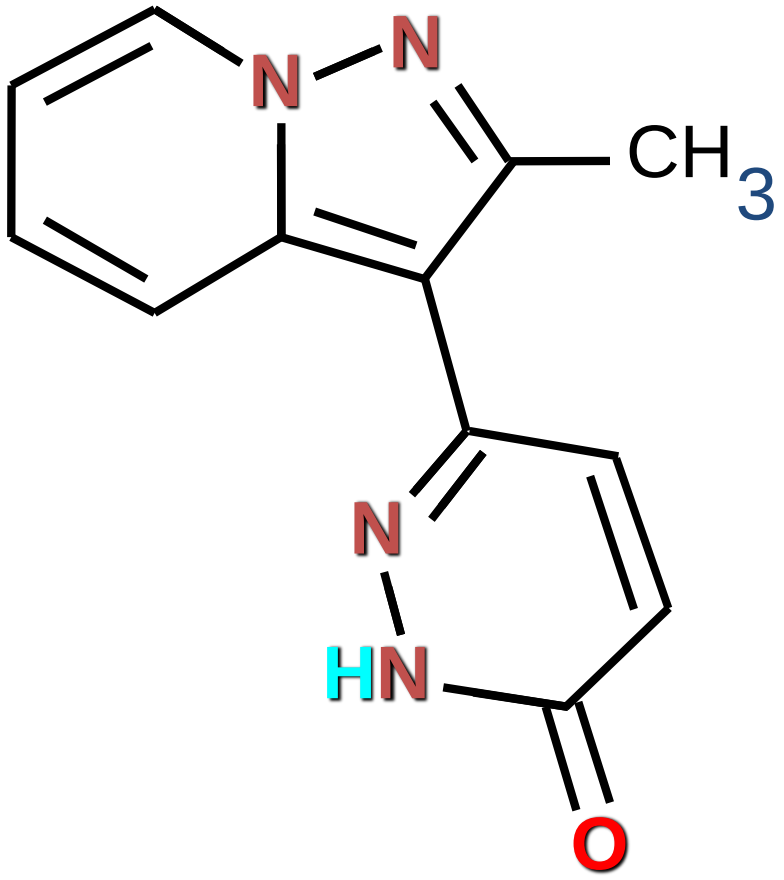
# 4-intramolecular hydrogen bonding

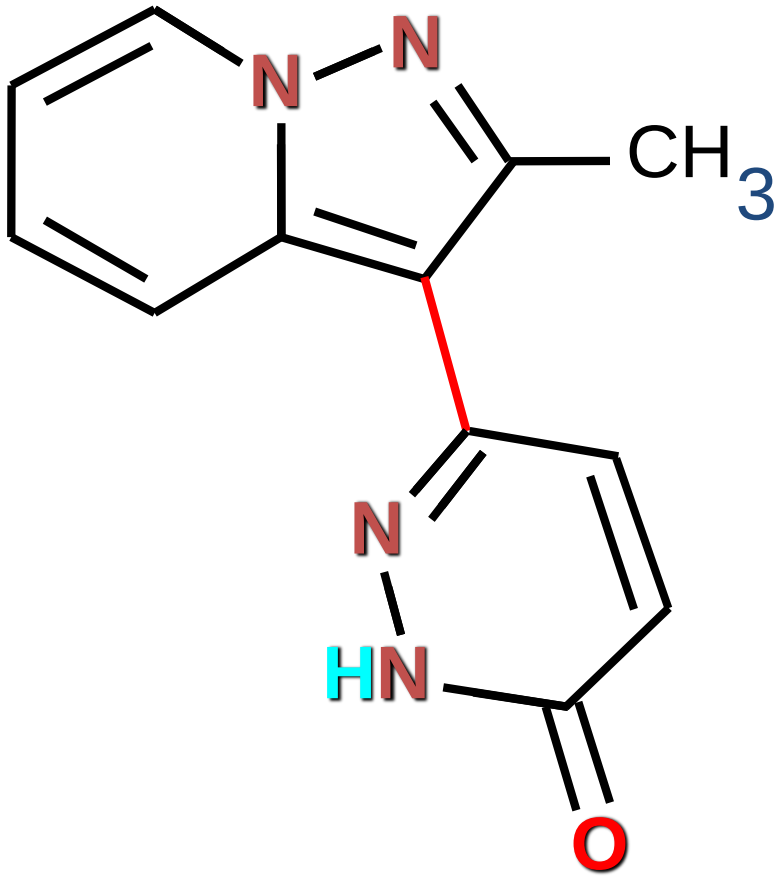


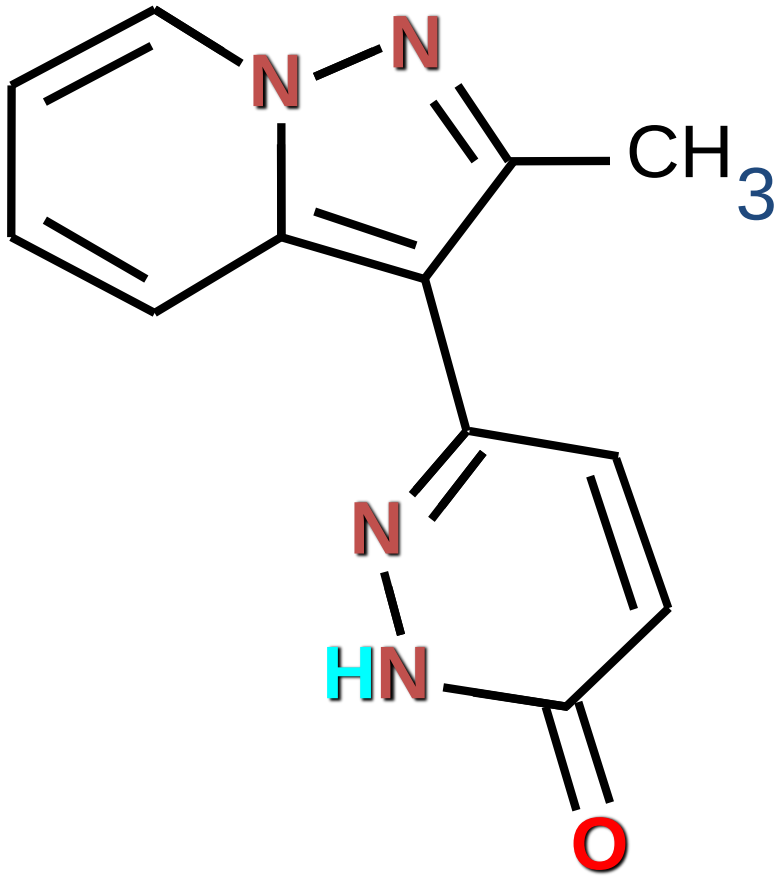
# Identification of an active conformation by rigidification




N
N
CH3
N
HN
O

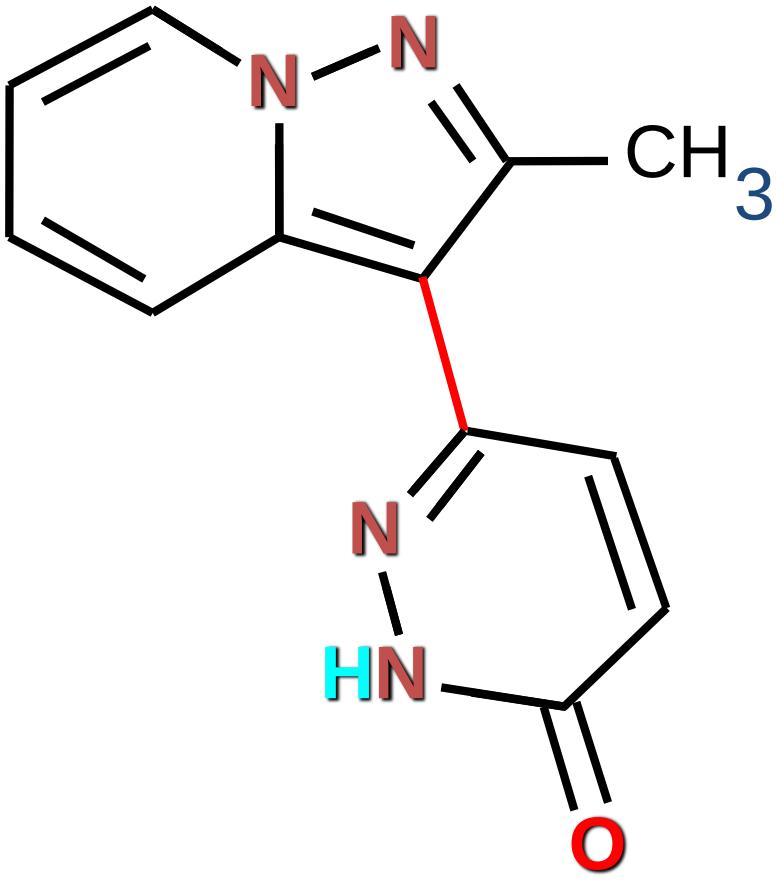

N
N
CH3
N
HN
O

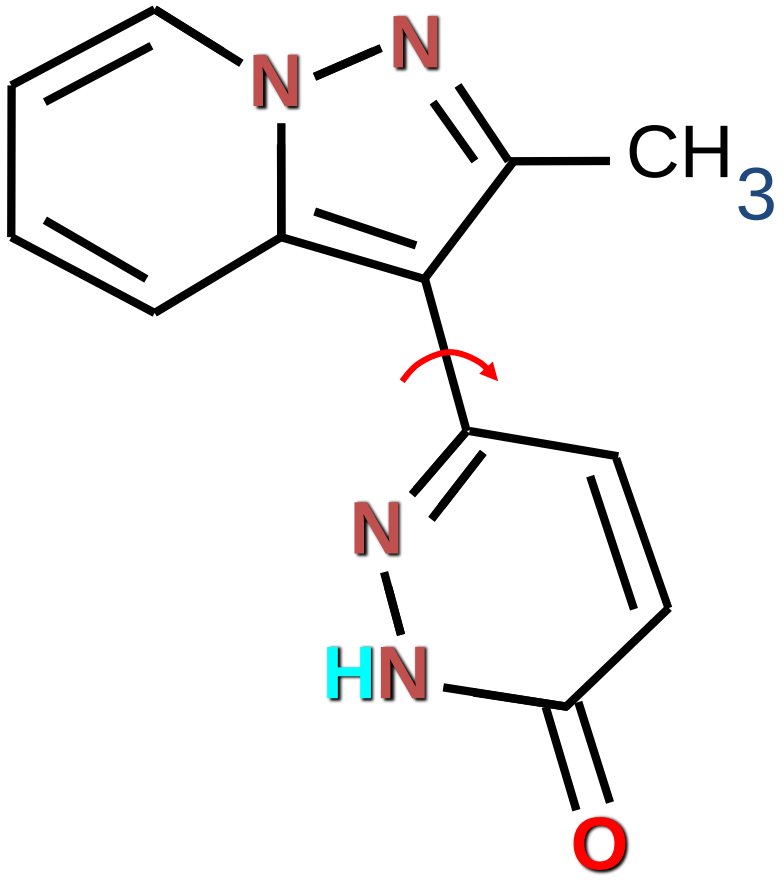

N
N
CH3
N
HN
O

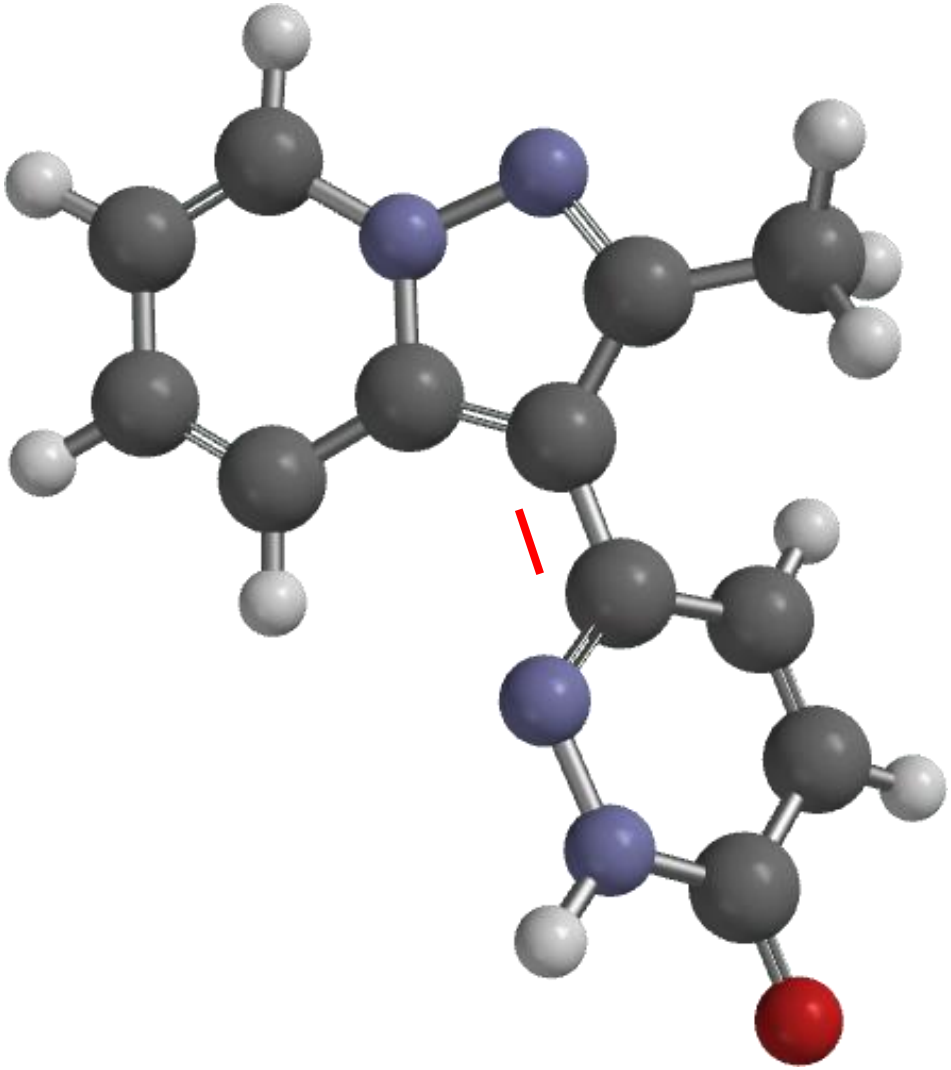

N
N
CH3
N
HN
O

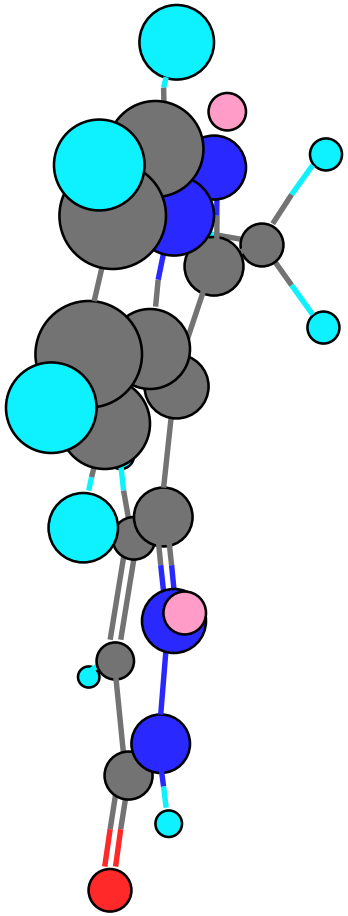
# 7.9 Rigidification

## Identification of an active conformation by rigidification

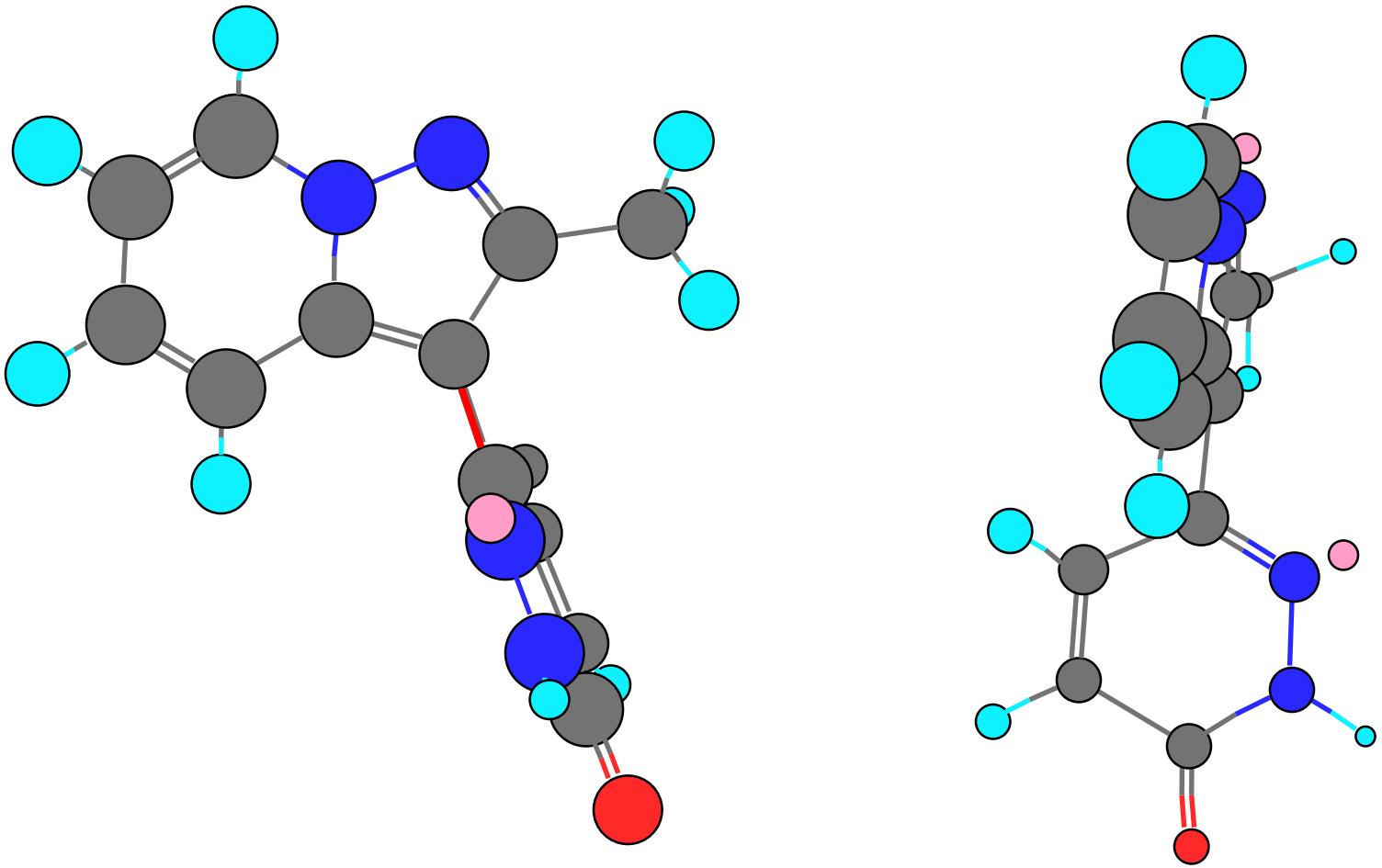



N
N
CH3
N
HN
O



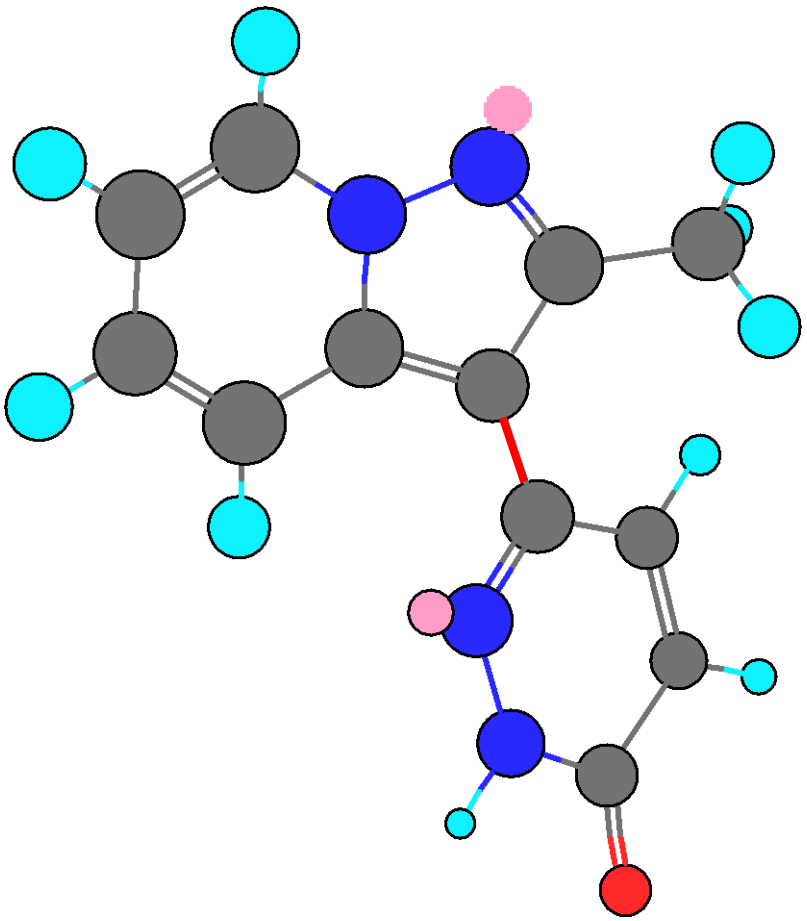


## Planar conformation



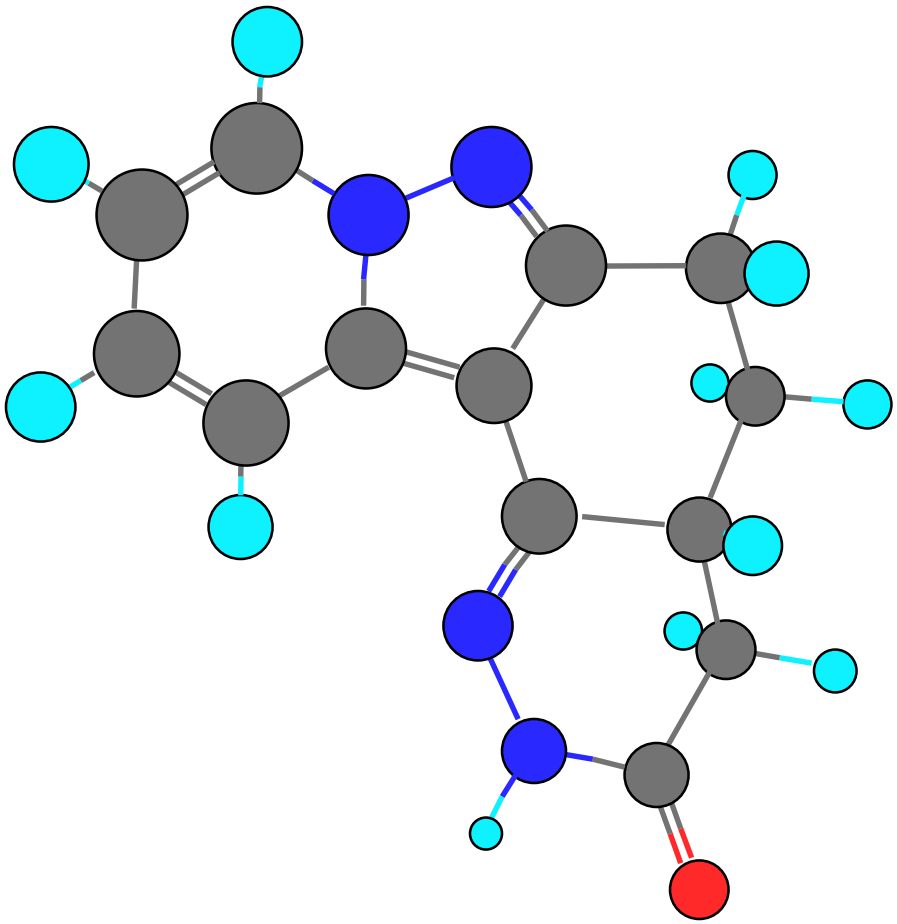
## Orthogonal conformation

Redo 3d diagrams



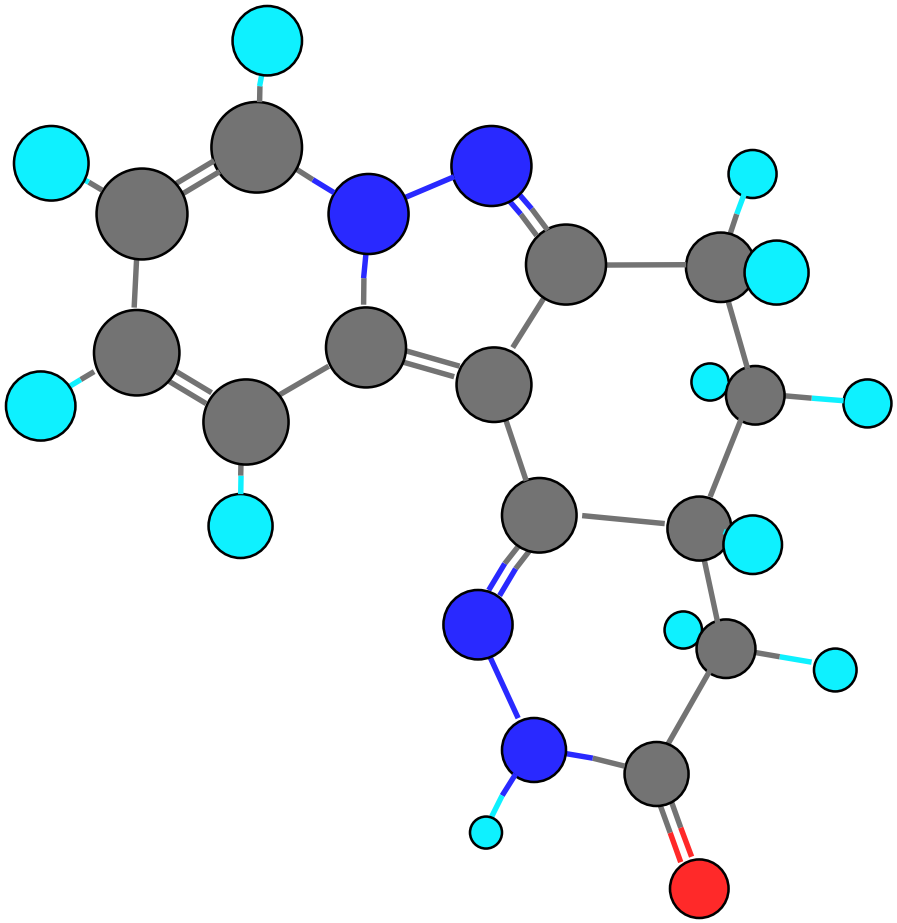
**CYCLISATION**

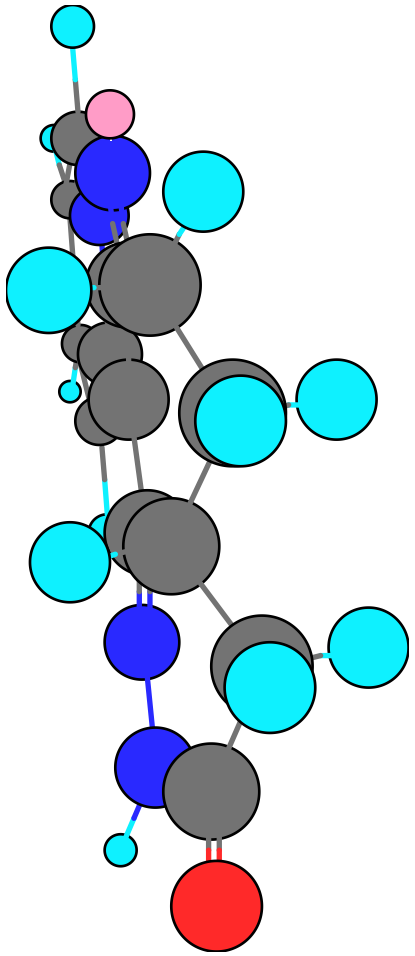
Redo 3d diagrams



# CYCLISATION

Redo 3d diagrams

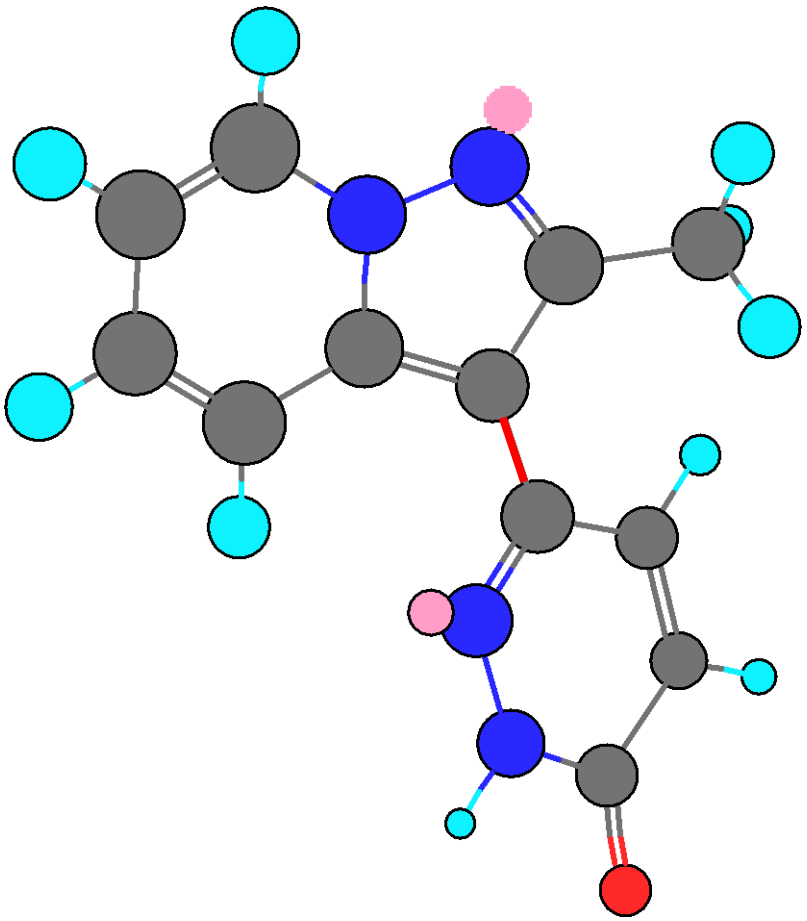




# CYCLISATION

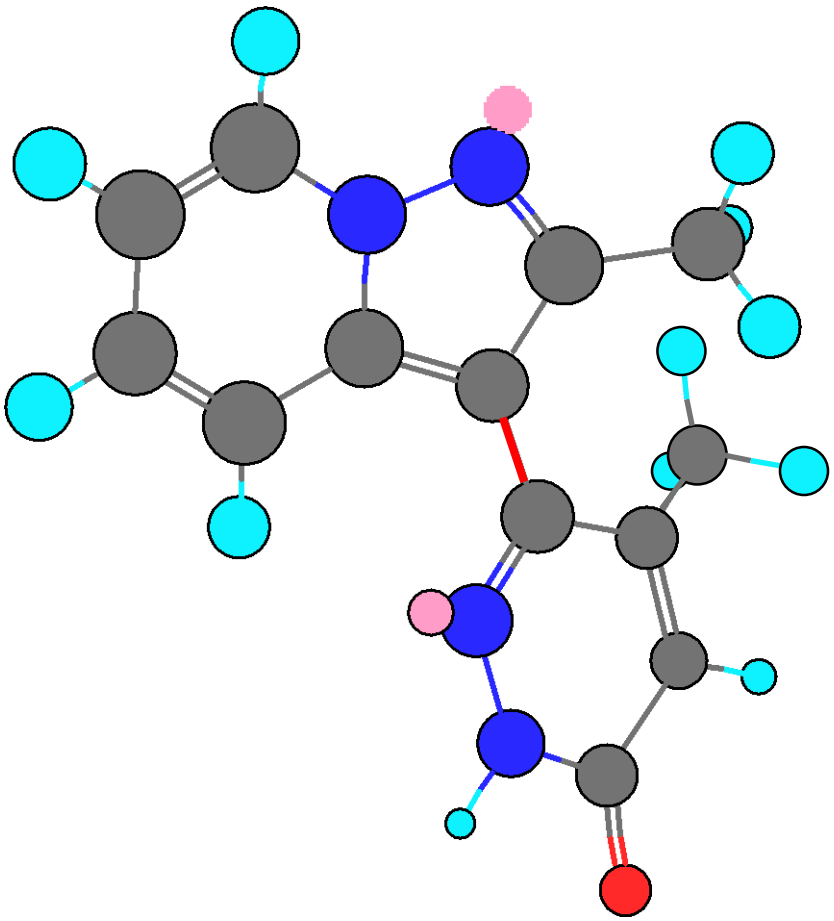
## Locked into planar conformation

Redo 3d diagrams



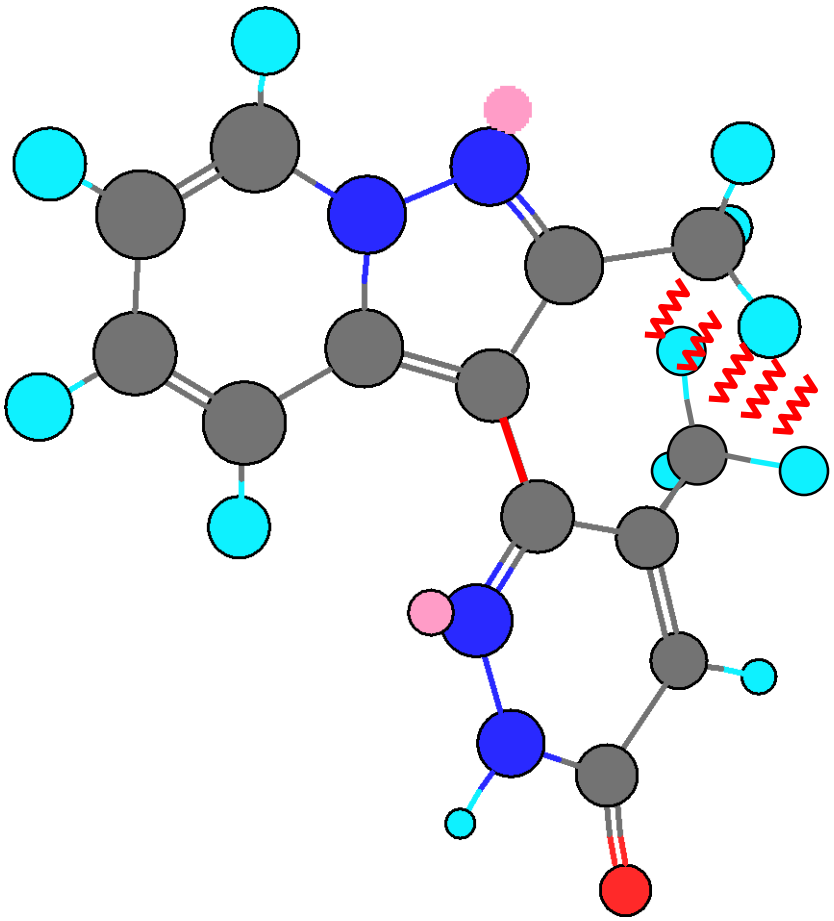
# STERIC HINDRANCE

Redo 3d diagrams



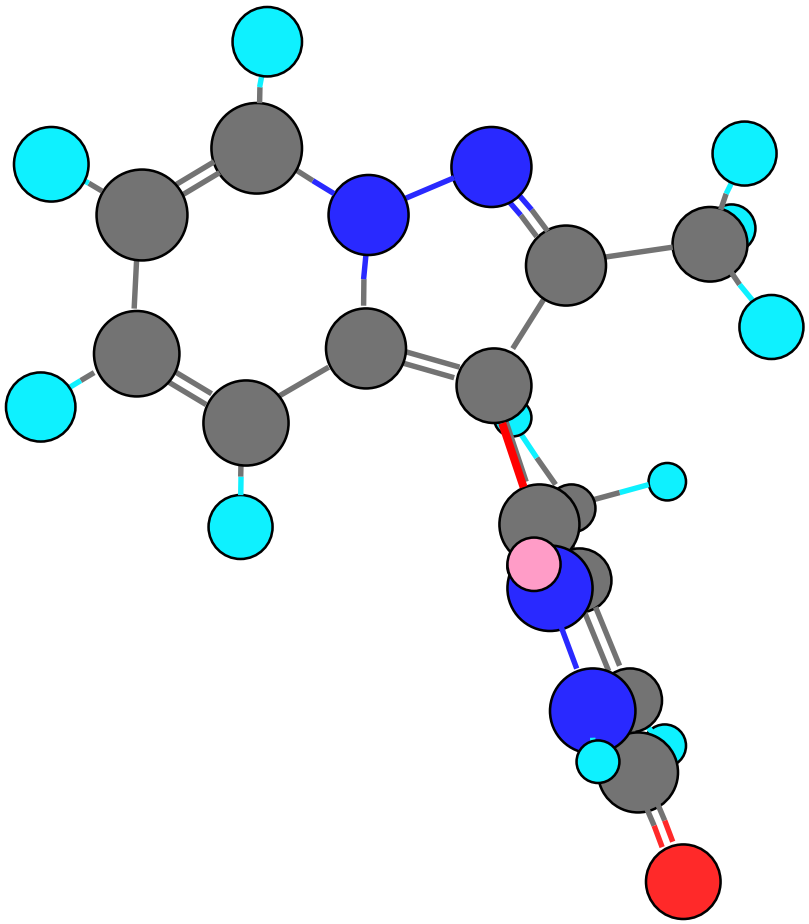
# STERIC HINDRANCE

Redo 3d diagrams



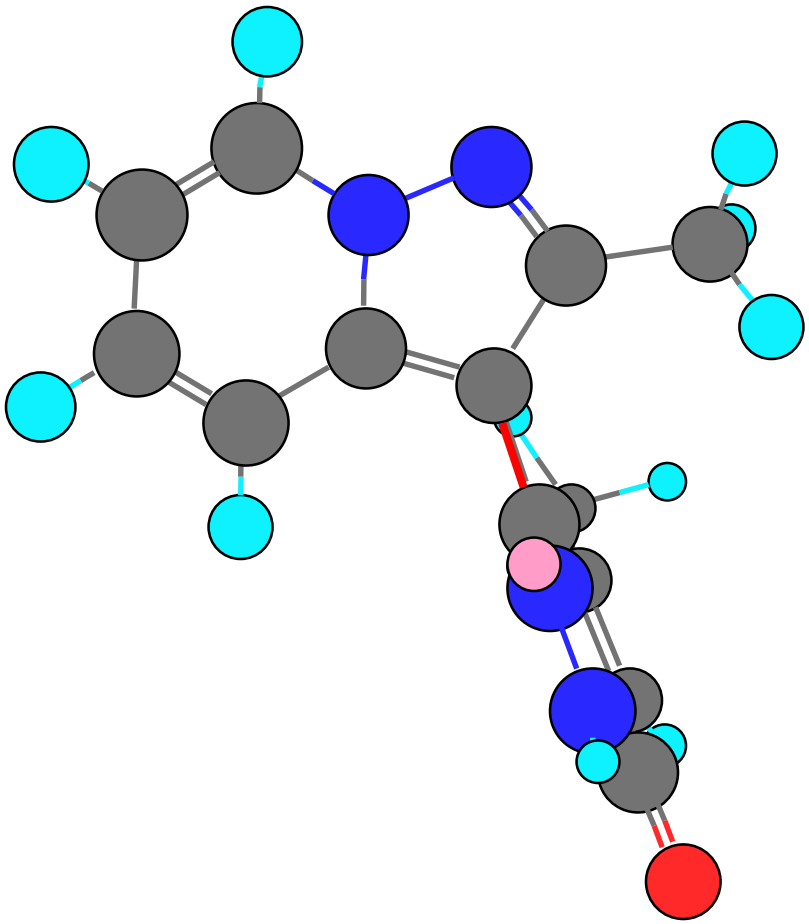
# STERIC HINDRANCE

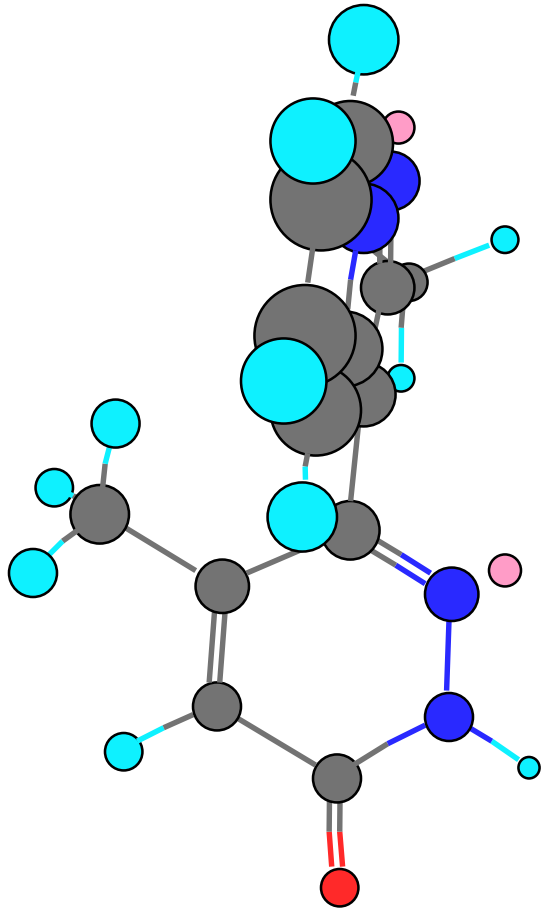
Redo 3d diagrams



# STERIC HINDRANCE

Redo 3d diagrams





# STERIC HINDRANCE

Redo 3d diagrams

**Locked into orthogonal conformation**

# END OF LECTURE 1